Signal Hoist: Reading the Signals of Illegal Fishing

A visual-analytics report on the IEEE VAST Challenge 2023 (Mini-Challenges 1, 2 and 3)

Entry Name: Signal Hoist
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Interactive version: https://amukherjee1991.github.io/Vast23/
Source code: https://github.com/amukherjee1991/Vast23
Answer sheets: MC1.md · MC2.md · MC3.md

This is an independent portfolio response written after the challenge, not an official 2023 entry. Oceanus, FishEye International and every entity named here are fictional and come from the challenge data.

Summary

FishEye International, an NGO that fights illegal, unreported and unregulated (IUU) fishing in the fictional island nation of Oceanus, hands analysts three knowledge graphs: entities and relations extracted from news text by NLP (MC1), seven years of customs shipment records with twelve bundles of AI-predicted links (MC2), and a corporate register of companies, owners and contacts (MC3). Each graph is noisy in a different way. Each one also contains a structure that looks like evidence but turns out to be an artefact of how the data was made.

The work produced three findings, each with a stated confidence:

1. **MC1.** 88 of the graph's 11,069 relations carry a weight below 0.8. They do not fall at random: they join into one 79-node web that random edges never produce ($p = 0.001$). Two-thirds of it is ownership, and three of the four tipped entities sit inside it. The web is a lead, not a verdict. The strongest structural suspect, **8327**, gets only *low* confidence, because almost every edge behind it is one the extractor itself scored as uncertain.
2. **MC2.** Only five of the twelve predicted-link bundles look like the real trade graph. Adding them changes no verdict. **Eight** pairs of fish traders fit the pattern of a company that stops trading and reappears under a new name, inheriting its partners. The clearest is receiver **−811 → −206** (Suspect, *high* confidence).
3. **MC3.** The register's 7,093-node "giant business group" is a name collision: remove the 651 multi-company people with common surnames and it falls to 306 nodes, which random removals never achieve. What stays anomalous is small: **BG00177** (three fishing-named firms sharing one owner and one contact across two countries) and **BG00001.061** are Suspect, with *medium* confidence.

Every number in this report comes from a deterministic Python pipeline over the raw challenge data. Every figure is a screenshot of the live, interactive site, where each claim sits beside the view that produced it.

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1. Approach

1.1 Principles

Four rules shaped the work:

- **Claims stay next to their evidence.** Each finding on the site is a sentence placed beside the chart, network or table that supports it. Every chart has a table view and names its source file.
- **Every structural claim is tested against chance.** A pattern counts as evidence only if a null model rarely produces it. The null models are random edge draws, degree-matched node sets, same-size random samples of real trade, and removing the same number of random people.
- **Verdicts come from rubrics written before they were applied.** The rubrics say which signals count, how many points each is worth, and where the thresholds sit. They are published in the JSON next to the scores, so a reviewer can change a weight and see what moves.
- **Confidence measures how solid the evidence is, not how bad the finding sounds.** There is no ground truth in this challenge, so every verdict states which edges and which assumptions it rests on.

1.2 The verdict scale

All three challenges use one scale, drawn as flags from the International Code of Signals. Each flag's meaning at sea matches the finding:

Flag	Verdict	Meaning at sea
L	Suspect	"Stop your vessel instantly"
U	Possible	"You are running into danger"
X	Insufficient evidence	"Watch for my signals"
N	Unlikely	"Negative"

Numeral pennants show confidence: one hoisted means low, two medium, three high. Every flag pairs a pattern with a colour, so the code still reads in grayscale and for colour-blind readers.

1.3 The interactive site

The site is a static React and D3 application with one chapter per mini-challenge, plus a home page and a method page. The visual world is a **night sea**. The page is dark water (#07111b), and light lives only in the data: bioluminescent teal, amber, rose-red and blue marks, drawn additively. The signal lamps carry the verdicts. Bodoni Moda sets the story, Archivo the interface, and JetBrains Mono the measurements.

Each chapter opens with its headline claim and the strongest verdicts, hoisted as lamps. Next comes a **scroll story**: a full-bleed stage pinned to the screen while six short text steps scroll past and drive it. These are a constellation of the MC1 graph, currents of trade into Oceanus for MC2, and a "galaxy" of companies and people for MC3 that breaks apart. Below the story, the chapter answers each official question (Q1…Qn) with interactive evidence views. It ends with a closing bridge to the next chapter. The keys **n** and **p** step between chapters, and **0–4** jump to one.

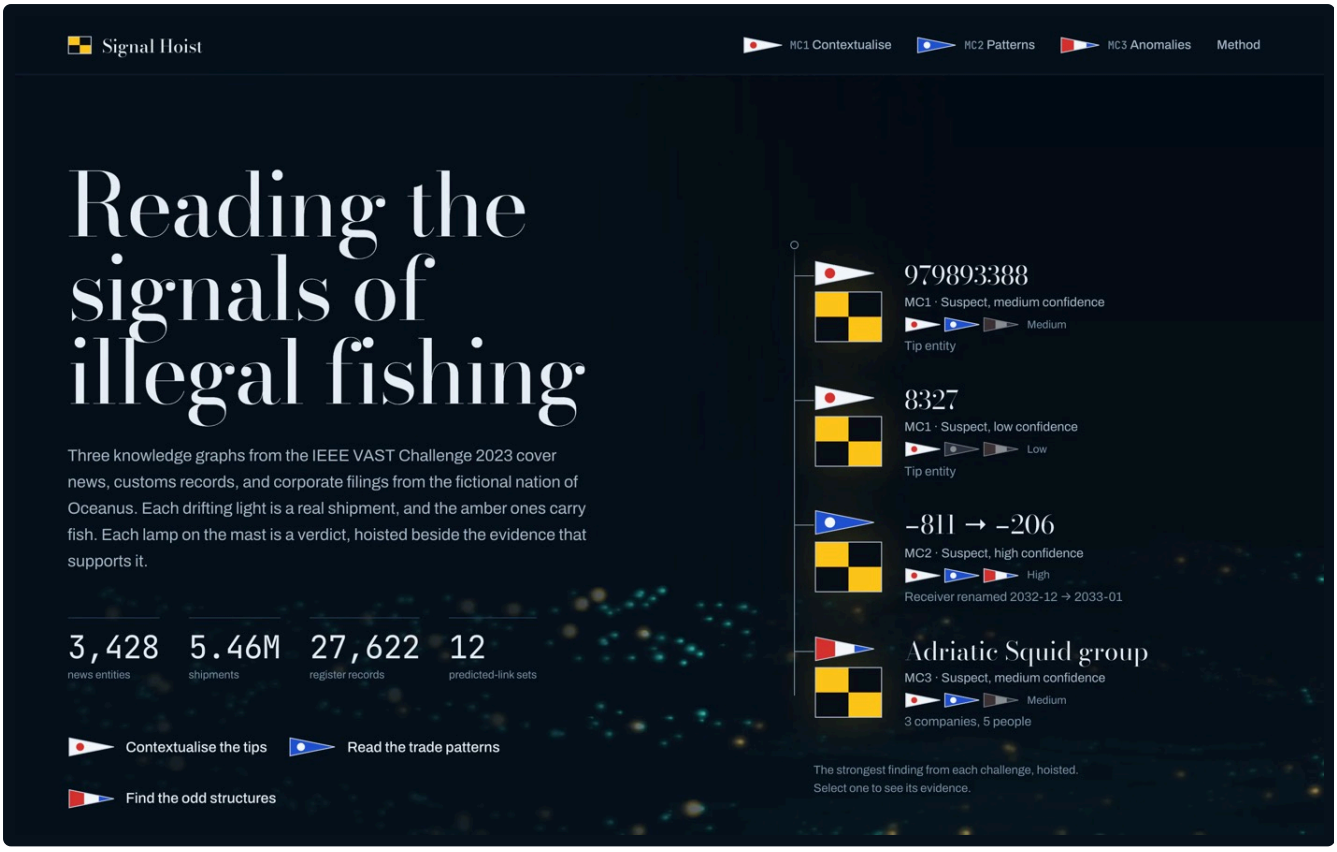


Figure 0.1. The home page. The mast on the right hoists the strongest finding from each challenge; the drifting lights below are real shipments, with fish in amber.

2. Data and pipeline

A Python 3.12 pipeline (`pipeline/build_all.py`, run by `npm run data`) reads the raw data once. It writes small JSON files that the browser loads, so the site never handles the 1.5 GB trade graph: 7 files and 0.77 MB for MC1, 7 files and 0.54 MB for MC2, and 6 files and 1.28 MB for MC3.

	MC1: news graph	MC2: trade graph	MC3: corporate register
Raw size	3,428 nodes, 11,069 edge rows	34,576 companies, 5,464,378 shipment edges, 12 bundles	27,622 nodes, 24,038 links
Cleaning	Integer and string ids kept apart (`77` ≠ `"77"`, 11 collisions); 322 duplicate rows and 124 multi-type pairs recorded rather than dropped	155,291 exact duplicates (2.8%) removed, leaving 5,309,087; HS codes repaired (lost leading zero, 5-digit codes in bundles)	807 multi-source links split; 374 duplicate company names merged; list-wrapped fields flattened
Model	Weighted multigraph; ego networks, paths, centrality, Louvain	245,807 shipper→receiver pairs × 84 months (2028-01 to 2034-12)	16,697 companies, 21,265 persons, 24,931 company–person edges; 13,587 components
Key tests	1,000 random 88-edge draws; degree-matched suspect sets; leave-one-out; 300 reweightings	200 same-size samples of real 2034 trade per bundle; k-means cross-check	random-person removal; bootstrap cluster stability; Isolation Forest

Determinism. Node order is sorted, seeds are fixed, and ties are broken by id, so two runs produce byte-identical files. The MC3 build is also stable across `PYTHONHASHSEED` values. That last property took work: `Graph.subgraph(set)` iterates in hash order, and Louvain returned 85 or 87 communities depending on the run until every graph was rebuilt with sorted insertion.

Data notes worth knowing. The MC1 data notes declare 3,721 nodes and 7,422 edges, but the file has 3,428 nodes and 11,069 edge rows, and one row misspells its weight field. The notes define neither `weight` nor the direction of ownership edges. In MC2, the graph's values are in USD and the bundles' in OMU, and only 281 graph edges carry OMU. In MC3, 77% of companies have no product text and people are identified by name alone.

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3. MC1: a hidden layer of weak links

FishEye's tip names four entities: **Mar de la Vida OJSC**, **979893388**, **Oceanfront Oasis Inc Carriers** and **8327**.

3.1 The low-weight layer

99.2% of the 11,069 edge rows have a weight between 0.80 and 0.99. The other 88 rows, weighted 0.03 to 0.77, are not scattered:

- **Size.** They form one connected component of 79 nodes. In 1,000 random draws of 88 edges, the largest component averaged 10.6 nodes and never exceeded 29 (p = 0.001).
- **Ownership.** 66% of the layer's edges are ownership, against 18% of all edges (p = 2e-23). The layer holds two circular-ownership loops and a five-hop directed chain.
- **FishEye.** The "FishEye International" node has 24 edges, 19 of them in this layer. The layer may encode FishEye's own reporting.
- **Suspects.** Three of the four suspects are inside the layer (p = 5e-5), and the fourth is one hop away.

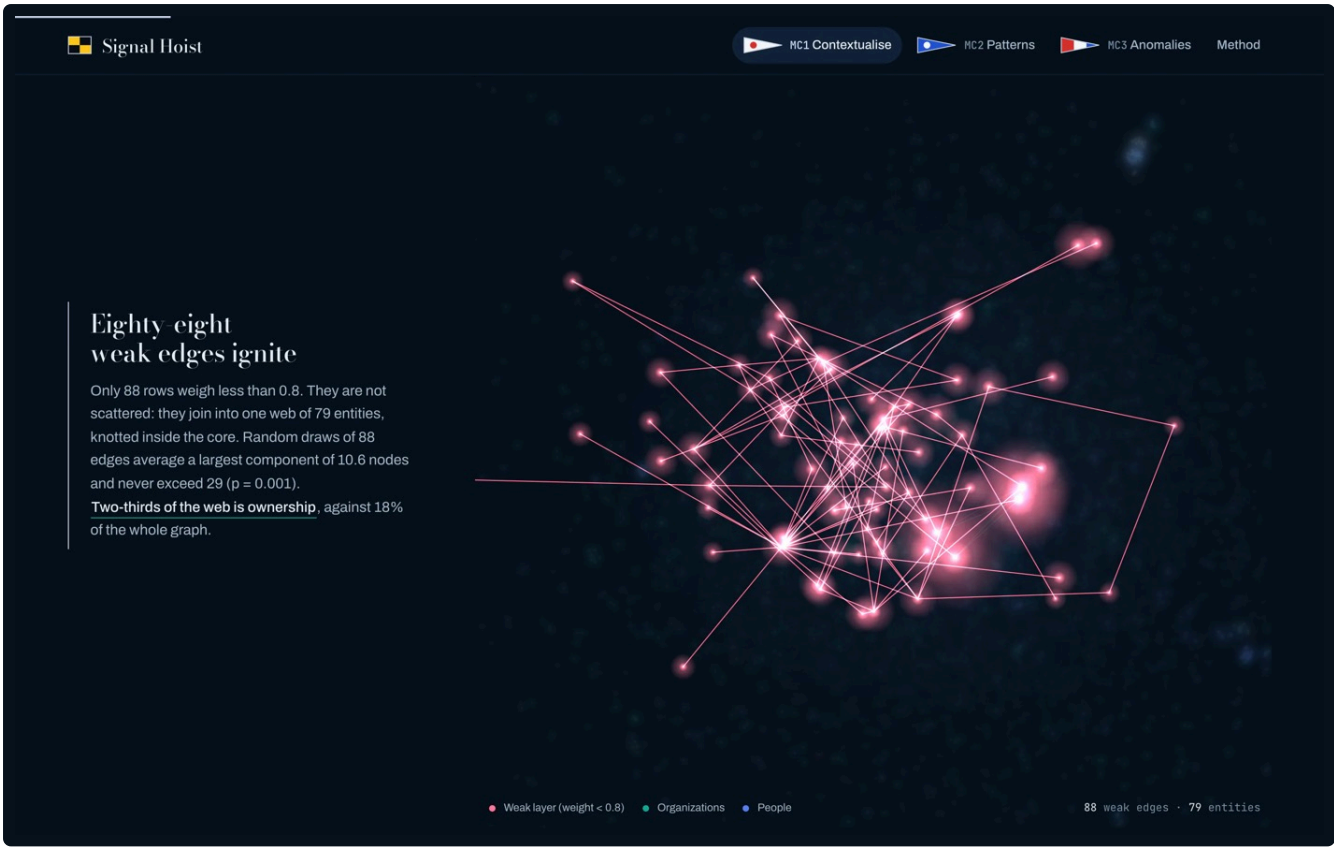


Figure 3.1. MC1 story step "Eighty-eight weak edges ignite". The 88 weak rows, lit rose over the faint full graph, knot into one 79-node web.

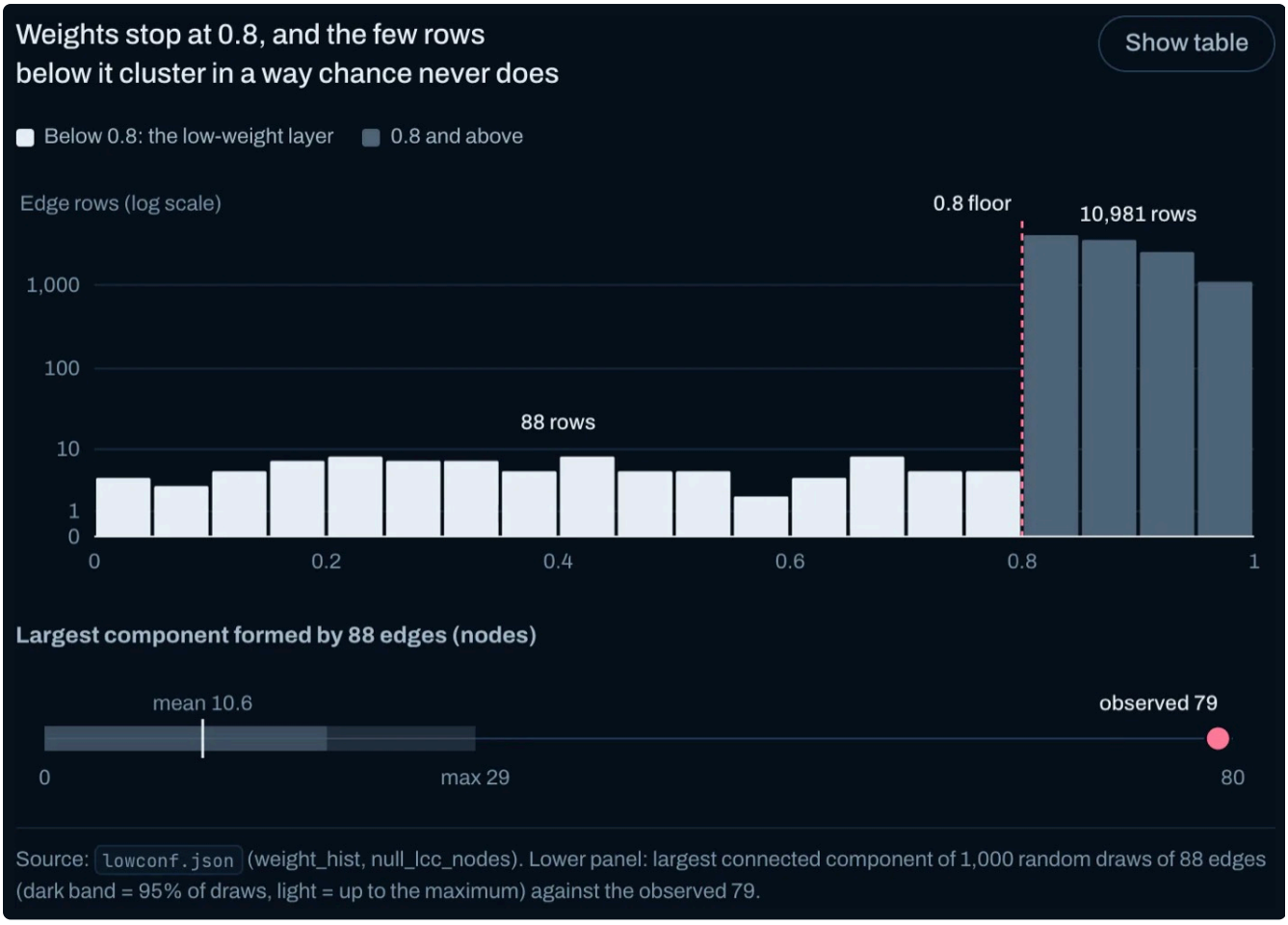


Figure 3.2. Weights stop at 0.8. Below the floor, 88 rows form a 79-node component, against a mean of 10.6 and a maximum of 29 in 1,000 random draws.

3.2 The four suspects

Verdicts come from a published seven-signal rubric: in the layer, FishEye tag, weak owner, ownership chain to another suspect, broker, shell marker, and keyword. A score of 4 or more points is Suspect, 2.5–3.5 Possible, 1–2 Insufficient evidence, and below 1 Unlikely. Confidence counts the full-point signals that rest on edges weighted ≥ 0.8 or on graph structure alone: 4 or more is High, 2–3 Medium, and 0–1 Low.

Entity	Verdict	Confidence	Points	Why
8327	Suspect	Low	5.0	Owned by Seabreeze Foods through a 0.12 edge, and FishEye tags Seabreeze (0.13). Heads a six-hop ownership chain through #40213337, Sea Harvest, Amanda Mckenzie and Mar del Oeste Pic to 979893388 and Oceanfront Oasis (weakest link 0.32). Membership edge to the fragment "Illegal". Nearly every signal rests on weak edges.
979893388	Suspect	Medium	4.0	Nameless hub: degree 177 and betweenness 0.096, both at the 99.8th percentile. One of a 10-hub dense core. Owns 10 and is owned by 12; its upstream tree reaches 307 entities. Owns Oceanfront Oasis through a strong 0.84 edge. "ipoa-iuu" and "poached" lie two hops out.
Mar de la Vida OJSC	Possible	Low	3.0	Untyped; 15 neighbours, mostly people and memberships. Its only weak edge is a FishEye membership (0.46). No ownership chain.
Oceanfront Oasis Inc Carriers	Possible	Low	3.0	Four incoming edges: FishEye (0.45), its owner 979893388 (0.84), and two family edges from nameless organizations (probably noise). A thin subsidiary of a suspect.

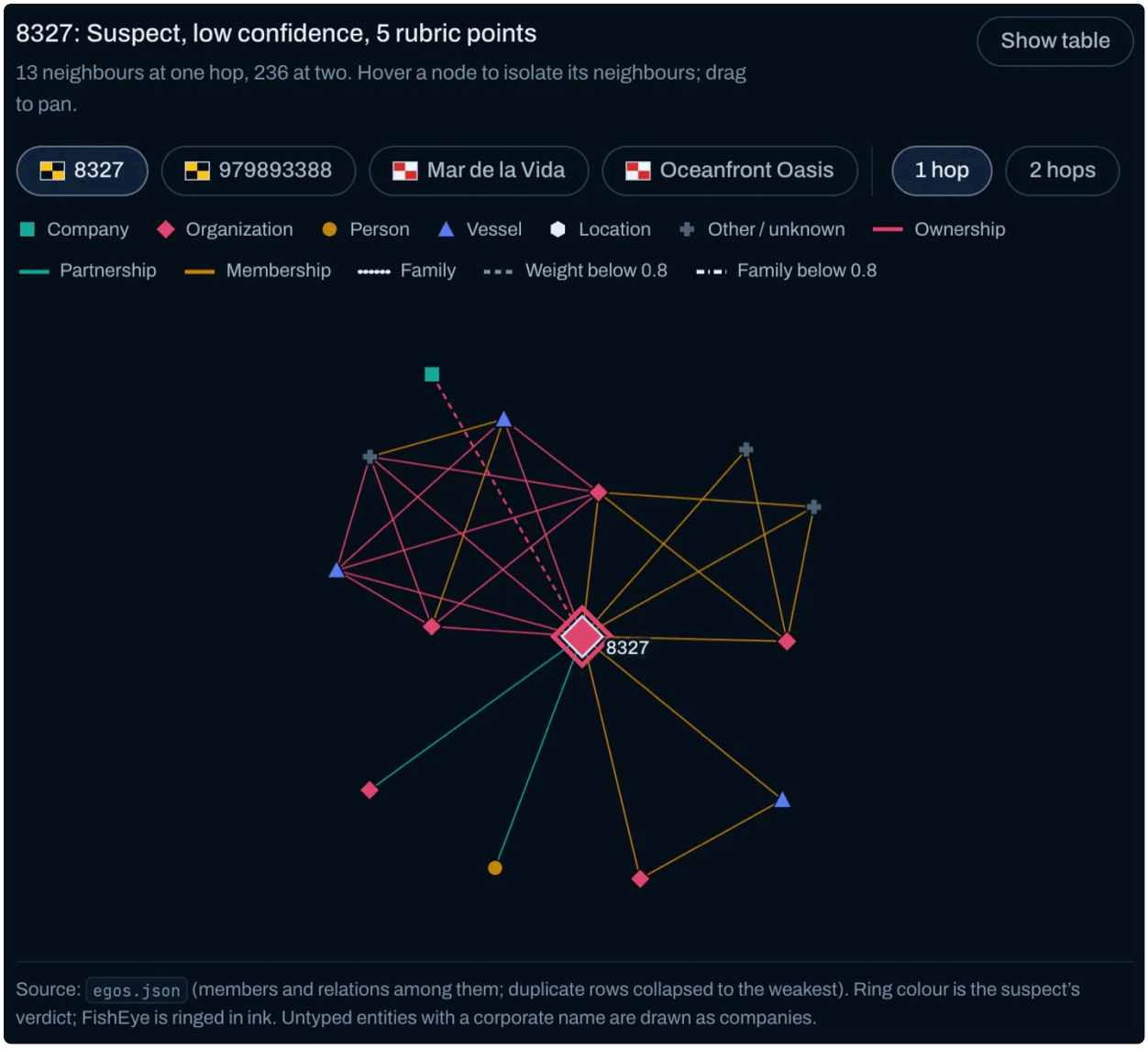


Figure 3.3. 8327 at one hop. Dashed edges weigh below 0.8, and the ring is its verdict lamp.

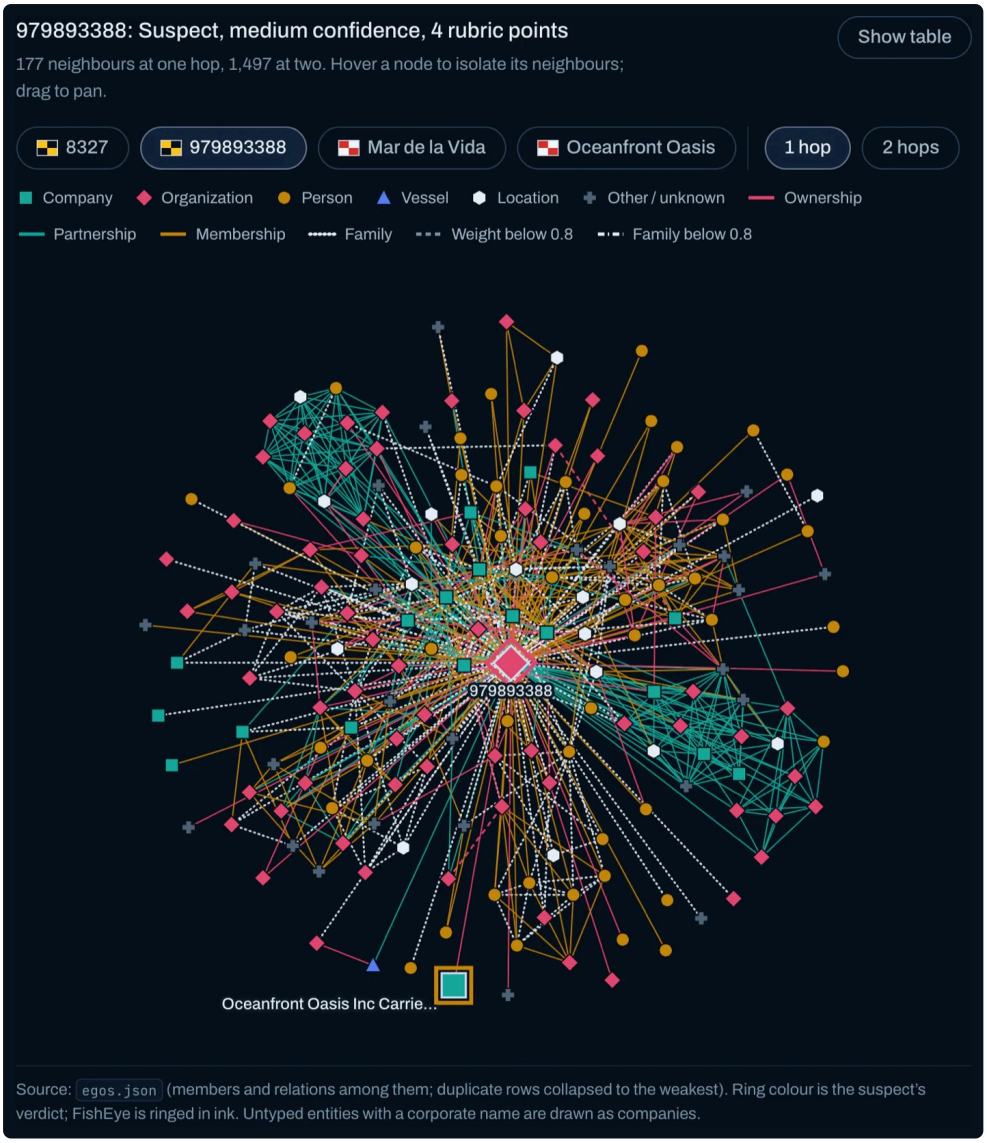


Figure 3.4. 979893388, a 177-neighbour hub. Its strong edge to Oceanfront Oasis is at the bottom.

Two pairs, not four suspects. The suspects split into two pairs. Mar de la Vida and Oceanfront Oasis are small named entities, each tied to the layer by one weak FishEye membership. 8327 and 979893388 are nameless ownership intermediaries that both own and are owned, and only they fire the rubric's shell marker.

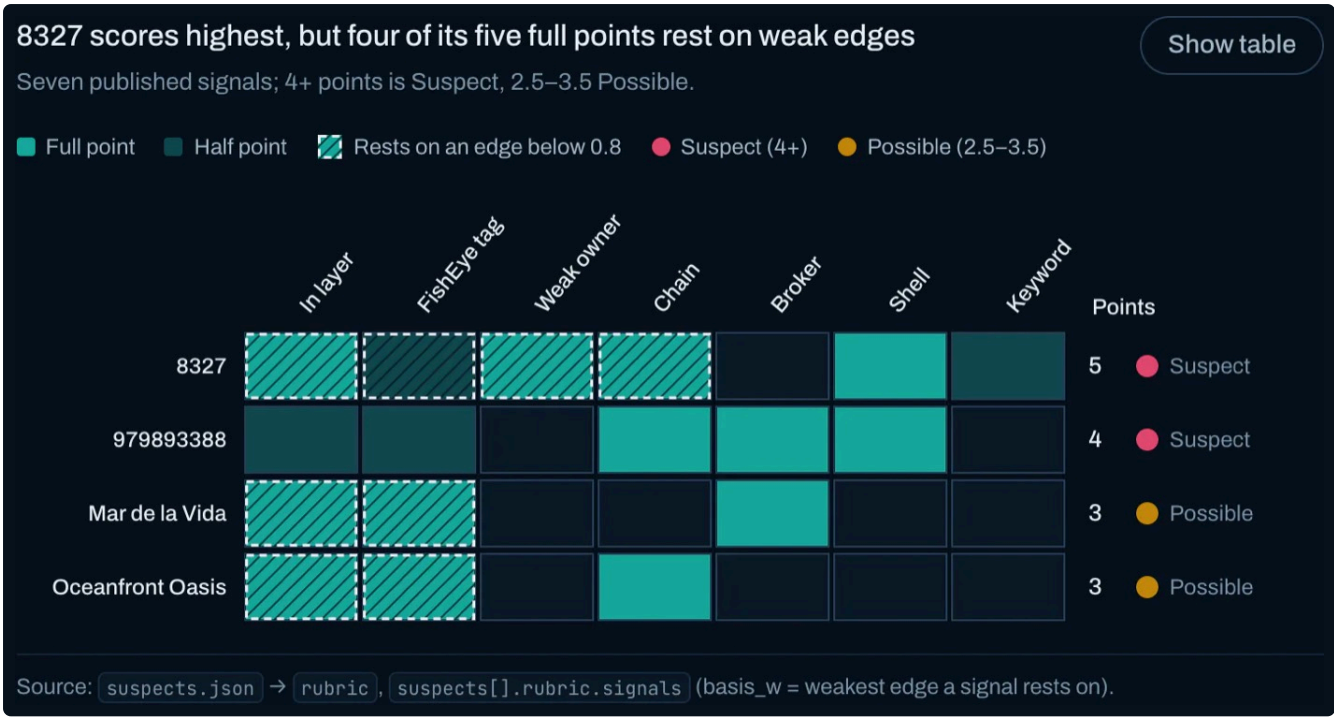


Figure 3.5. 8327 scores highest, but four of its five full points rest on edges below 0.8 (hatched).

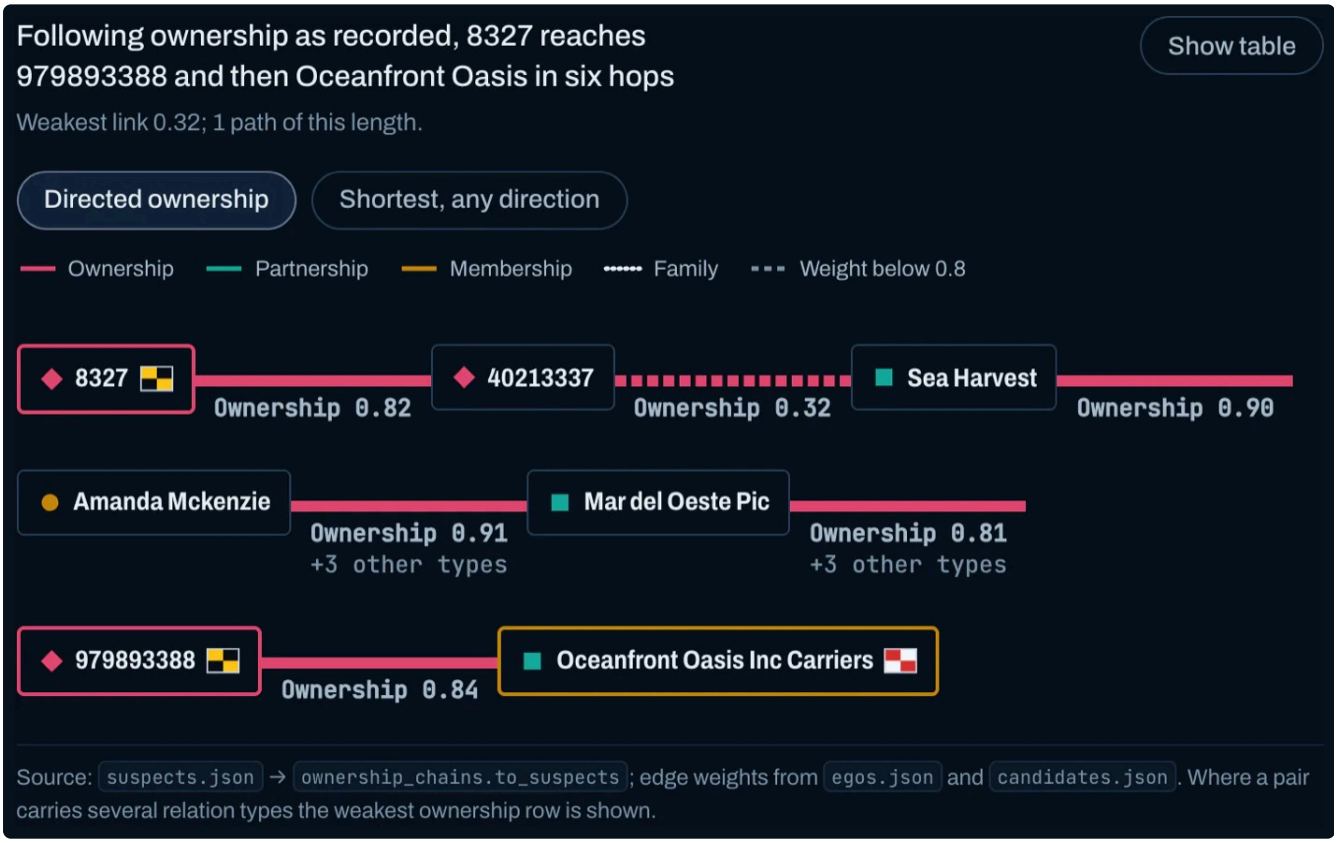


Figure 3.6. Following ownership as recorded, 8327 reaches 979893388 and then Oceanfront Oasis in six hops.

What is uncertain.

- **Distance.** The suspects sit 2.17 hops apart on average, against 3.84 for random pairs. But 8% of random four-node sets with the same degrees are as close, so their closeness is mostly what their degrees predict.
- **Communities.** Louvain puts them in three communities (modularity 0.69), and across five seeds only 0.33–0.63 of each suspect's community neighbours stay with it.

- **Ownership direction.** Read literally, the chain has the company Sea Harvest owning the person Amanda Mckenzie, so the direction cannot be trusted.

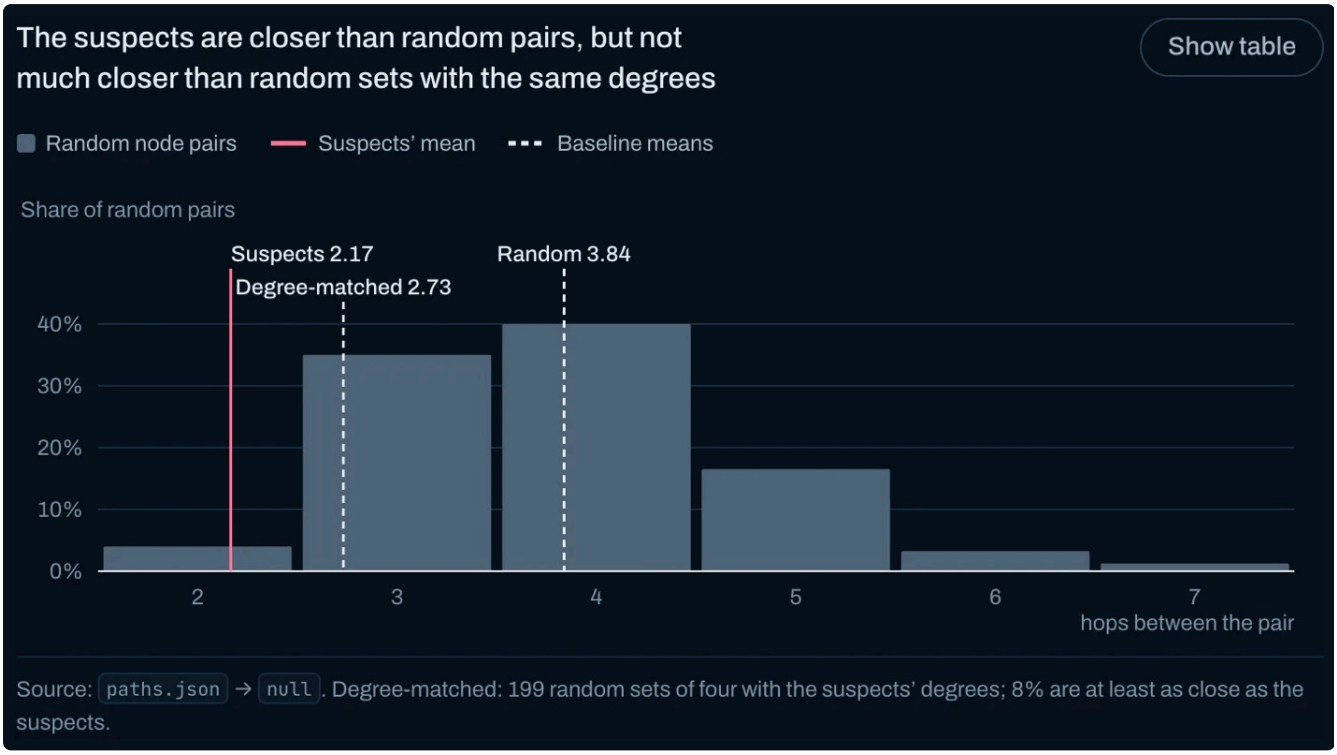


Figure 3.7. The suspects are closer than random pairs, but not much closer than random sets with the same degrees.

3.3 Who else FishEye should investigate

All 1,209 companies and organizations are scored on seven structural features, each converted to a percentile before weighting:

Feature	Weight
Low-weight edges	0.25
Proximity to suspects	0.20
Shared neighbours	0.15
Ownership counterparties	0.15
Relation-type profile similarity	0.10
Repeated or mixed-type edges	0.10
FishEye tag	0.05

Validation.

- **Leave-one-out.** Held out of the seeds, each suspect ranks 11th, 51st, 41st and 35th of 1,210 (top 1–4%). Degree alone places them 96th, 7th, 528th and 128th.
- **Weight perturbation.** Across 300 random reweightings, the mean Spearman correlation is 0.98 (minimum 0.89).
- **Ablation.** Dropping the low-weight feature keeps only 8 of the top 25, so the list is mainly a way to rank the layer. All 25 top candidates sit in the layer.

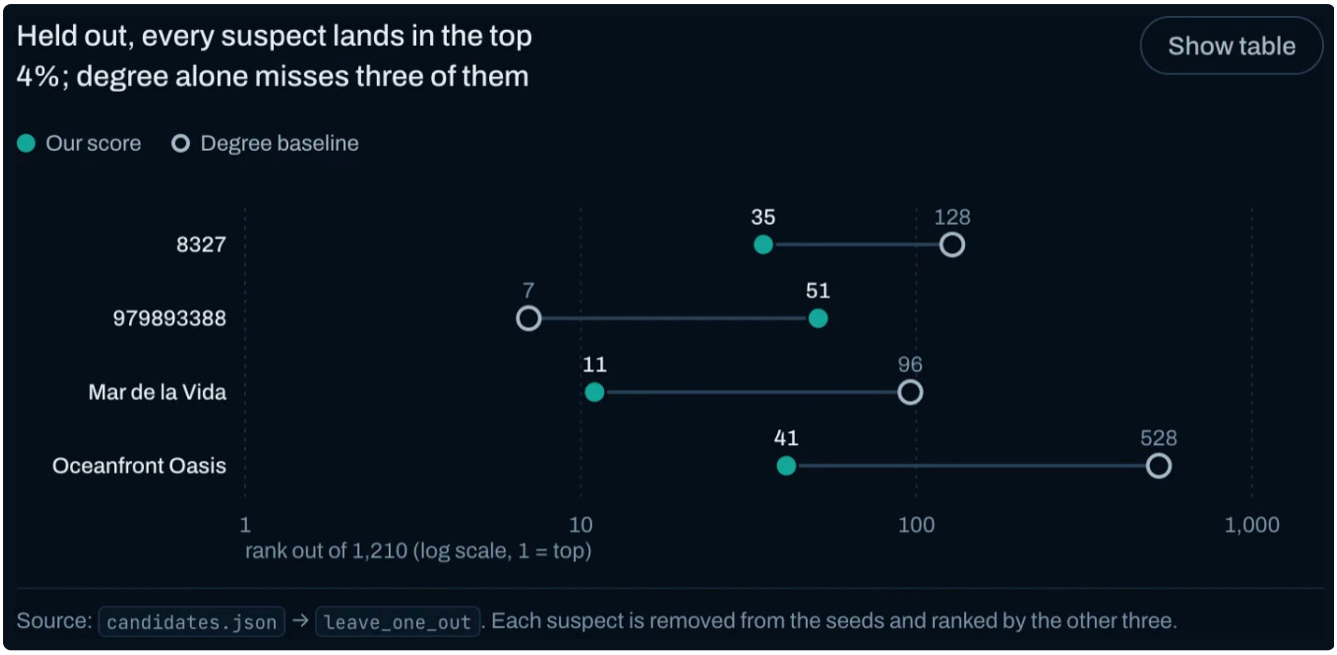


Figure 3.8. Held out, every suspect lands in the top 4%. Degree alone misses three of them.

Investigate first (rubric Suspect, medium or high confidence):

- **#903311212** (rank 1, score 0.97): a hub with 234 neighbours that owns 43 entities and has 42 owners. FishEye tags it, and it reaches 979893388 through Mar del Oeste Pic.
- **#48** (rank 18): the only *high*-confidence candidate. FishEye tags it through a 0.95 edge, and its ownership chain reaches 979893388.
- **#40213337** (rank 7, the holding 8327 owns), **#196159** (11) and **#83** (2).
- The hub-core members **Mar del Oeste Pic** (5) and **png xi Line** (6), and **sh rn y Corporation** (8).

The structures matter as much as the names: two ownership loops (#21538 → #52193 → #958085 → #55674, and American → #5421 → #81 → "l y S.A. de C.V. International" → #59), a five-hop chain from Darin Torres to #4942, and nine people joined by weak family edges who own layer companies.

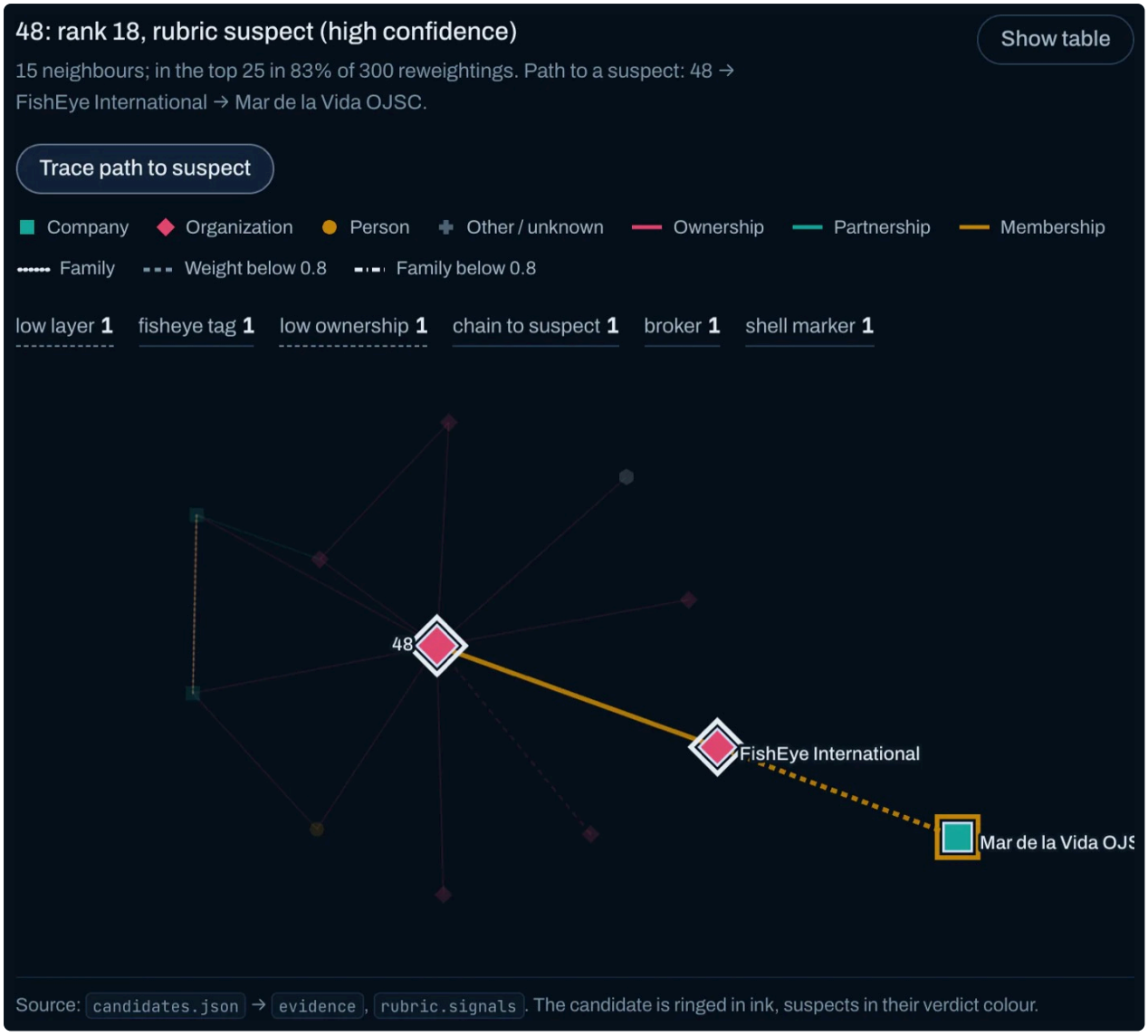


Figure 3.9. #48, the only high-confidence candidate, with its path to a suspect traced.

Confidence. Medium for the layer as a whole, which is statistically clear. **Low to medium** for each individual company, because its strongest evidence is also its least reliable.

4. MC2: trade patterns, predicted links and renamed traders

4.1 Temporal patterns

The market. Trade runs one way, into Oceanus: 62 of the 80 largest country-to-country flows start or end there, and Marebak → Oceanus alone carries 1.86 million shipments. Yearly volume peaks in 2030 at 831k shipments and falls to 688k in 2034. Fish (HS 0301–0309, 1604, 1605) holds at 96k–116k a year, so its share rises from 12.7% to 15.6%. Fish is mildly seasonal: an index of 0.84–0.89 from February to April and 1.05–1.08 from July to December.



Figure 4.1. MC2 story step "Trade runs one way: into Oceanus". Each light is about 200 shipments travelling a current between two countries.

Companies. Shippers are durable: a median span of 59 months, and 4,875 of 12,217 trade for 73–84 months. Receivers are fleeting: a median of 3 months, and 44% appear in a single month. Fish-majority shippers churn more: 2,139 of 4,299 are transient, against 4,740 of all 12,217 shippers.



Figure 4.2. MC2 story step "Shippers stay lit; receivers flicker out". Each trail is about 50 companies, lit while they trade.

Relationships. Every shipper→receiver pair's monthly series is classified by rules applied in order:

Pattern	Pairs	Share of shipments	Signature
One-off	108,067 (44%)	2.6%	a single month
Intermittent	60,968	14.5%	recurring but irregular
Short burst	31,604	3.0%	≤ 6 months, then nothing
Fading	16,729	21.9%	second half ≤ ½ of the first
Ramping	13,385	16.0%	second half ≥ 2× the first
Abrupt stop	5,994	8.3%	stops at full strength ≥ 12 months before the data ends
Steady	4,634 (1.9%)	32.6%	≥ 24 months, ≥ 60% of months active
Seasonal	4,426	1.2%	≥ 75% of trade in the same 3 calendar months

The network is a small core of steady pairs carrying a third of all shipments, ringed by a large fringe of one-off links. Abrupt stops are slightly *rarer* among fish pairs (1.7% vs 2.4%), so an abrupt stop alone is not a sign of IUU fishing. As a cross-check, k-means (k = 8) on 101,115 recurring pairs agrees with the rules at a weighted purity of 0.57.

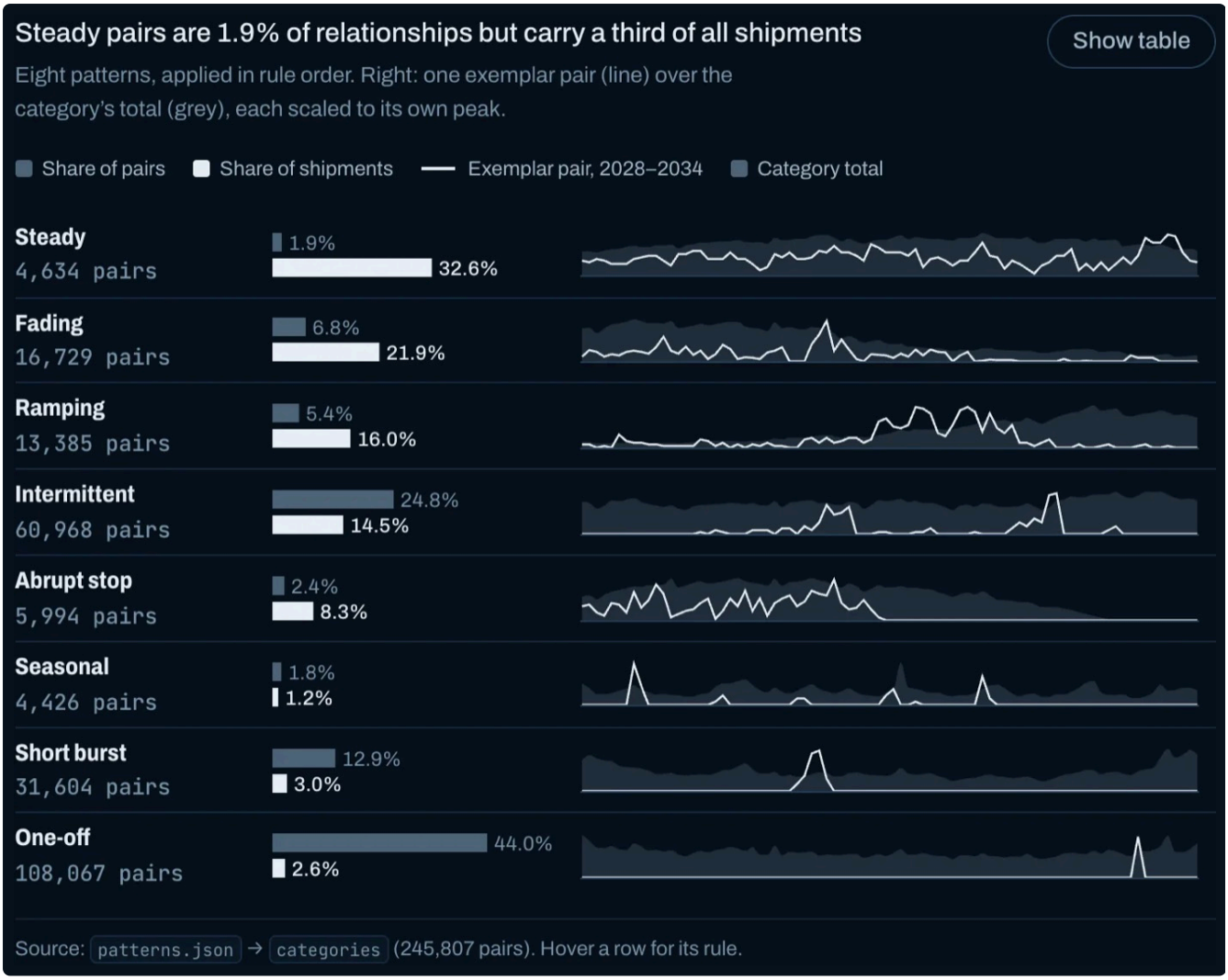


Figure 4.3. Steady pairs are 1.9% of relationships but carry a third of all shipments.

Successors. Company pairs with near-identical partner and product profiles mostly *coexist*: they are sister firms. But a second, narrower peak sits at gaps of 0–6 months after a stop, at 1.7 pairs per gap-month against 0.3 just before. That is the signature of a company stopping and a look-alike starting straight afterwards. It is real but small.

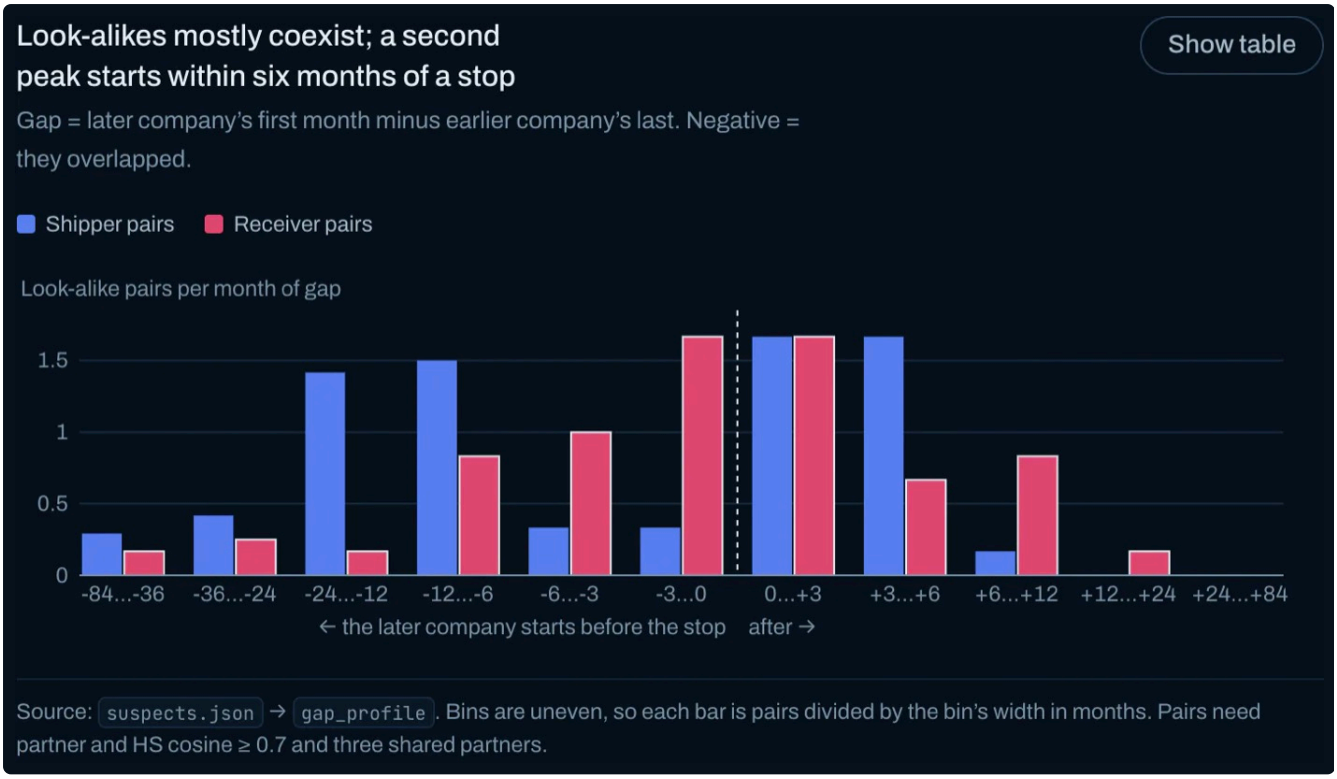


Figure 4.4. Look-alikes mostly coexist; a second peak starts within six months of a stop.

Confidence: **High** for the aggregate patterns, which are counts over the full graph.

4.2 Which predicted links to trust

Each bundle is scored on 11 components against **200 same-size random samples of real 2034 edges**, so small bundles are not punished for noise. Gates override the score: under 25% distinct links, fewer than 20 distinct links, or over 95% exact copies of graph edges force *Unreliable*. Under 50% of links with a weight, or over 5% outside the date window, cap the verdict at *Questionable*.

Bundle	Score	Verdict	Confidence	Decisive evidence
carp	0.99	Reliable	High	165 distinct links, 44% anchored to real records, value bias +0.09 dex, 95% country match
herring	0.95	Reliable	High	clean; month mix slightly off
chub_mackerel	0.90	Reliable	Medium	52% of rows repeat, but 155 distinct links pass every test
pollock	0.88	Reliable	Medium	34% repeats; top target 17%
salmon_wgl	0.87	Reliable	Medium	57% repeats; 92 distinct links are realistic
catfish	0.56	Questionable	Medium	no weights, 8.7% of dates in 2035, 49% country match
mackerel	0.52	Questionable	Medium	no weights, 32% of dates in 2035
salmon	0.78*	Unreliable	High	132 links are 13 records
cod2	0.75*	Unreliable	High	193 links are 4 records
shark	0.62	Unreliable	High	19 distinct records, all copies of graph edges
lichen	0.61	Unreliable	High	23 distinct records, all copies; value bias +0.26 dex
tuna	0.34	Unreliable	High	no weights, 0% anchored, 18% distinct, 22% country match

* A gate overrides the score.

Novelty is two-sided. An exact match with a real edge (same shipper, receiver, HS code and day) is almost impossible by chance. Some matches anchor a bundle to real records, but all matches mean it adds nothing. The five diverse bundles independently converge on 42–59% matches.

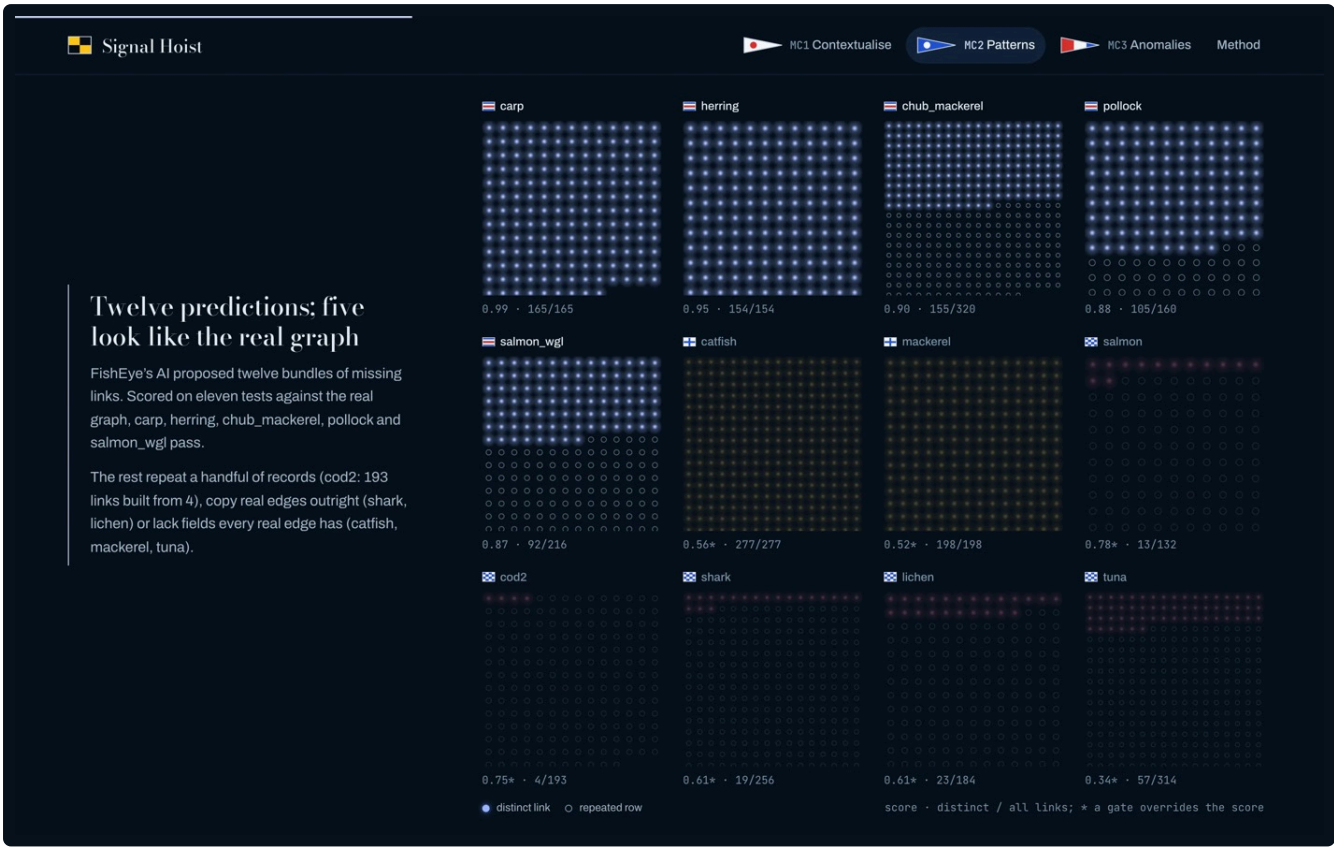


Figure 4.5. MC2 story step "Twelve predictions; five look like the real graph". Each bundle's links form a grid; repeated rows dim.



Figure 4.6. The scorecard. carp and herring pass every test; the gated bundles fail on repetition or missing fields.

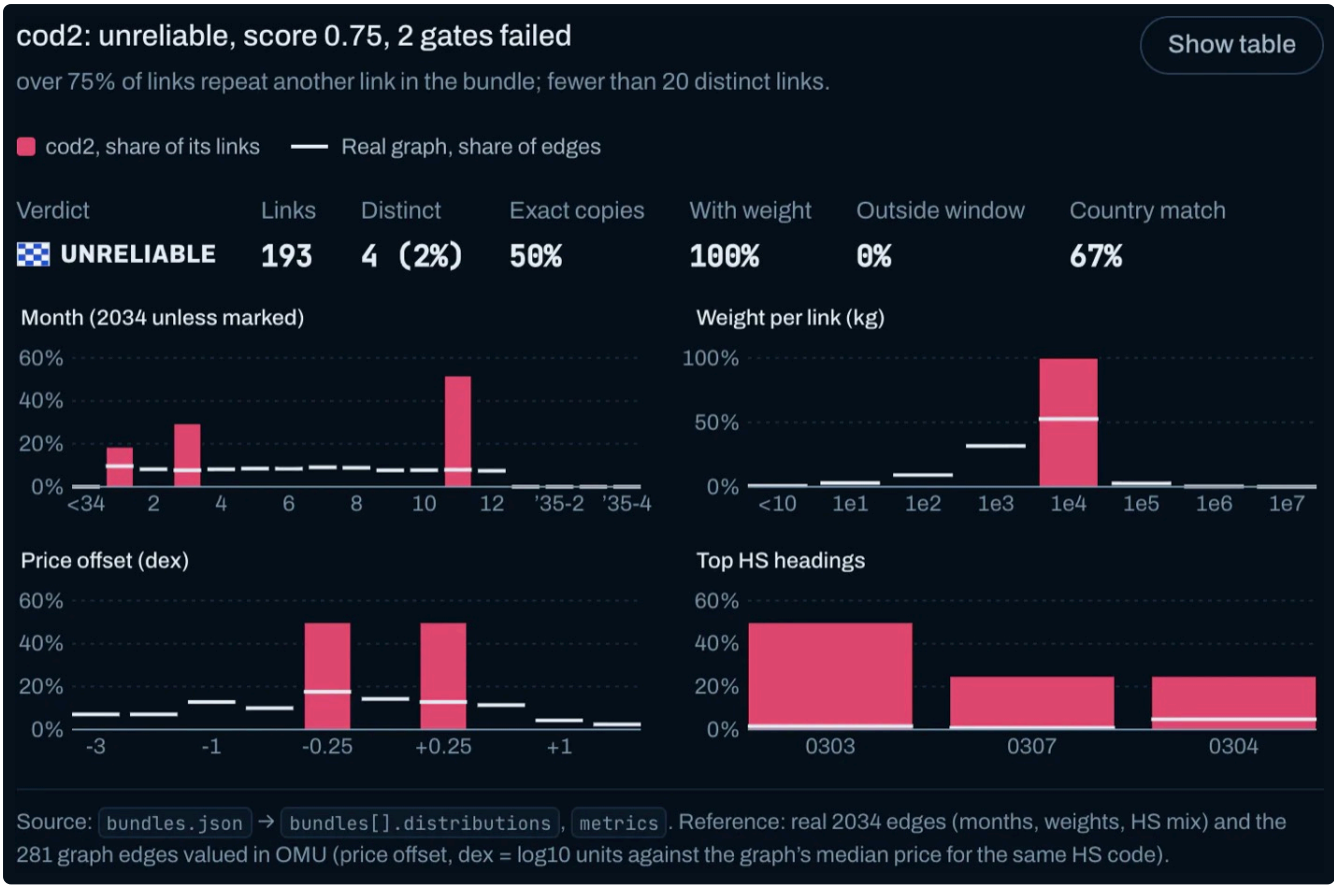


Figure 4.7. cod2: 193 links collapse to 4 distinct records.

4.3 What the reliable links change

The five Reliable bundles hold 1,015 links, but only **347** are usable (distinct, new and inside the window): under 0.1% of 2034 volume. 261 existing pairs gain links, 58 new pairs appear, and only 15 pairs change pattern. Three things stand out:

- **Three stops were missing data.** Predicted links keep three "stopped" companies trading for five to eight more months: *Chhattisgarh Salt spray OAO United*, *Bahía Azul Ltd Holdings* and *Okavango River S.A. de C.V. Solutions*. So the successor detector requires 6 silent months, and it marks a pair Unlikely when predicted links revive the predecessor.
- **Verdicts are stable.** All 68 flagged successor pairs (8 Suspect, 60 Possible) keep their verdicts on the augmented graph, and again after the two Questionable bundles (701 links) are added.
- **Hubs get richer.** The predictions densify the core rather than reveal hidden companies. The largest gains go to *hăi dăn Corporation Wharf* (33 links) and *Sailors and Surfers Incorporated Enterprises* and *Mar del Este CJSC* (11 each).

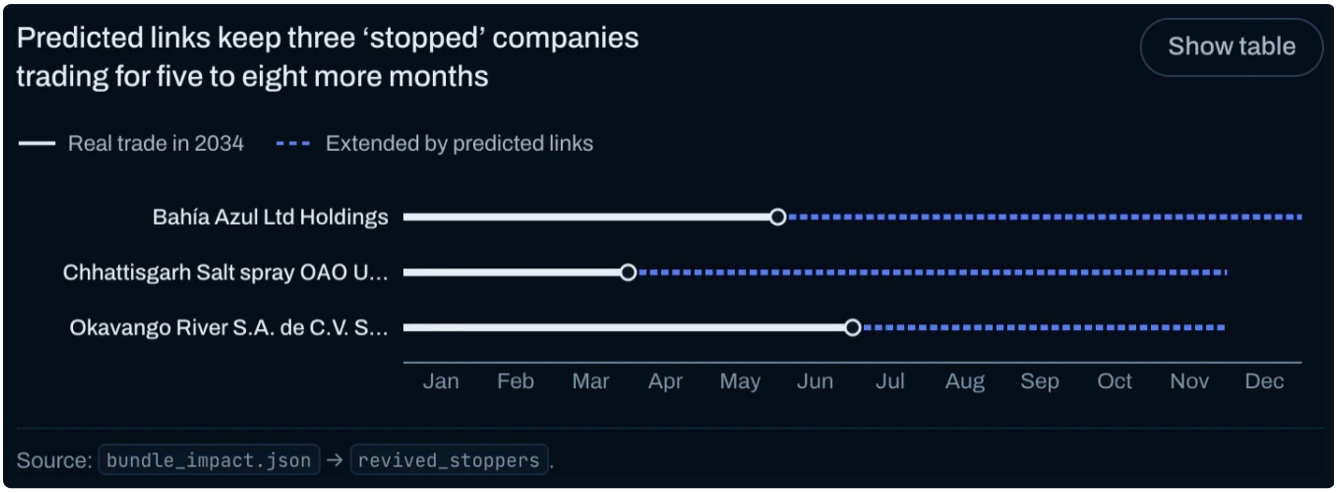


Figure 4.8. Predicted links (dashed) keep three "stopped" companies trading.

Confidence: **High** that the impact is small. **Medium** on the revivals, which are three cases, each from one bundle.

4.4 Companies that fit the IUU pattern

The matcher pairs 2,780 shippers and 3,888 receivers that stop ≥ 6 months before the data ends with companies that start between −6 and +12 months of the stop. Similarity is TF-IDF cosine plus Jaccard on partners and HS codes, so shared hubs count little, and shared partners must actually switch over. Of 872 candidate pairs, 8 are Suspect, 60 Possible, 572 Insufficient evidence and 232 Unlikely.

Predecessor → successor	Verdict	Confidence	Evidence
receiver −811 → −206	Suspect	High	fillets (0304); Dec 2032 → Jan 2033; 5 shared shippers supply 78% of −206's trade
Beachcomber's Bounty S.A. de C.V. Express → Karelians Limited Liability Company International	Suspect	Medium	Karelians also succeeds five more fish shippers that stopped in 2030
Saltwater Surf Club Sp → Chhattisgarh GmbH & Co. KG Sea spray	Suspect	Medium	dried fish (0305) only; all 3 shared partners switch
Asveja Limited Liability Company → Marine Mavericks Marine biology	Suspect	Medium	molluscs (0307); 95% of the successor's trade is inherited
receiver −812 → −556	Suspect	Medium	prepared crustaceans (1605); 5 shared shippers
receiver −513 → −690	Suspect	Medium	identical partner set
The Salty Dog Corporation → Gulf of Guinea Sp	Suspect	Low	13-shipment predecessor
Estrella del Sol CJSC → Mandalore S.A. de C.V. Logistics	Suspect	Low	12-shipment predecessor

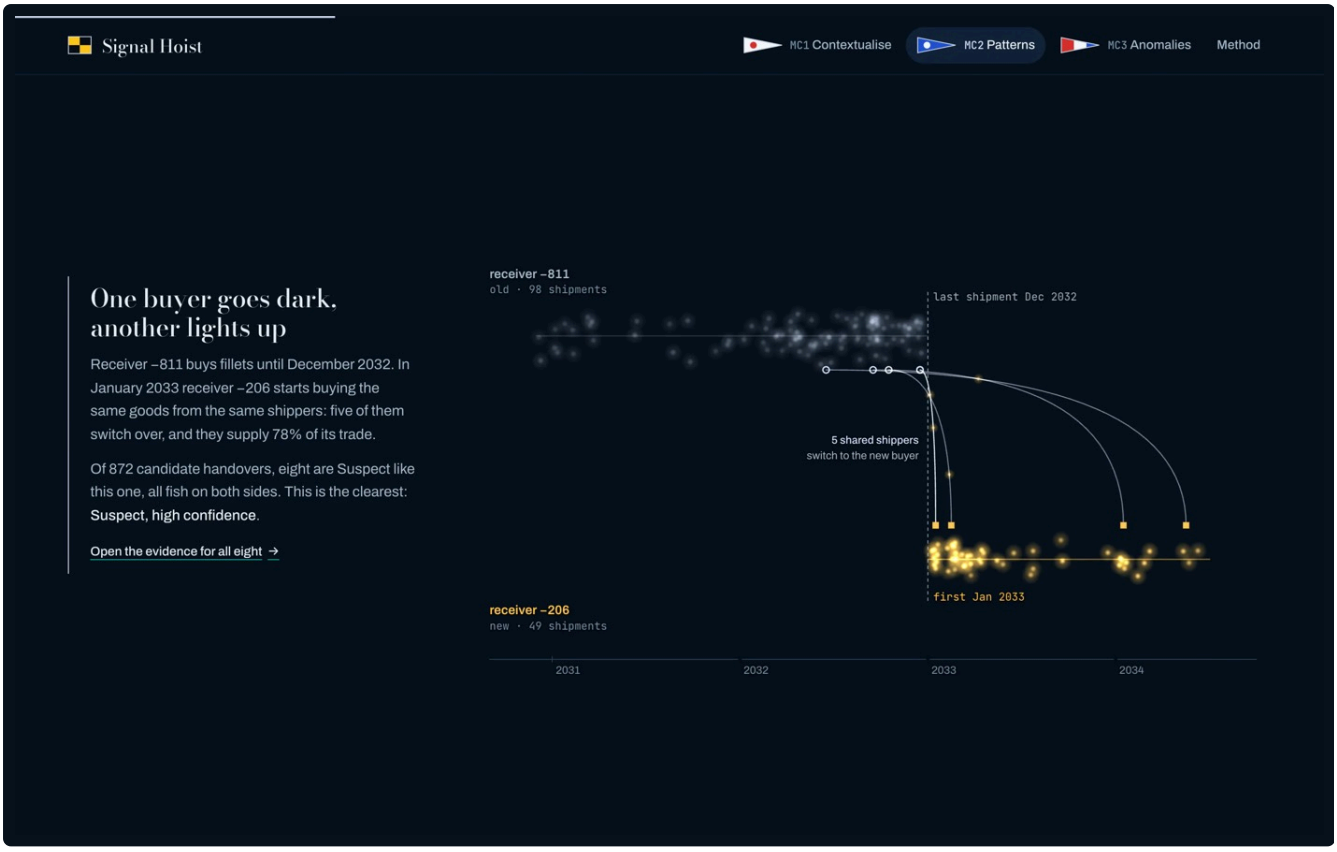


Figure 4.9. MC2 story step "One buyer goes dark, another lights up". –811's shipments end in December 2032, and five of its shippers arc down to –206.

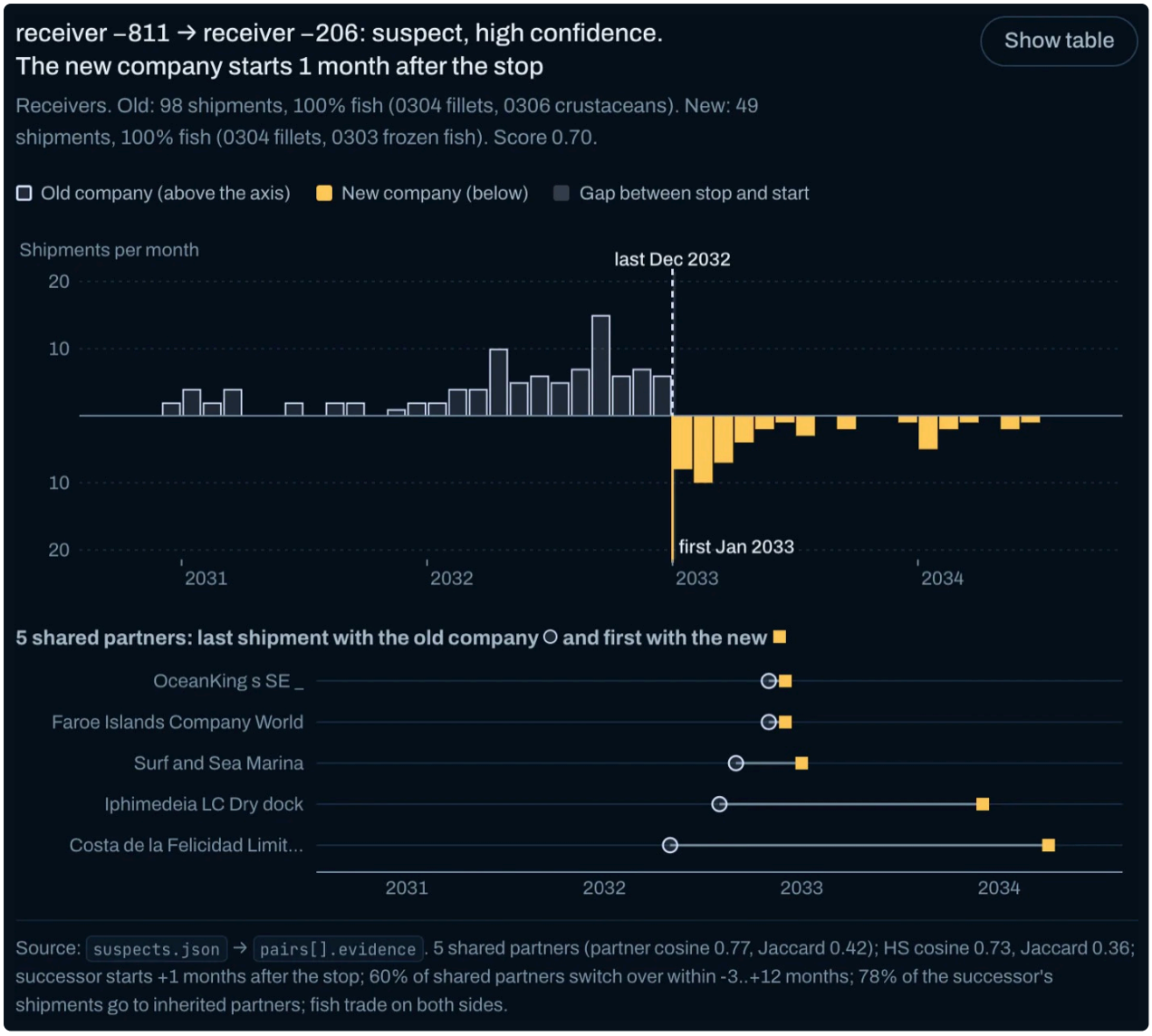


Figure 4.10. The –811 → –206 handover: monthly trade mirrored around the stop, and each shared shipper's last trade with the old id and first with the new.

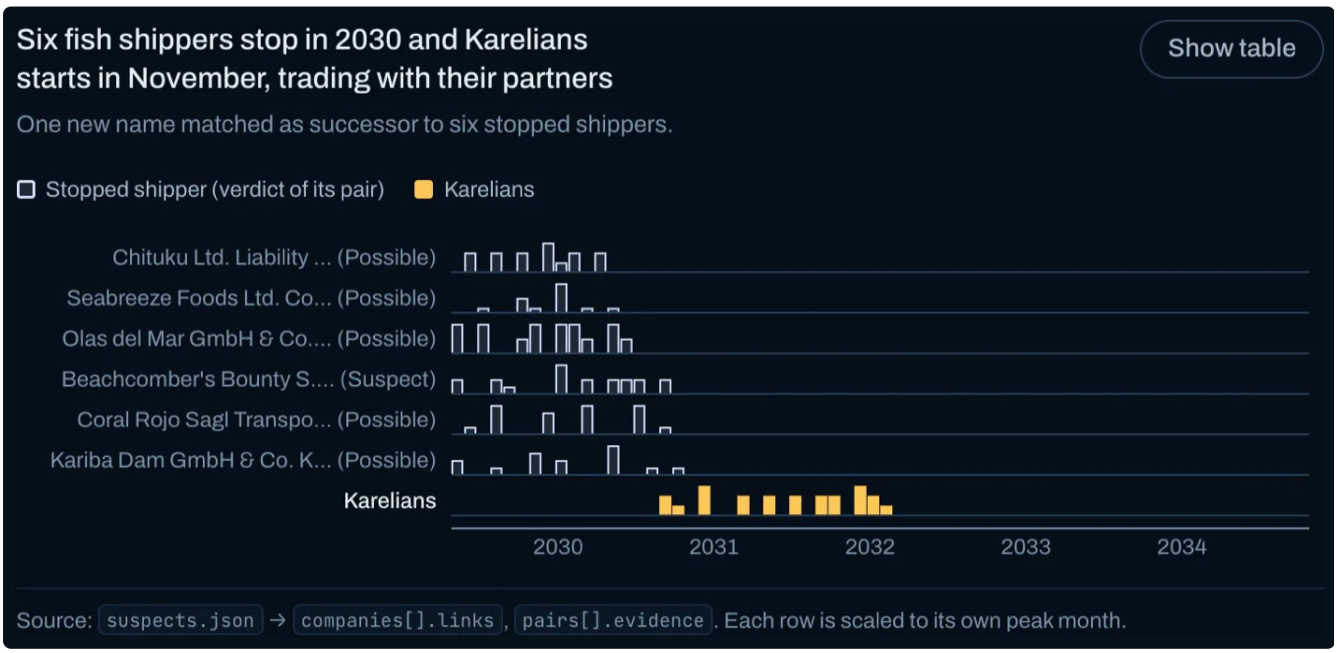


Figure 4.11. Six fish shippers stop in 2030, and Karelians starts in November, trading with their partners.

Confidence: **Medium** overall. Each pair has only 3–5 shared partners, there is no ground truth, and 24 high-similarity pairs proved Unlikely because their partners did not switch.

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5. MC3: the giant network is a name collision

Every link joins a company to a person. A *business group* is a connected component. There are 13,587 of them, and the giant component is split by seeded Louvain into 86 slices for ranking.

5.1 Anomalies

Name collisions create the giant component. People are identified only by name. Of people linked to one company, 21.9% carry one of the 50 most common surnames; of those linked to five or more, 92.6% do. The giant component holds 7,093 nodes (1,506 companies and 5,587 people). Remove the 651 multi-company people with common surnames and it falls into 1,083 pieces: 306 nodes stay connected, and the next largest piece holds 288. Remove 651 *random* multi-company people instead, and 495–1,918 nodes stay joined across 10 trials. Confidence: **High**.

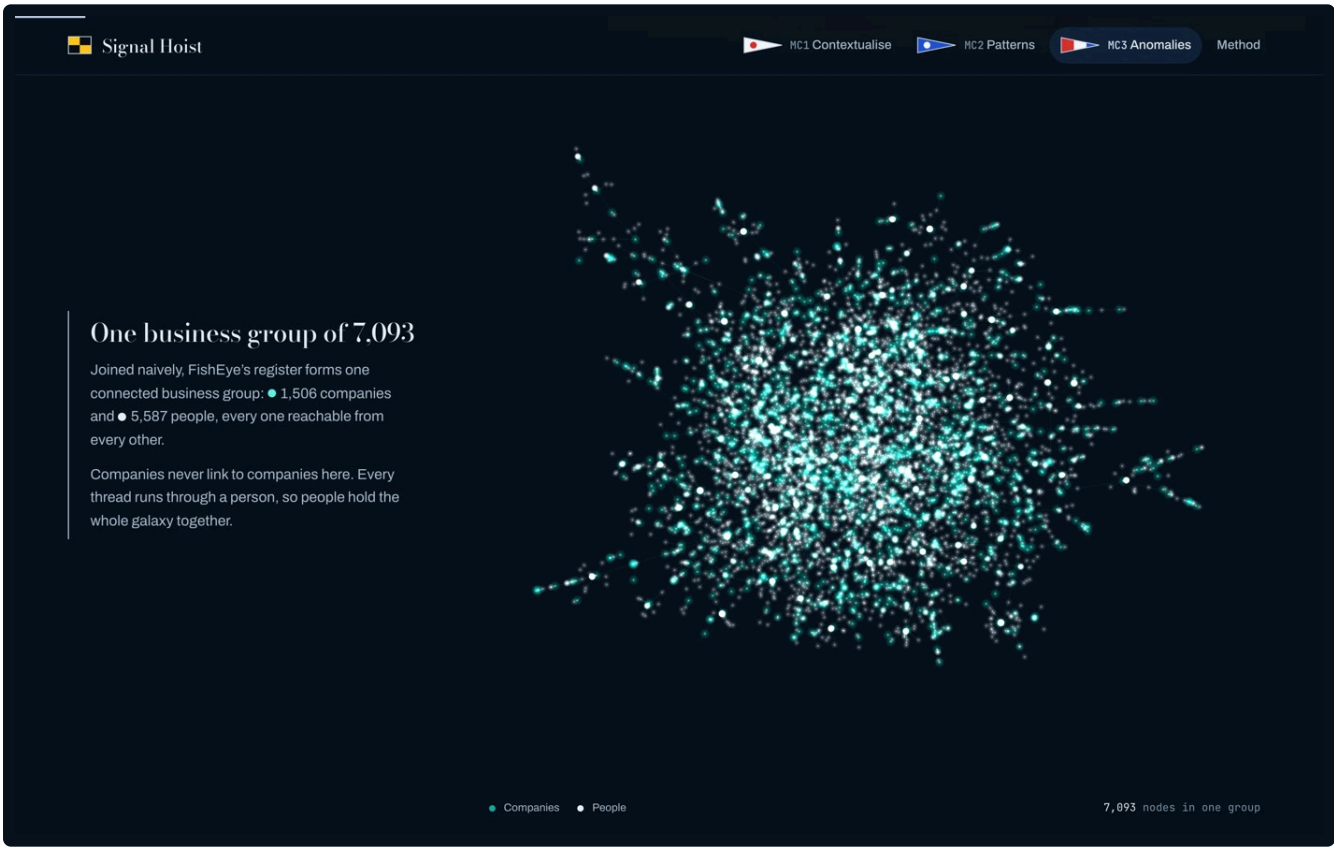


Figure 5.1. MC3 story step "One business group of 7,093". Joined naively, the register forms one connected galaxy.



Figure 5.2. Mid-morph: the galaxy breaking apart as the 651 common-surname people are removed.

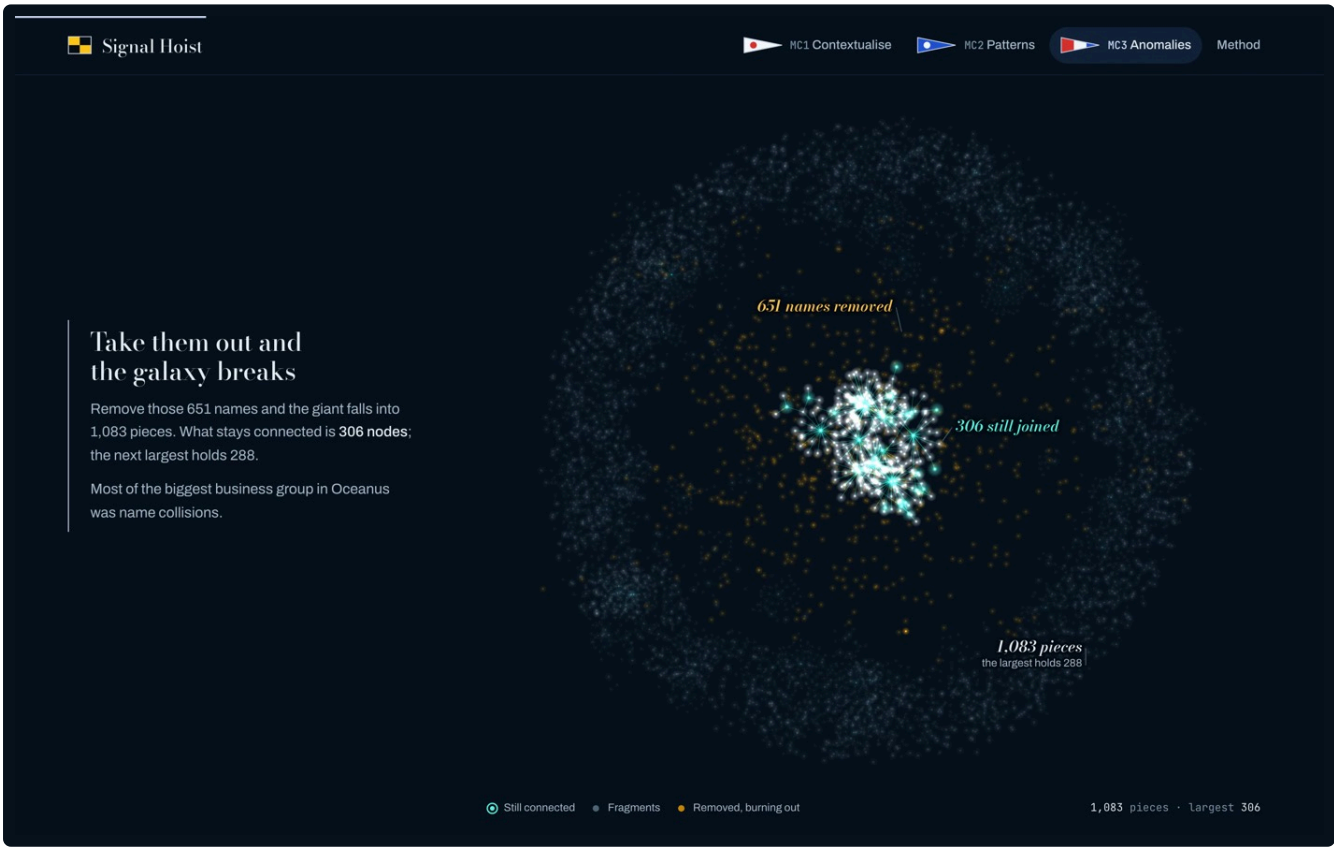


Figure 5.3. MC3 story step "Take them out and the galaxy breaks". 1,083 pieces remain, and 306 nodes stay joined.

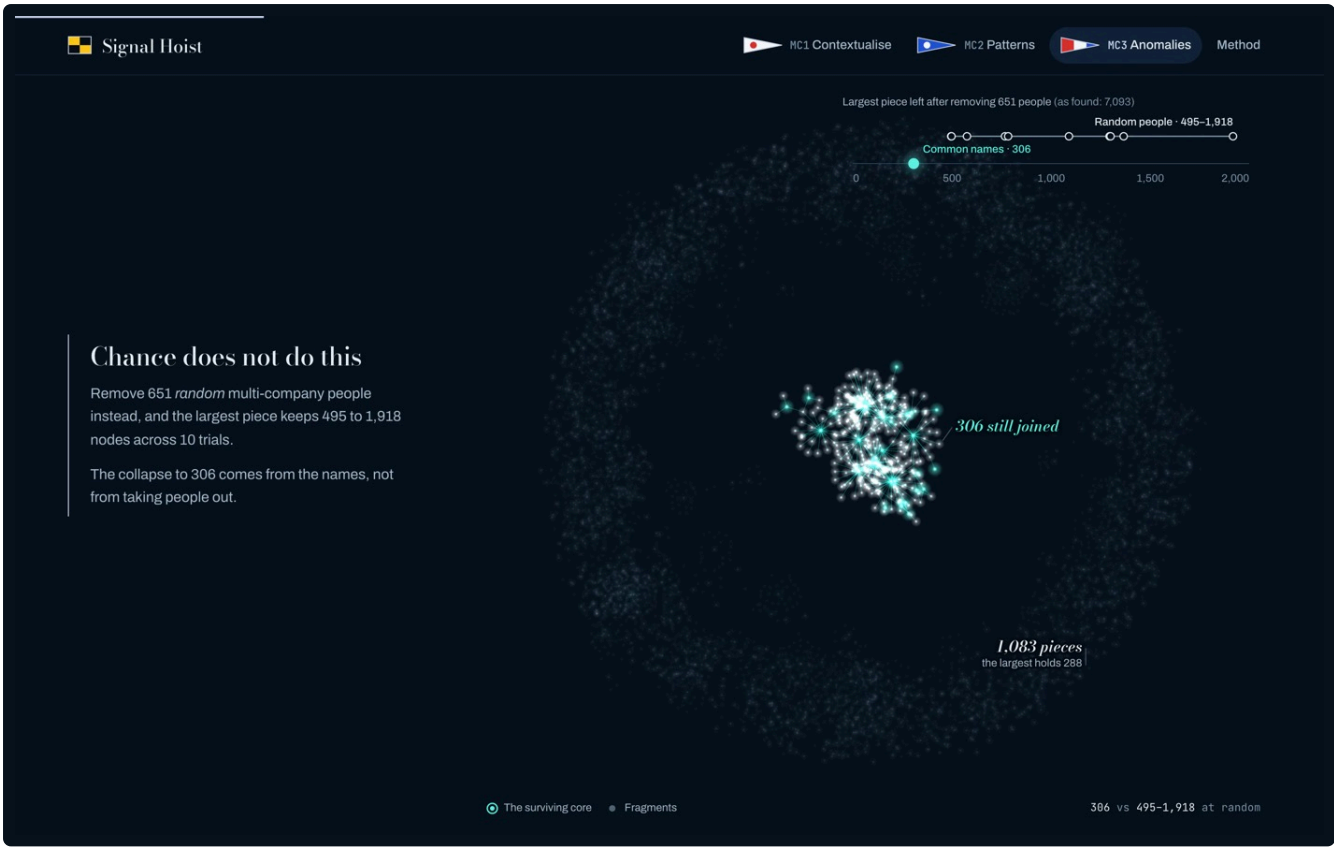


Figure 5.4. MC3 story step "Chance does not do this". Random removals leave 495–1,918 nodes joined, never 306.

The other anomalies:

- **Ownership fragmentation.** 107 companies have 20 or more beneficial owners and 38 have 30 or more. *Vespuci Sandbar Sp Brothers* has 119. 82 groups carry this flag. Confidence: **High**.
- **Serial owners with uncommon names.** These are the person links least likely to be collisions. *Anna Wheeler* owns 7 companies and *Jason Carrillo* is the contact for 6 of them (BG00148, the top anomaly of 2,347 scored groups). Confidence: **Medium**.
- **Registry inconsistencies.** 112 groups contain a distinctive company name registered in several countries. 166 groups contain a fishing-named firm that declares an unrelated product (274 companies). *Adriatic Squid* has both "Footwear" and "Seafood" records. Confidence: **Medium**.
- **Data-generation artefacts, not IUU signals.** 1,391 names have a stripped word ("Greek ermen"); 113 product texts end in "source: freelance researcher"; 11,420 Faker-style names all sit in country "ZH".

Groups with three or more nodes are scored by an Isolation Forest over 16 size-normalised features. Each score is explained by robust z-scores and 8 rule flags.



Figure 5.5. 13 of the 14 people with the most companies carry a top-50 surname. Anna Wheeler is the exception.

5.2 Grouping similar businesses

Product text is classified into *fishing* (1,043), *adjacent* (623), *other* (2,209) and *unknown* (12,822). Fishing words in names are a second, independent signal on 1,201 names, including pinyin such as *yú* and *diao yu*.

Because 77% of companies have no product text, there are two similarity spaces. The **text space** (3,875 companies) is LSA of TF-IDF terms with weight 0.8, plus structural features with weight 0.2. The **structure space** (12,822 companies) uses structural features only. k is chosen by silhouette × bootstrap stability, which gives k = 20 for text and k = 13 for structure. Four text clusters are seafood segments (T09 species, T12 processed, T16 generic fish products, T20 frozen): 861 companies, each cluster at least 84% fishing.

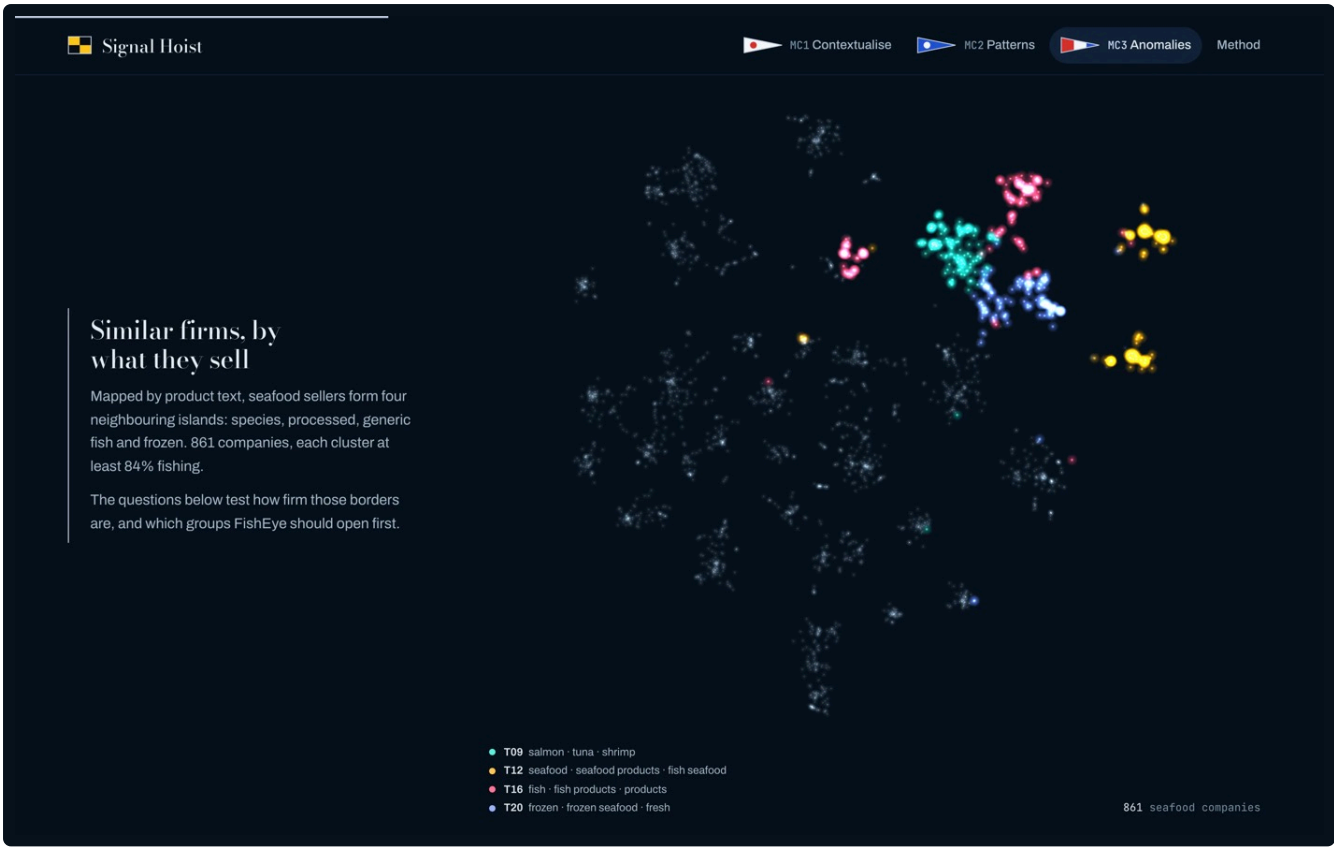


Figure 5.6. MC3 story step "Similar firms, by what they sell". The four seafood clusters are lit on the text map.

5.3 Confidence in the groupings

Three independent measures are used: cohesion (within-cluster cosine and silhouette), stability (clusterwise Jaccard over ten 80% subsamples), and each company's margin between its nearest and second-nearest centroid. *High* needs stability ≥ 0.85 and silhouette ≥ 0.25; *Medium* needs stability ≥ 0.6 and silhouette ≥ 0.05. The text space has 2 High, 10 Medium and 8 Low clusters (mean stability 0.60). The structure space has 8 High, 2 Medium and 3 Low (silhouette 0.63, mean stability 0.82), but it is easy to cluster rather than rich. T12 (processed seafood) is solid, while T16 (generic "fish products") blurs into its neighbours.



Figure 5.7. Only 2 of the 20 text clusters reach high confidence.

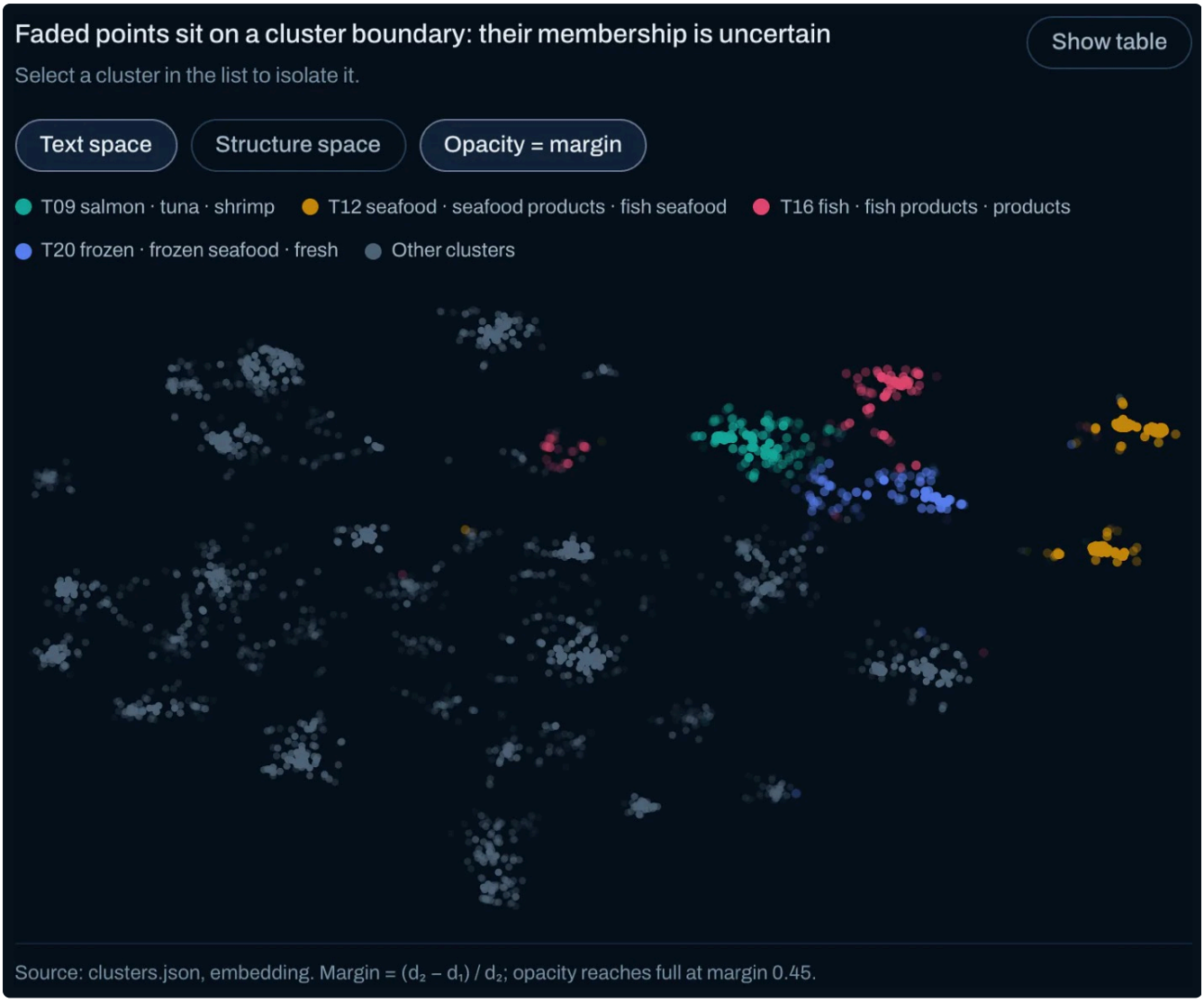


Figure 5.8. Opacity encodes margin, so companies on a cluster boundary fade.

5.4 Are anomalous companies fishing illegally?

Anomaly and fishing overlap only moderately. Of the 119 most anomalous groups (the top 5%), 13% reach a fishing relevance of 0.5, against 17% of the rest. Anomaly alone does not point to fishing, so the ranking combines them: 0.45 × anomaly percentile + 0.30 × flag score + 0.25 × fishing relevance. The product-text share inside fishing relevance is shrunk towards the corpus rate of 26.9%. Of the 30 ranked groups, 2 are Suspect, 20 Possible, 7 Insufficient evidence and 1 Unlikely.

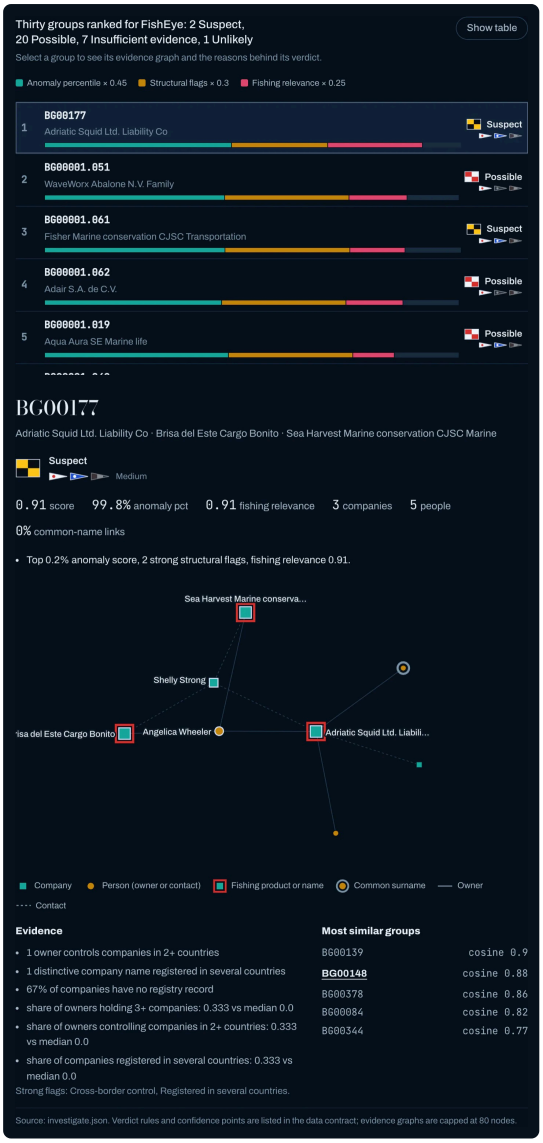


Figure 5.9. Thirty groups ranked for FishEye, with BG00177 open.

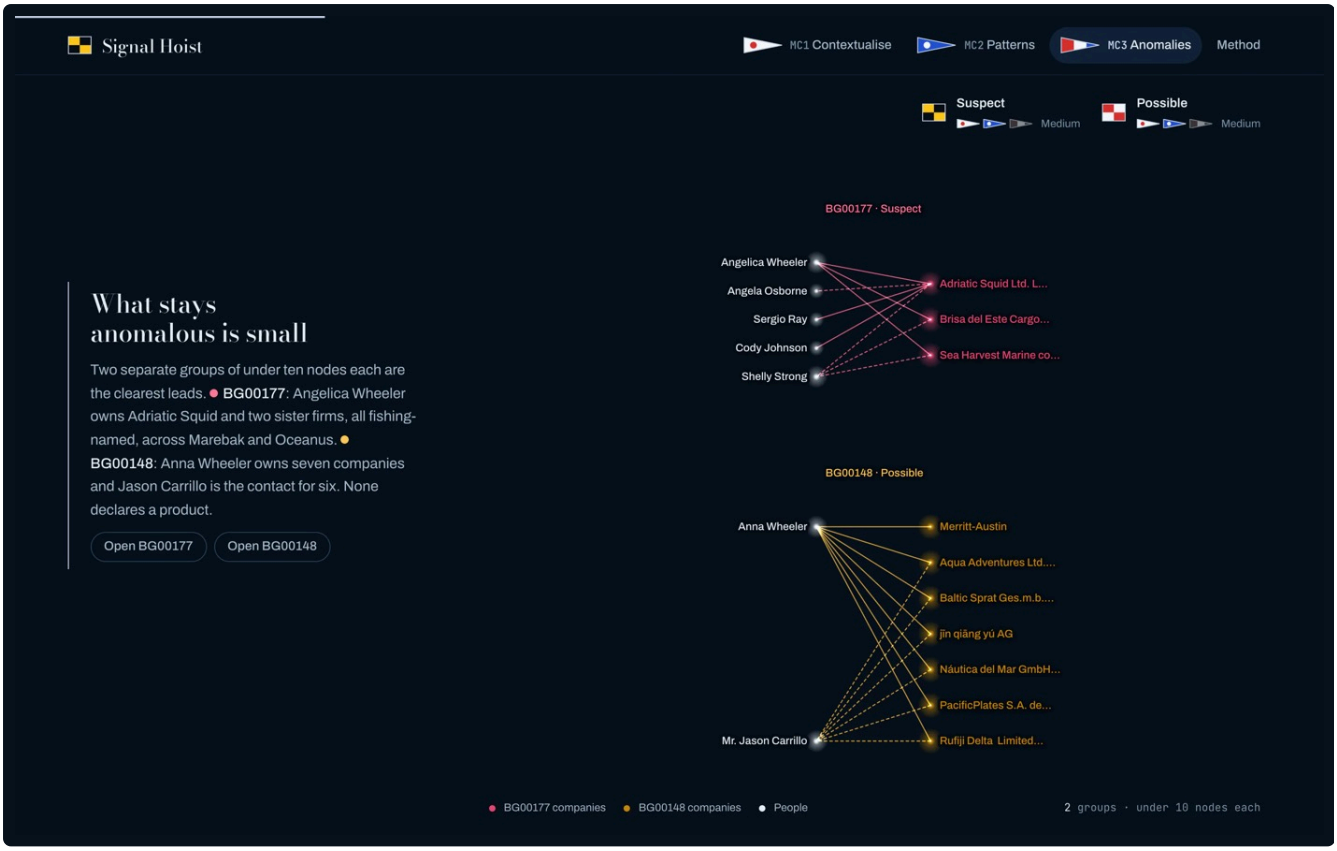


Figure 5.10. MC3 story step "What stays anomalous is small": BG00177 and BG00148.

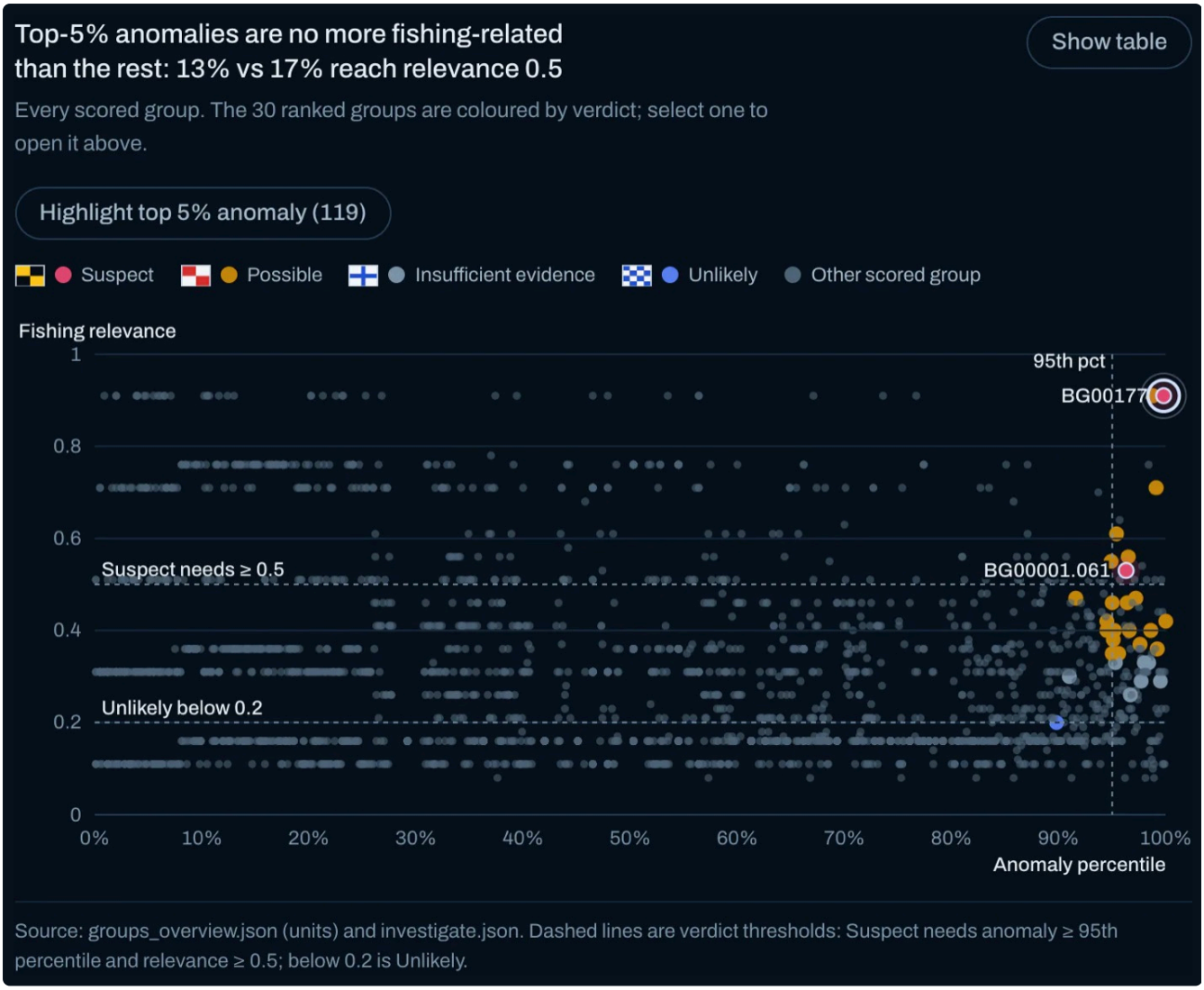


Figure 5.11. Top-5% anomalies are no more fishing-related than the rest.

Confidence: **Medium** that the Suspect and Possible groups are a better-than-random lead list. **Low** that any single group is engaged in IUU fishing.

6. Verdict summary

MCI: the tipped entities

Entity	Verdict	Confidence	Why
8327	Suspect	Low	Strongest structural signature (5.0 points), but almost all of it rests on edges weighted 0.12–0.32
979893388	Suspect	Medium	99.8th-percentile hub; owns Oceanfront Oasis through a strong edge; evidence on strong edges and structure
Mar de la Vida OJSC	Possible	Low	Tied to the layer only by a weak FishEye membership (0.46)
Oceanfront Oasis Inc Carriers	Possible	Low	Thin subsidiary of 979893388 with a weak FishEye tag (0.45)
#48 (candidate)	Suspect	High	FishEye tag on a 0.95 edge; ownership chain to 979893388

Entity	Verdict	Confidence	Why
#903311212 (candidate)	Suspect	Medium	Rank 1; FishEye-tagged hub one hop from 979893388

MC2: successor pairs and bundles

Entity	Verdict	Confidence	Why
receiver –811 → –206	Suspect	High	Next-month handover; 5 shared shippers supply 78% of the new trade
Beachcomber's Bounty → Karelians	Suspect	Medium	Karelians succeeds six stopped fish shippers
Saltwater Surf Club → Chhattisgarh GmbH & Co. KG Sea spray	Suspect	Medium	Dried fish only; all shared partners switch
Asveja → Marine Mavericks	Suspect	Medium	95% of the successor's trade inherited
receiver –812 → –556	Suspect	Medium	Prepared crustaceans; 5 shared shippers
receiver –513 → –690	Suspect	Medium	Identical partner set
The Salty Dog → Gulf of Guinea Sp	Suspect	Low	Small predecessor
Estrella del Sol → Mandalore	Suspect	Low	Small predecessor
carp, herring	Reliable	High	Pass every realism test
chub_mackerel, pollock, salmon_wgl	Reliable	Medium	Realistic distinct links, unexplained repetition
catfish, mackerel	Questionable	Medium	No weights; dates past the window
salmon, cod2, shark, lichen, tuna	Unreliable	High	Repetition, wholesale copying or missing fields

MC3: business groups

Group	Verdict	Confidence	Why
BG00177 (Adriatic Squid and two sister firms)	Suspect	Medium	One owner and one contact across three fishing-named firms in two countries; conflicting registry records
BG00001.061 (Fisher Marine conservation CJSC)	Suspect	Medium	16 companies; one with 26 owners; 62% unregistered; a Louvain slice
BG00148 (Anna Wheeler's seven companies)	Possible	Medium	Top anomaly; no products declared, so the fishing link rests on names
BG00069, BG00071	Possible	Low	Multi-country clusters, but tied by common-surname people
BG00861, BG00386	Possible	Medium	Single firms registered in several countries
BG00003	Insufficient evidence	High	Strong structure, no verifiable fishing link
BG00016	Unlikely	Medium	Fishing relevance 0.20

7. Validation

Check	Where	Result
Random-edge null (1,000 draws of 88 edges)	MC1 layer	Largest component mean 10.6, max 29, observed 79 (p = 0.001)

Check	Where	Result
Ownership share in layer vs graph	MC1	66% vs 18% (p = 2e-23)
Suspects inside the layer	MC1	3 of 4 (p = 5e-5)
Degree-matched suspect sets (199)	MC1 distances	8% are at least as close: closeness is mostly degree
Louvain across five seeds	MC1 communities	0.33–0.63 retention; no stable shared community
Leave-one-out	MC1 candidates	Held-out suspects rank 11–51 of 1,210; degree ranks 7–528
300 random reweightings	MC1 candidates	Mean Spearman 0.98, min 0.89
Feature ablation	MC1 candidates	Low-weight feature essential (8 of 25 kept); others 19–24
200 same-size samples of real 2034 trade	MC2 bundles	Distribution tests scaled to bundle size
k-means (k = 8) vs rules	MC2 patterns	Weighted purity 0.57
Re-detection on augmented graphs	MC2 successors	68 of 68 flags unchanged, also with Questionable bundles
Random-person removal (10 trials)	MC3 giant	495–1,918 nodes vs 306 for common surnames
Bootstrap stability (ten 80% subsamples)	MC3 clusters	Mean 0.60 (text), 0.82 (structure)
Determinism	All pipelines	Byte-identical outputs across runs

8. Limits

- **No ground truth.** Every verdict is a ranked lead with stated confidence, not an accusation.
- **Undocumented semantics.** MC1 edge weight is read as extraction confidence, and ownership direction is unknown. Both are assumptions.
- **Hand-set rubrics.** Weights and thresholds are judgement calls. They are published with the scores so they can be challenged.
- **Small evidence.** MC2 successor pairs rest on 3–5 shared partners, and stops late in 2034 cannot be told apart from missing data.
- **Identity by name.** In MC3, collisions inflate shared control. A keyword fishing classifier misses euphemisms, and 77% of companies have no product text.
- **Louvain slices.** Groups cut from the giant component have boundaries that depend on community detection.

9. Reflection

The hardest part of each challenge was deciding what to trust. In MC1, extraction errors formed structure that looked like evidence. In MC2, two HS-code formats would have silently zeroed out every chapter-03 fish link in the bundles. In MC3, common names like "John Smith" fused 1,506 companies into one group. In each case, a null model turned a suspicion into a measurement. Null models also reversed early conclusions: raw cosine had rated catfish and mackerel reliable, and a single clustering space reported a reassuring but hollow silhouette of 0.49.

The open-source Python stack and D3 were enough. A data dictionary for MC1's weights and edge directions would have helped most. So would unique person identifiers in MC3, documentation of the MC2 currency and HS conventions, and even a few labelled cases to calibrate the rubrics. With more time, the next steps would be to

resolve people by co-occurrence, validate the fishing labels against a hand-labelled sample, test how sensitive the rankings are to their weights, and link entities across the three graphs.

Hours spent: [AUTHOR TO FILL: hours] · *Permission to post in the Visual Analytics Benchmark Repository:* [AUTHOR TO FILL: Yes / No] · *Video:* Not provided

VAST Challenge 2023: Mini-Challenge 1

Entry Name: Signal Hoist

Team Member(s): [Name] — [Affiliation], GitHub @amukherjee1991

Tools Used: Python (pandas, NetworkX, scikit-learn, SciPy, PyArrow), React, D3, Vite, Playwright

Approximately how many hours were spent working on this submission in total? [AUTHOR TO FILL: hours]

May we post your submission in the Visual Analytics Benchmark Repository after VAST Challenge 2023 is complete? [AUTHOR TO FILL: Yes / No]

Video: Not provided

Interactive version: https://amukherjee1991.github.io/Vast23/#/mc1 (every figure below is a screenshot of this page in its night-sea design: either a step of the scroll story that opens the chapter, or an evidence view from the question sections; all are interactive there)

Questions

Figures are numbered *question.figure*: Figure 1.3 is the third figure in the answer to Question 1. Verdicts use one scale throughout: **Suspect**, **Possible**, **Insufficient evidence**, **Unlikely**, each with a confidence of **High**, **Medium** or **Low**.

The four entities named in FishEye's tip are Mar de la Vida OJSC, 979893388, Oceanfront Oasis Inc Carriers and 8327.

Question 1

> *Use visual analytics to dynamically display and explore context around the suspected entities listed above. What did you learn about each one? Can you connect them to illegal fishing? Provide evidence for or against the case that each entity is involved in illegal fishing and use visual analytics to express confidence in your conclusions. Limit your response to 600 words and 6 images.*

The key context is a layer of weak links. 99.2% of the 11,069 edge rows have a weight between 0.80 and 0.99. The other 88 rows, weighted 0.03 to 0.77, form one connected component of 79 nodes (Figures 1.1, 1.2). This is not chance:

- **Size.** In 1,000 random draws of 88 edges, the largest component averaged 10.6 nodes and never exceeded 29 ($p = 0.001$).
- **Ownership.** 66% of the layer's edges are ownership, against 18% of all edges ($p = 2e\text{-}23$). It holds two circular-ownership loops.
- **FishEye.** "FishEye International" has 24 edges, 19 of them in this layer, so the layer may encode FishEye's own reporting.
- **Suspects.** Three of the four suspects are inside the layer ($p = 5e\text{-}5$); the fourth is one hop away.

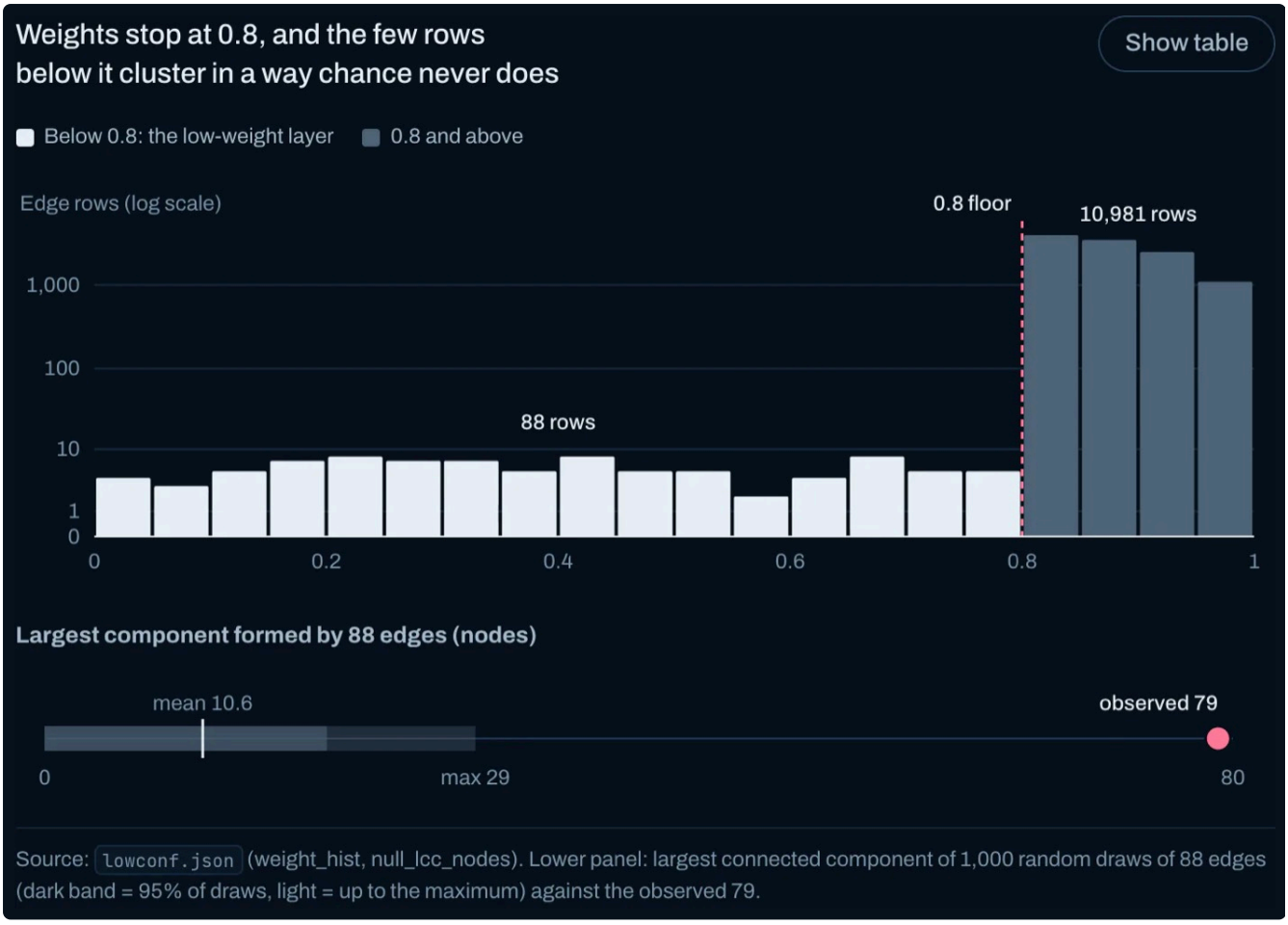


Figure 1.1. Weights stop at 0.8; the 79-node component of the 88 rows below lies far outside 1,000 random draws (max 29).

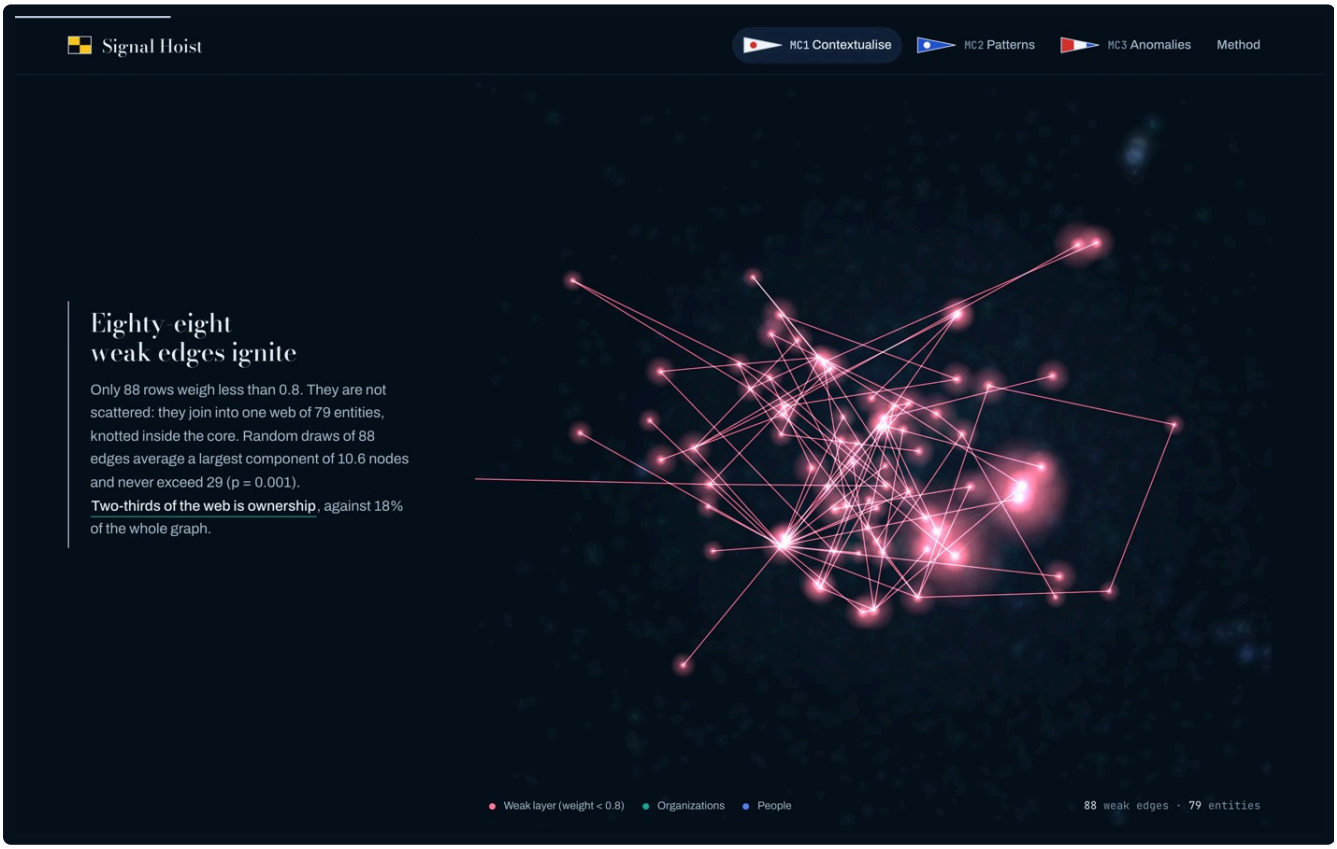


Figure 1.2. Story step: the 88 weak rows, lit rose, knot into one 79-node web.

Verdicts come from a published seven-signal rubric (4+ points Suspect, 2.5–3.5 Possible). Confidence counts full-point signals resting on edges weighted ≥ 0.8 or on structure alone (4+ High, 2–3 Medium, 0–1 Low). Ego views (Figures 1.3–1.6) ring each suspect in its verdict colour and dash weak edges.

8327: Suspect, Low confidence (5.0 points; Figure 1.3). A nameless organization with 13 neighbours. "Seabreeze Foods Ltd. Corporation United" owns it through one of the ten weakest edges in the graph (w = 0.12), and FishEye tags Seabreeze (w = 0.13). 8327 owns #40213337, which has six low-weight ownership rows and owns Sea Harvest, Sea Products and #58623386. 8327 has a membership edge (w = 0.82) to the garbled fragment "Illegal". A directed ownership chain runs from 8327 through #40213337, Sea Harvest, Amanda Mckenzie and Mar del Oeste Pic to 979893388, then to Oceanfront Oasis: six hops, weakest edge 0.32. Strongest structural signature of the four, but nearly every signal rests on edges below 0.8.

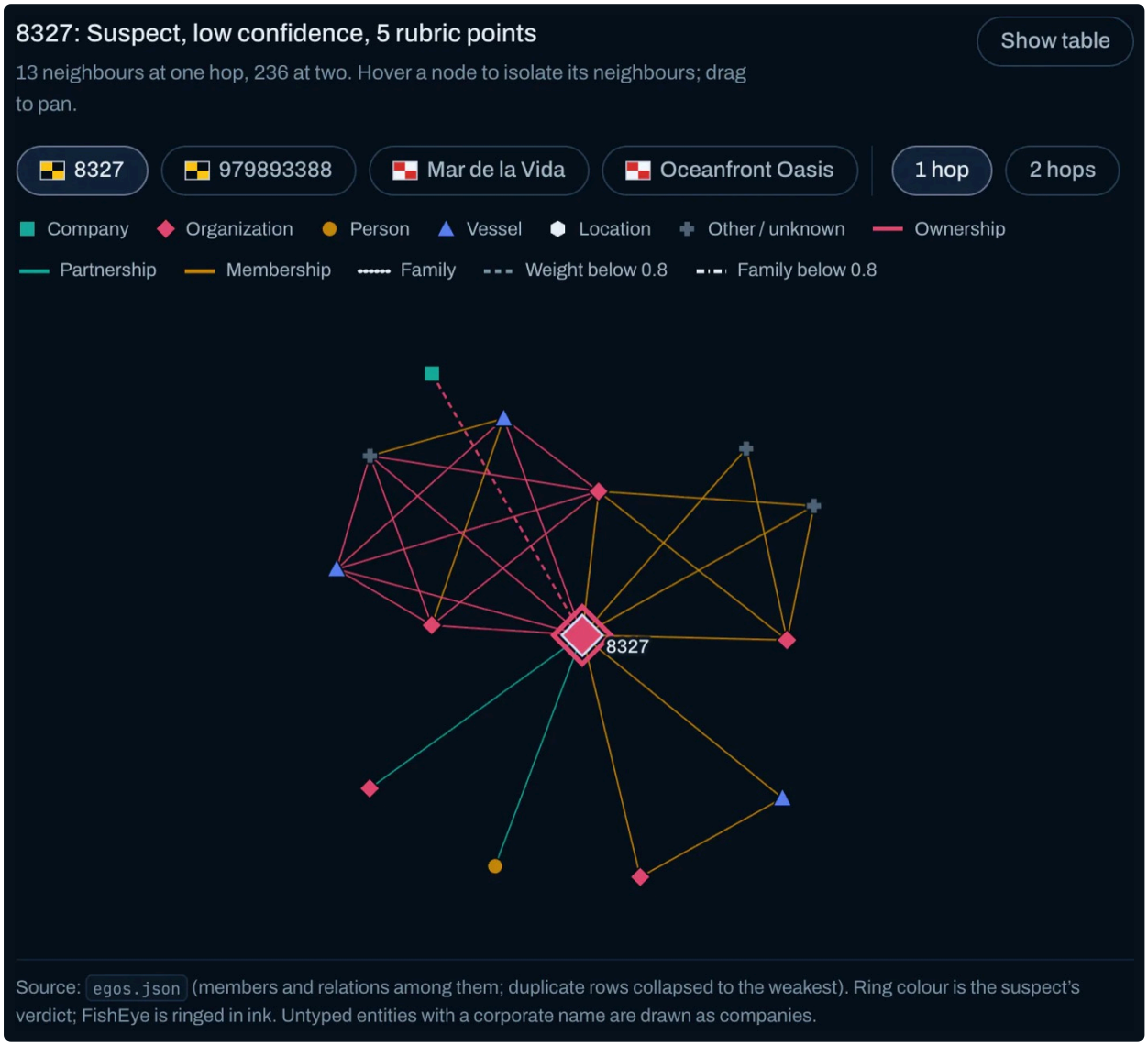


Figure 1.3. 8327 at one hop (13 neighbours); dashed edges weigh below 0.8.

979893388: Suspect, Medium confidence (4.0 points; Figure 1.4). A nameless hub: degree 177 and betweenness 0.096, both at the 99.8th percentile. It is one of 10 hubs forming a dense core (density 0.84). It owns 10 entities and is owned by 12; its upstream ownership tree reaches 307 entities within six hops. It owns Oceanfront Oasis through a high-weight edge (0.84), and its two-hop neighbourhood holds the fragments "ipoa-iuu" and "poached". It is not in the layer but sits next to two layer hubs. It reads as a holding layer, or an entity the extractor over-linked.

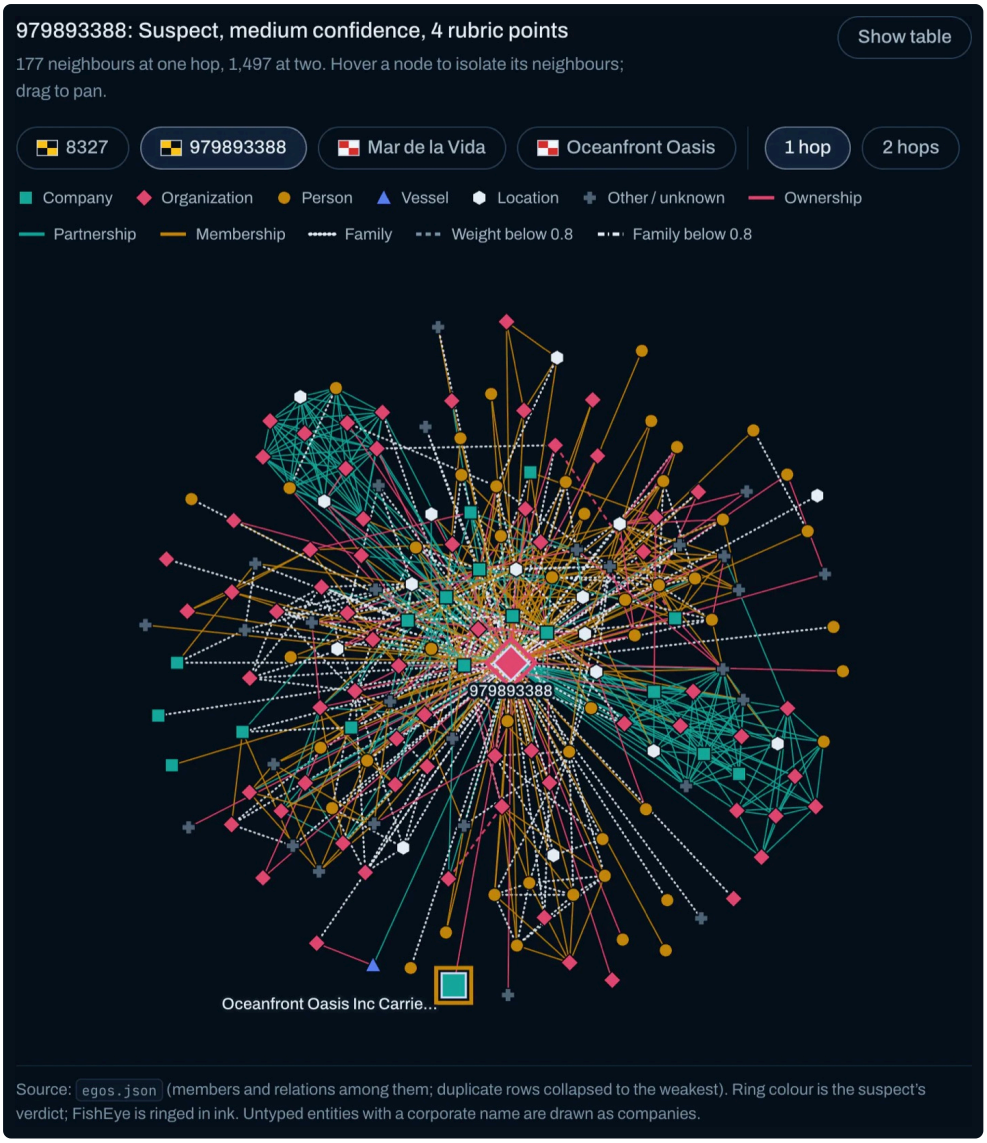


Figure 1.4. 979893388, a 177-neighbour hub, owns Oceanfront Oasis (bottom) through a 0.84 edge.

Mar de la Vida OJSC: Possible, Low confidence (3.0 points; Figure 1.5). Untyped, 15 neighbours, mostly memberships, 6 of them people. Its only low-weight edge is a FishEye membership (0.46). It owns one person, Jon White, and has no ownership chain. At two hops it reaches "Officer Pleads Guilty" and "Special Action Group Combat Organized Crime"; betweenness is at the 95th percentile. The weak FishEye tag is the main evidence.

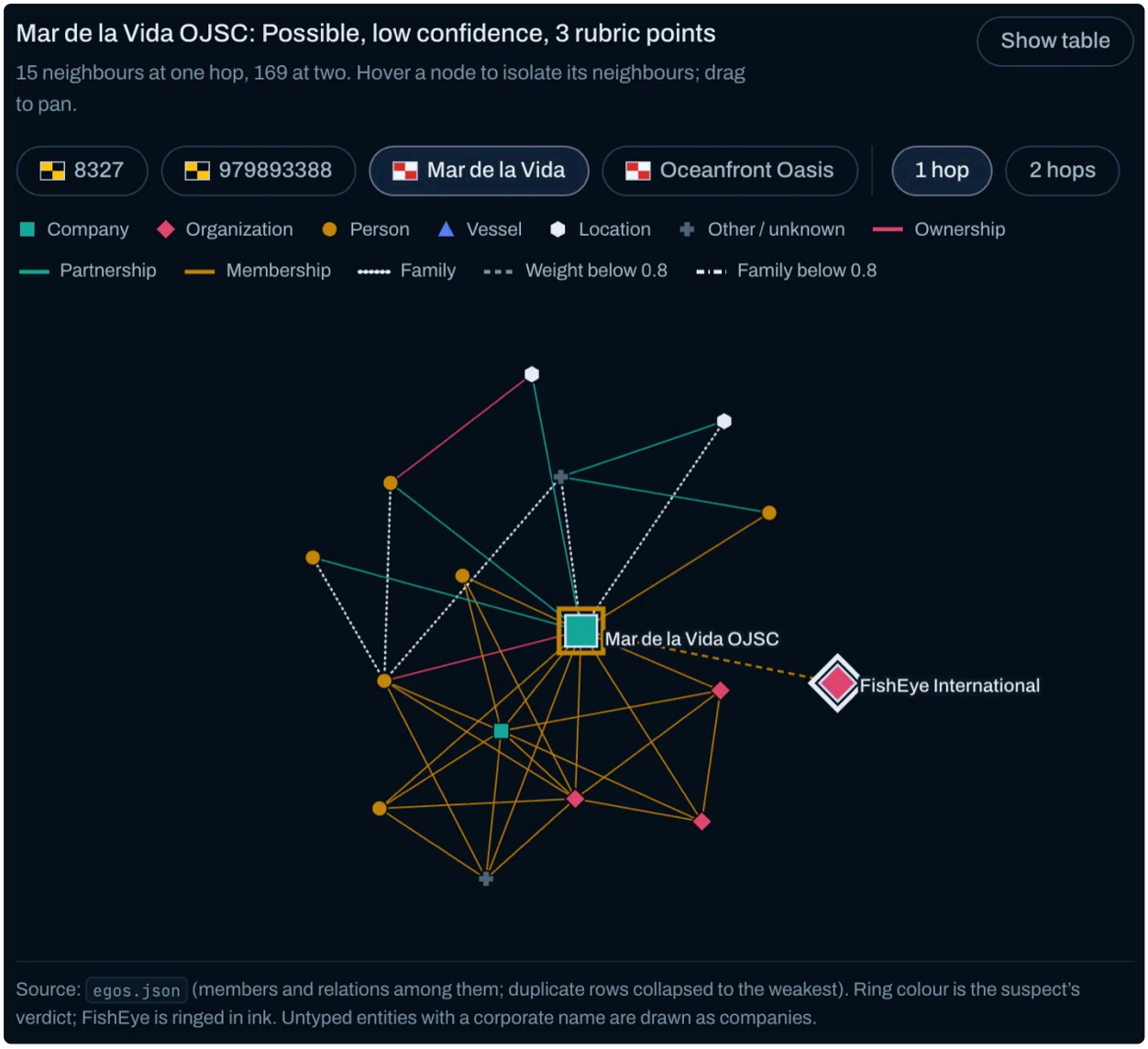


Figure 1.5. Mar de la Vida: its only weak edge is the dashed FishEye membership (0.46).

Oceanfront Oasis Inc Carriers: Possible, Low confidence (3.0 points; Figure 1.6). Four edges, all incoming: a FishEye membership (0.45), ownership by 979893388 (0.84), and two family edges from nameless organizations, probably noise. A thin subsidiary of a suspect.

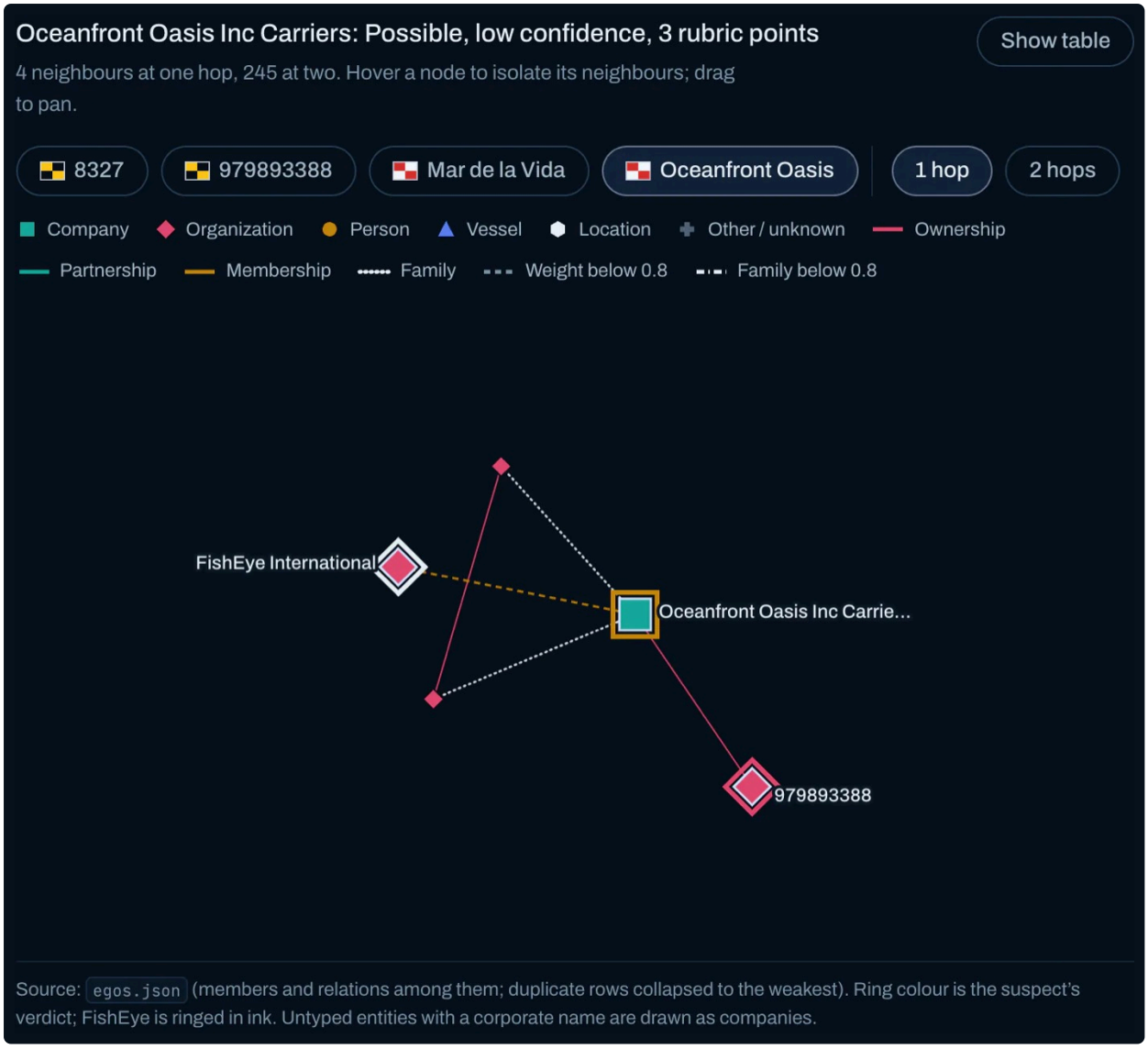


Figure 1.6. Oceanfront Oasis: four incoming edges, FishEye (0.45) dashed.

Caveats. The data notes define neither `weight` nor ownership direction, so reading weight as extraction confidence is an assumption. 356 nodes (10%) are NLP debris.

Question 2

> *Use your visual analytics tool to compare and contrast what you learned about the suspect entities. Are there patterns that may indicate illegal activity? Use visual analytics to express confidence that a pattern exists and where uncertainty may be affecting this confidence. Limit your response to 400 words and 4 images.*

The suspects fall into two pairs (Figure 2.1).

Tagged small entities: Mar de la Vida and Oceanfront Oasis. Named, untyped, with 15 and 4 neighbours. Each touches the layer only through one weak FishEye membership (0.46, 0.45), and FishEye is their only shared neighbour.

Ownership intermediaries: 8327 and 979893388. Nameless numeric organizations that both own and are owned (3 and 2; 10 and 12). Only they fire the rubric's shell marker. They differ in scale: 979893388 is at the 99.8th percentile of degree, 8327 at the 91st.

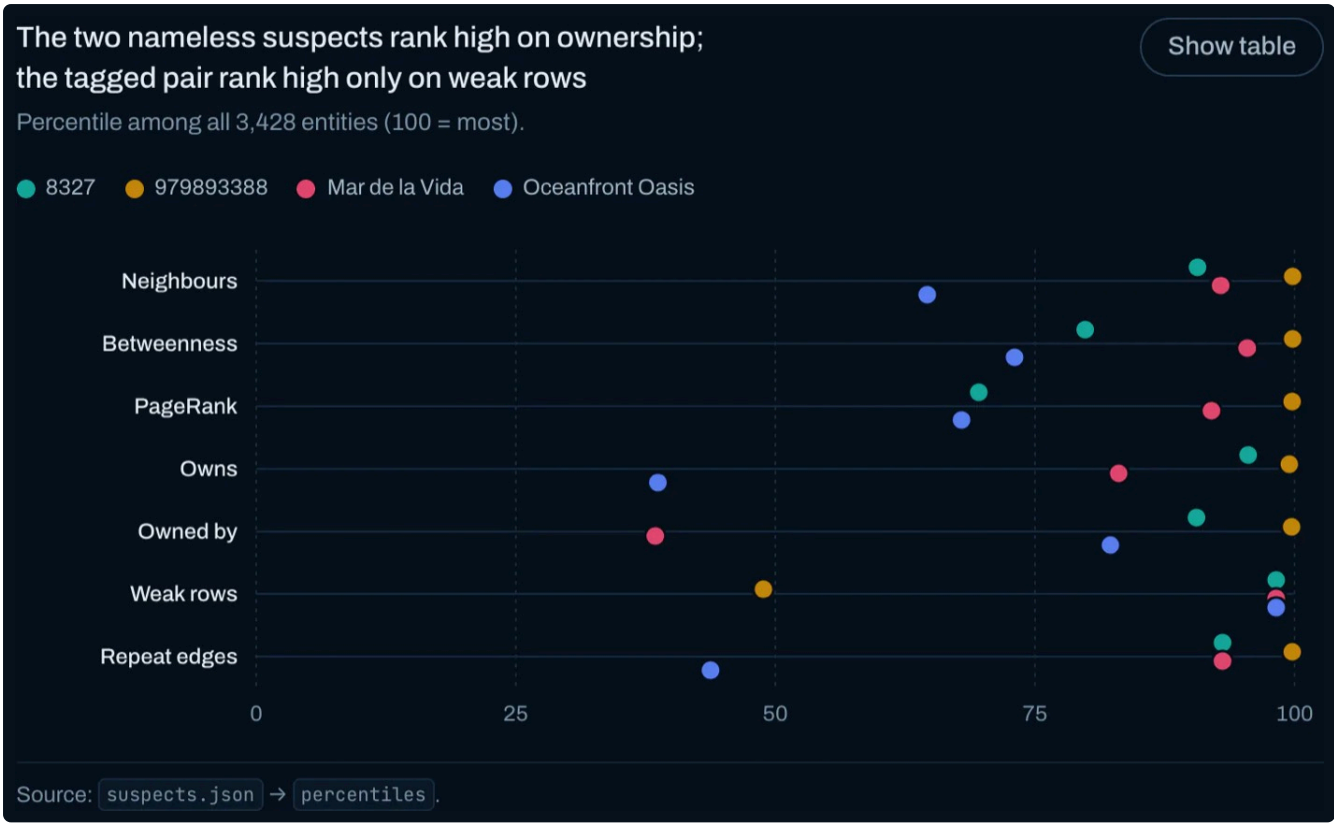


Figure 2.1. Percentiles among 3,428 entities: the nameless pair rank high on ownership, the tagged pair only on weak rows.

Patterns that point to illegal activity:

1. **Ownership in the weak layer.** Two-thirds of its edges are ownership, with two circular loops (4 and 5 entities) and a five-hop directed chain.
2. **One chain through three suspects.** The six-hop chain links 8327, 979893388 and Oceanfront Oasis (Figure 2.3). Mar de la Vida has no ownership path to another suspect.
3. **FishEye tags.** Two suspects are tagged directly; 8327 through its owner, 979893388 through a neighbour.
4. **Nameless intermediaries.** 34 of the layer's 79 nodes are nameless numeric organizations.

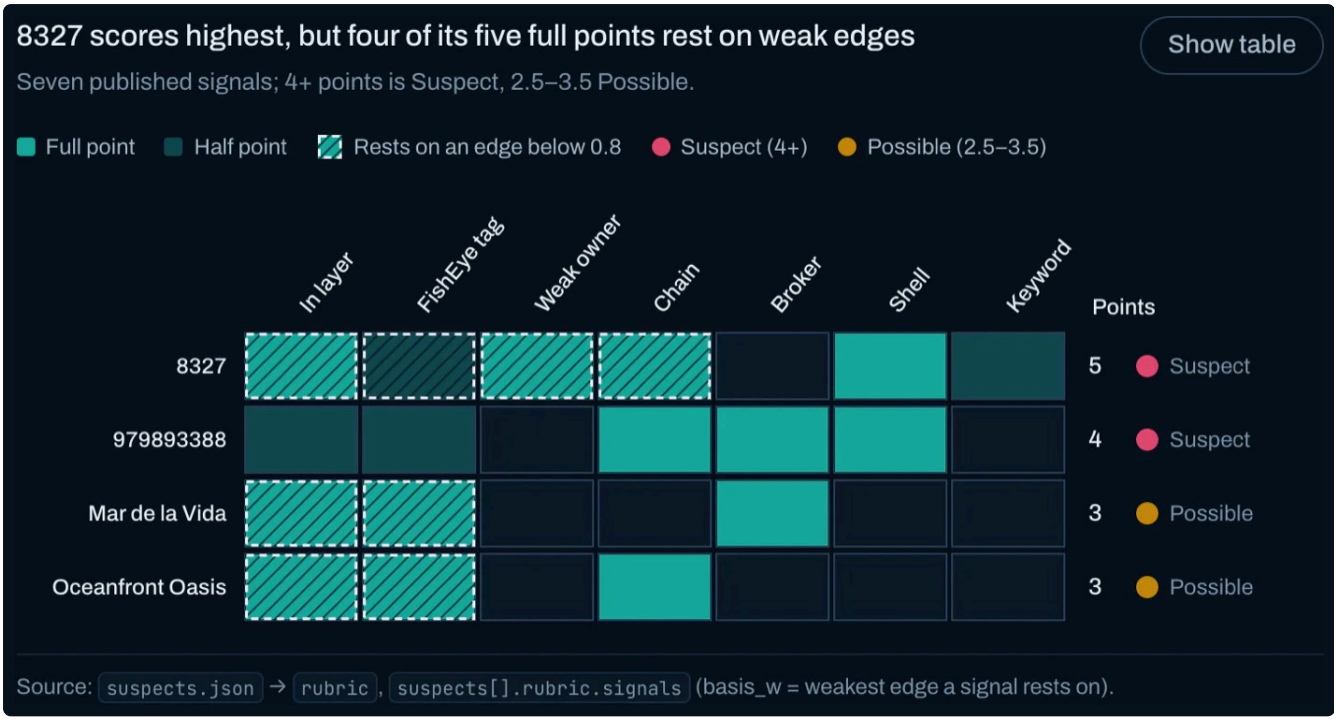


Figure 2.2. 8327 scores highest, but four of its five full points are hatched: they rest on edges below 0.8.

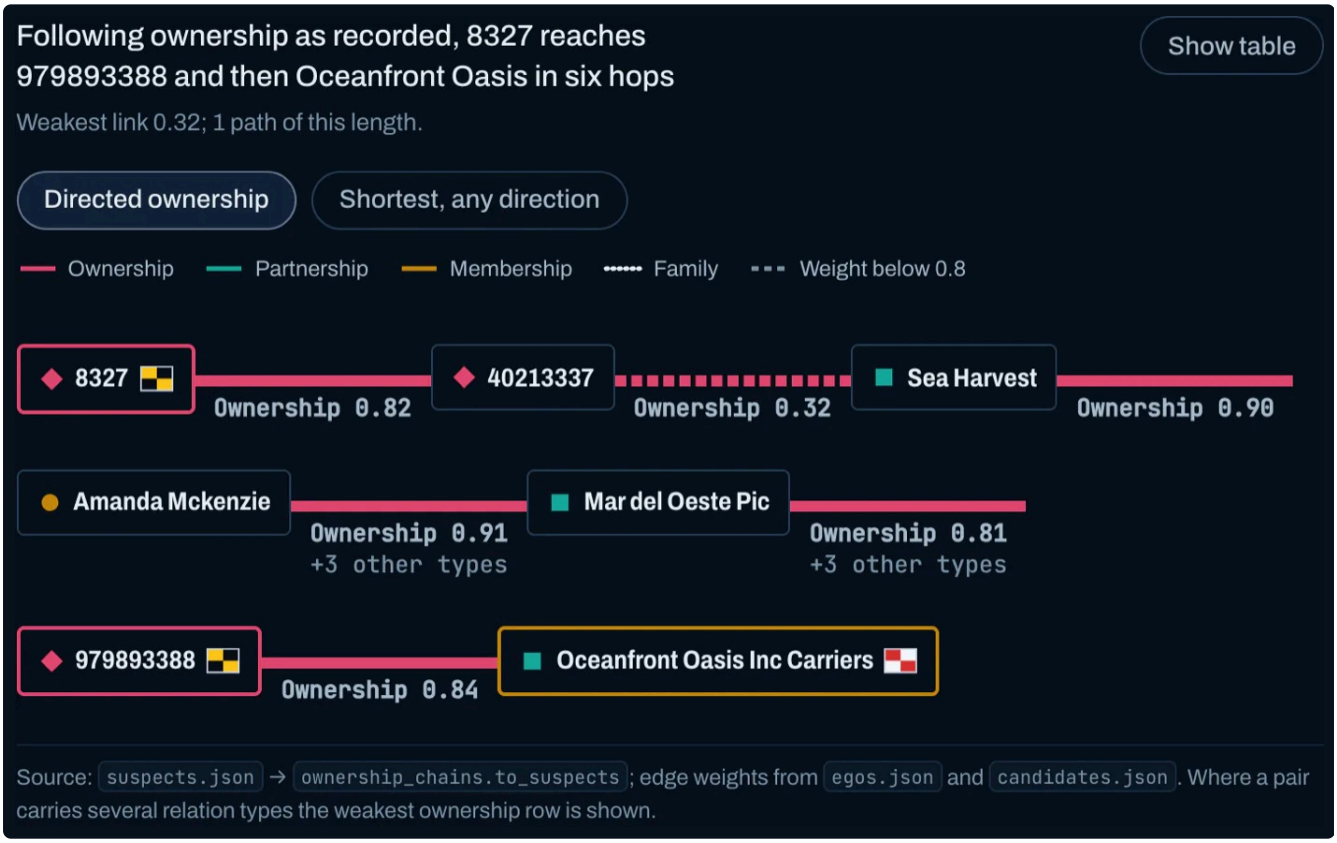


Figure 2.3. Ownership as recorded: 8327 reaches 979893388 and Oceanfront Oasis in six hops (weakest link 0.32).

Uncertainty, and how the views show it:

- **Distances.** The suspects sit 2.17 hops apart on average, against 3.84 for random pairs, but 8% of random degree-matched sets are as close (Figure 2.4). Their closeness is mostly what their degrees predict. Pattern confidence: Low.

- **Communities.** Louvain places the suspects in three communities (modularity 0.69), and across five seeds only 0.33–0.63 of each suspect's community neighbours stay with it. No stable shared community.
- **Ownership direction.** Read literally, the chain has the company Sea Harvest owning the person Amanda Mckenzie. Direction cannot be trusted.
- **Edge weights.** The two Suspect verdicts differ in confidence: 979893388 rests on high-weight edges and structure (Medium), 8327 on edges from 0.12 to 0.32 (Low; Figure 2.2).

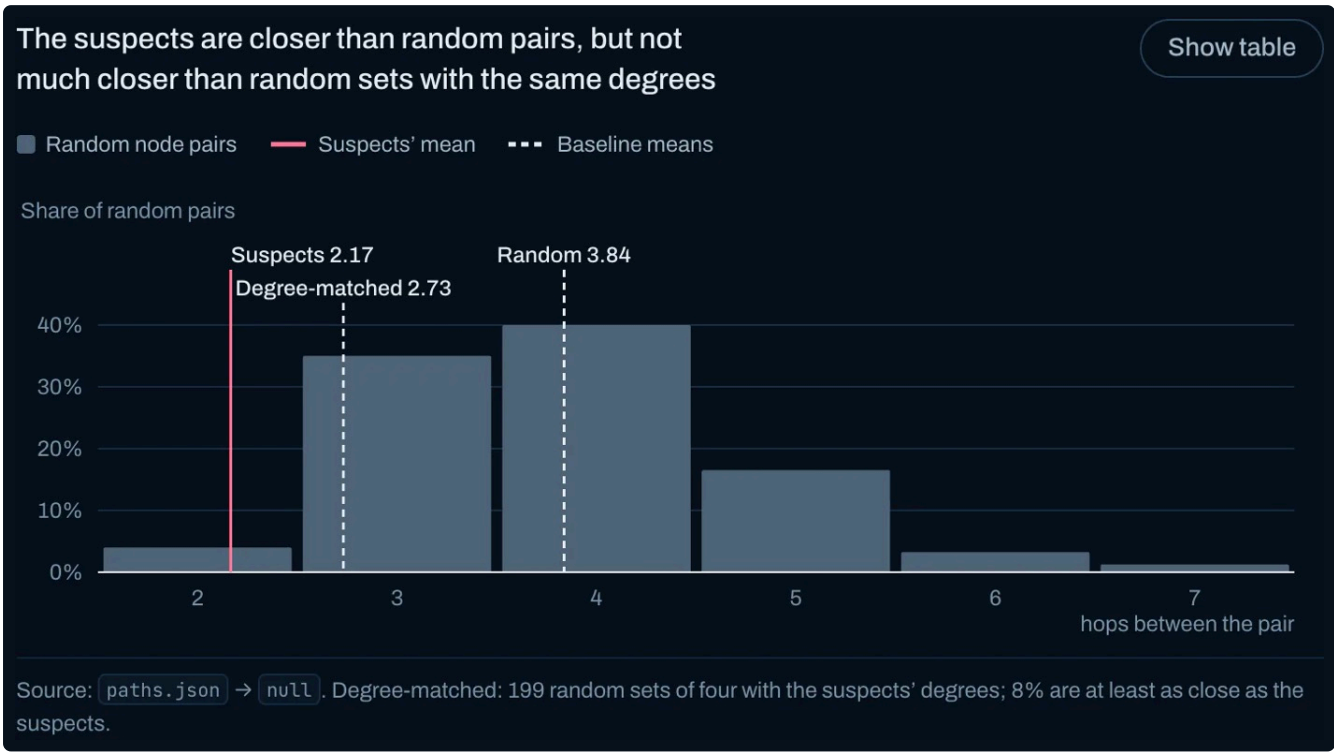


Figure 2.4. Suspects average 2.17 hops apart, random pairs 3.84, degree-matched sets 2.73.

Overall confidence that a pattern exists: **Medium** for the weak ownership layer, **Low** for any tie between specific suspects.

Question 3

> *What other companies should FishEye investigate for illegal fishing? Show how your visual analytics can be used to find entities that are worthy of further investigation. Limit your response to 600 words and 6 images.*

How the tool surfaces them. It scores all 1,209 companies and organizations on seven structural features, each converted to a percentile before weighting: low-weight edges (0.25), proximity to suspects (0.20), shared neighbours (0.15), ownership counterparties (0.15), relation-type profile similarity (0.10), repeated or mixed-type edges (0.10) and a FishEye tag (0.05). Each candidate comes with its feature contributions (Figure 3.1), an evidence subgraph of at most 22 nodes, and the verdict rubric used on the suspects (Figures 3.4, 3.5).

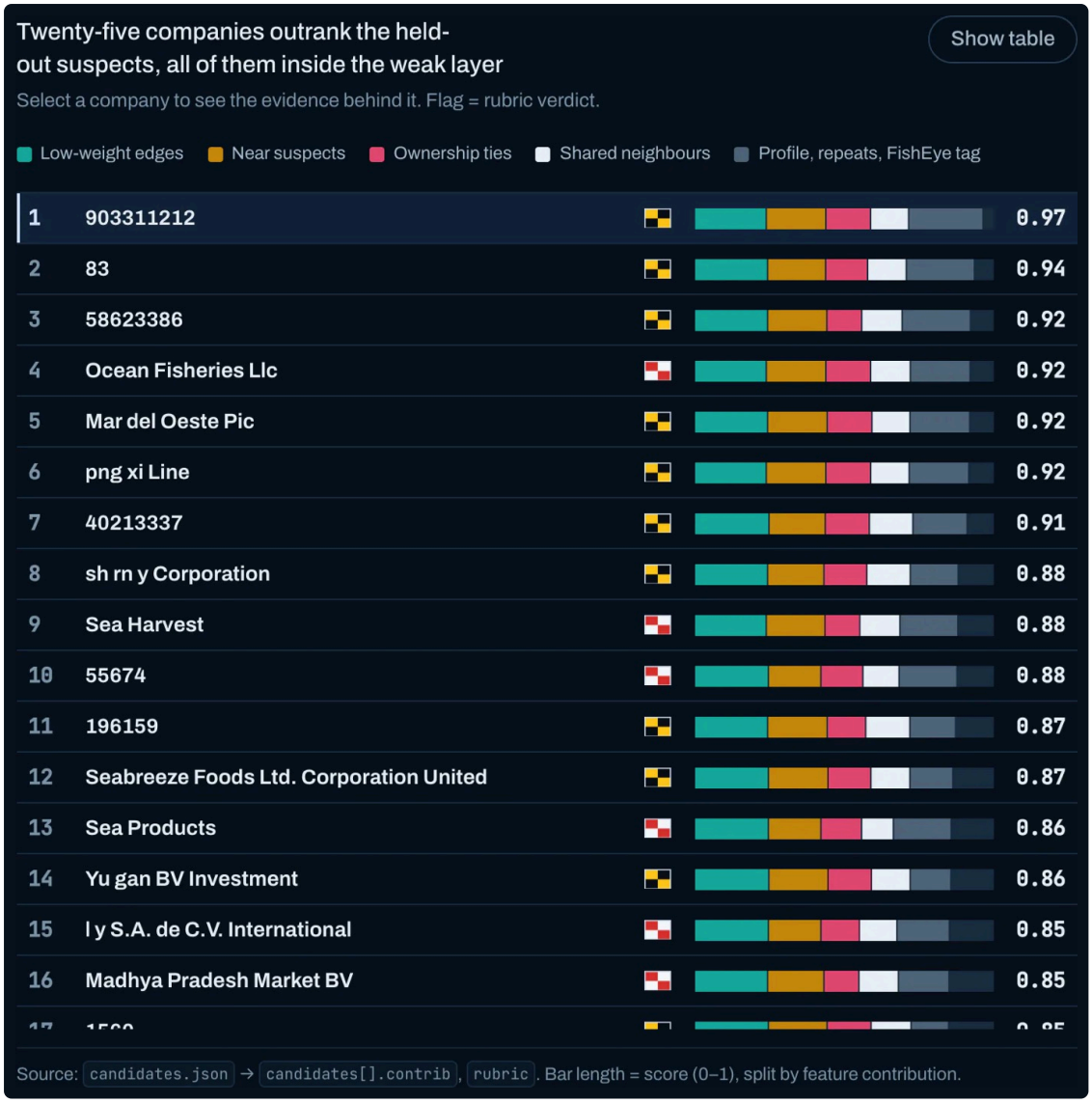


Figure 3.1. Ranked candidates; bars split by feature contribution, flags give the rubric verdict.

Validation.

- **Held-out suspects.** Holding each suspect out and seeding with the other three, the held-out one ranks 11th, 51st, 41st and 35th of 1,210 (top 1–4%). Ranking by degree places them 96th, 7th, 528th and 128th (Figure 3.2).
- **Weight changes.** Across 300 random reweightings, the ranking keeps a mean Spearman correlation of 0.98 (minimum 0.89).
- **Ablation.** Without the low-weight feature only 8 of the top 25 stay (Figure 3.3). The list is mainly a way to rank the layer.

All 25 top candidates sit in the weak layer; 4 are hubs.

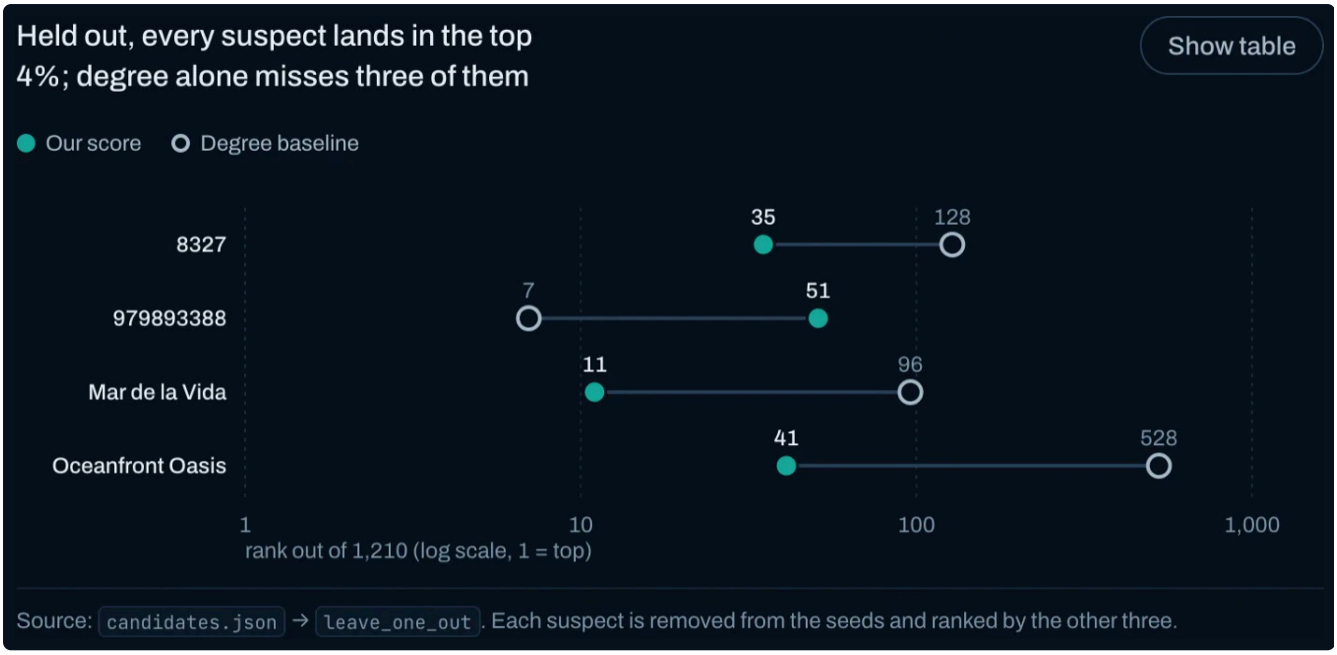


Figure 3.2. Held out, each suspect ranks 11th–51st (dots); degree (rings) misses three.

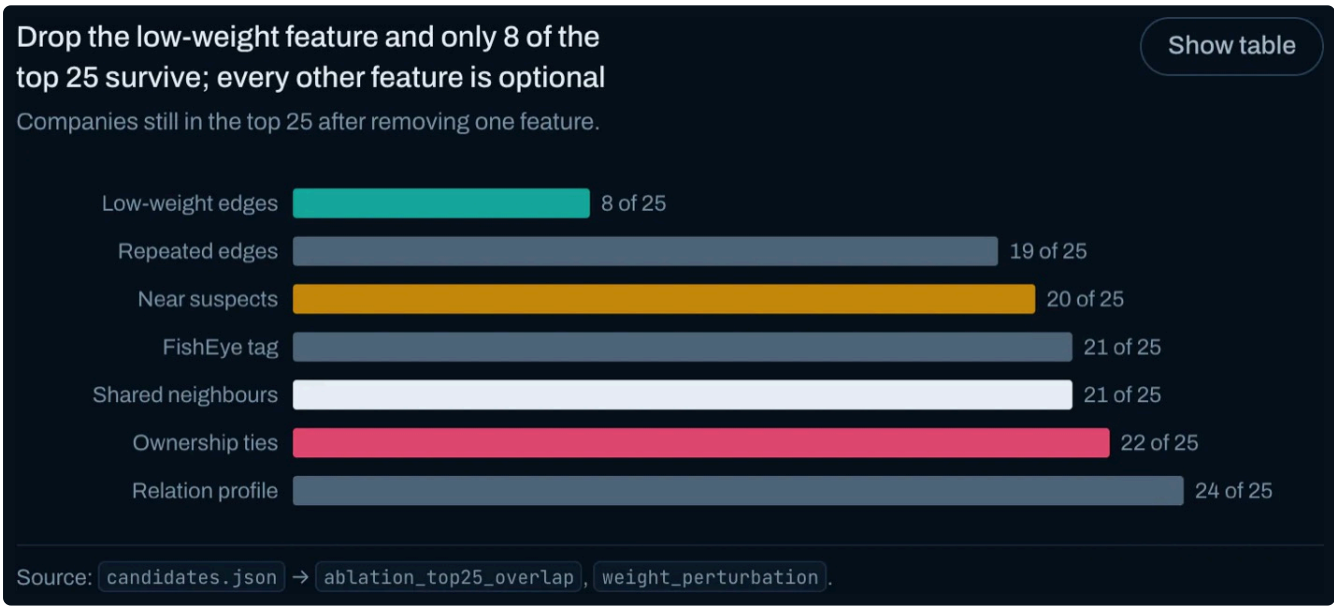


Figure 3.3. Without low-weight edges only 8 of the top 25 survive.

Investigate first: rubric Suspect with Medium or High confidence.

- **#903311212** (rank 1, score 0.97; Figure 3.4). A hub with 234 neighbours, 43 owned and 42 owners. FishEye tags it, it has a low-weight ownership edge, it links to the "poached" fragment, and it reaches 979893388 through Mar del Oeste Pic.
- **#48** (rank 18, the only **High**-confidence result; Figure 3.5). FishEye tags it with a high-weight edge (0.95). It owns 5 and is owned by 4; an ownership chain reaches 979893388 through png xi Line and Ocean Fisheries Llc.
- **#40213337** (rank 7). The holding 8327 owns, with six low-weight ownership rows; it owns Sea Harvest, Sea Products and #196159.
- **#196159** (rank 11). Owned through #40213337 and next to a FishEye-tagged entity.
- **#83** (rank 2). FishEye-tagged; owns 4, owned by 1, with a chain to 979893388.

- **Mar del Oeste Pic** (rank 5), **png xi Line** (rank 6), **sh rn y Corporation** (rank 8). The first two are hub-core members that own or connect to 979893388; the third has six ownership ties and shares neighbours with the suspects.

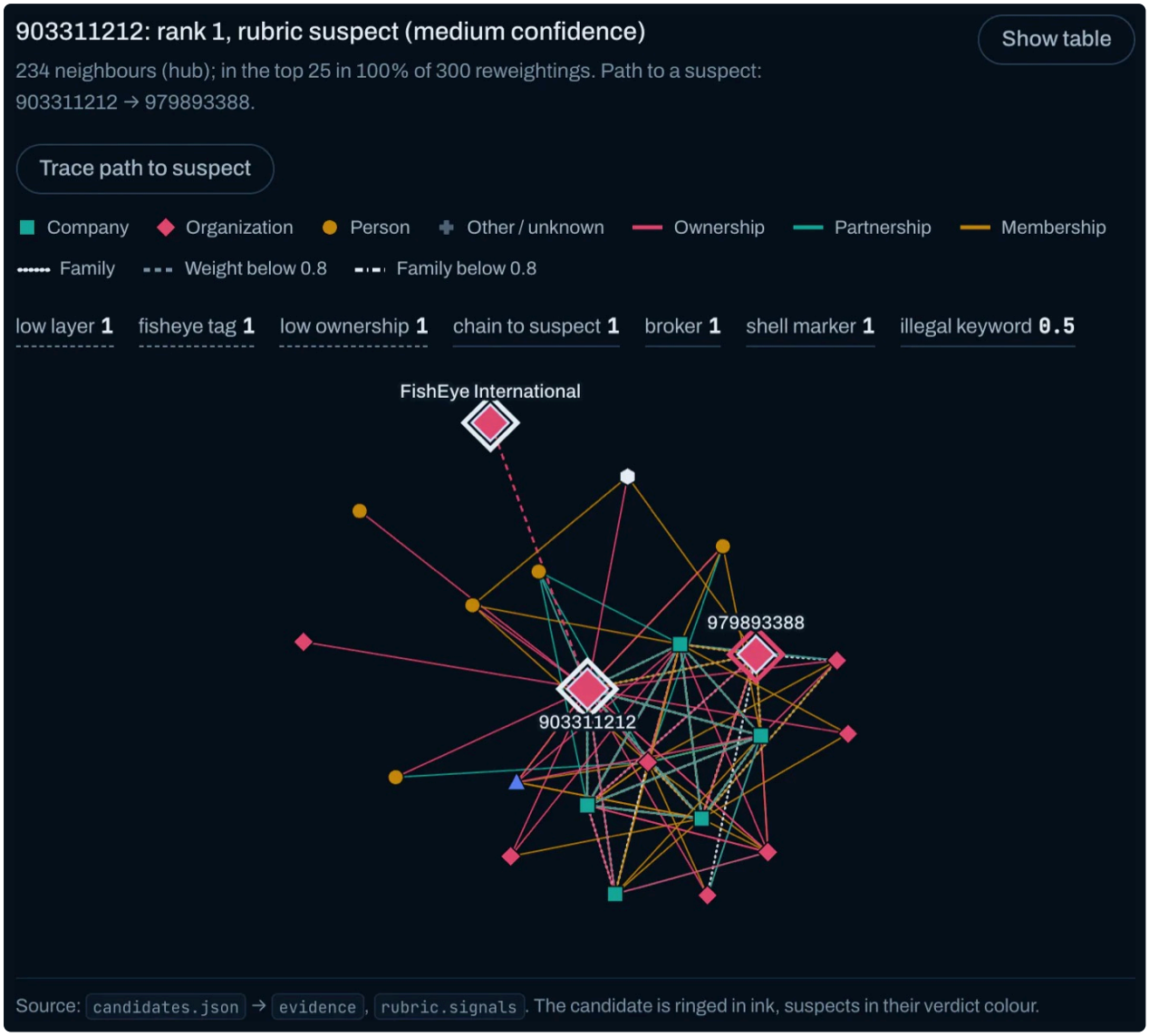


Figure 3.4. #903311212 (rank 1, Suspect, medium): a hub one hop from 979893388.

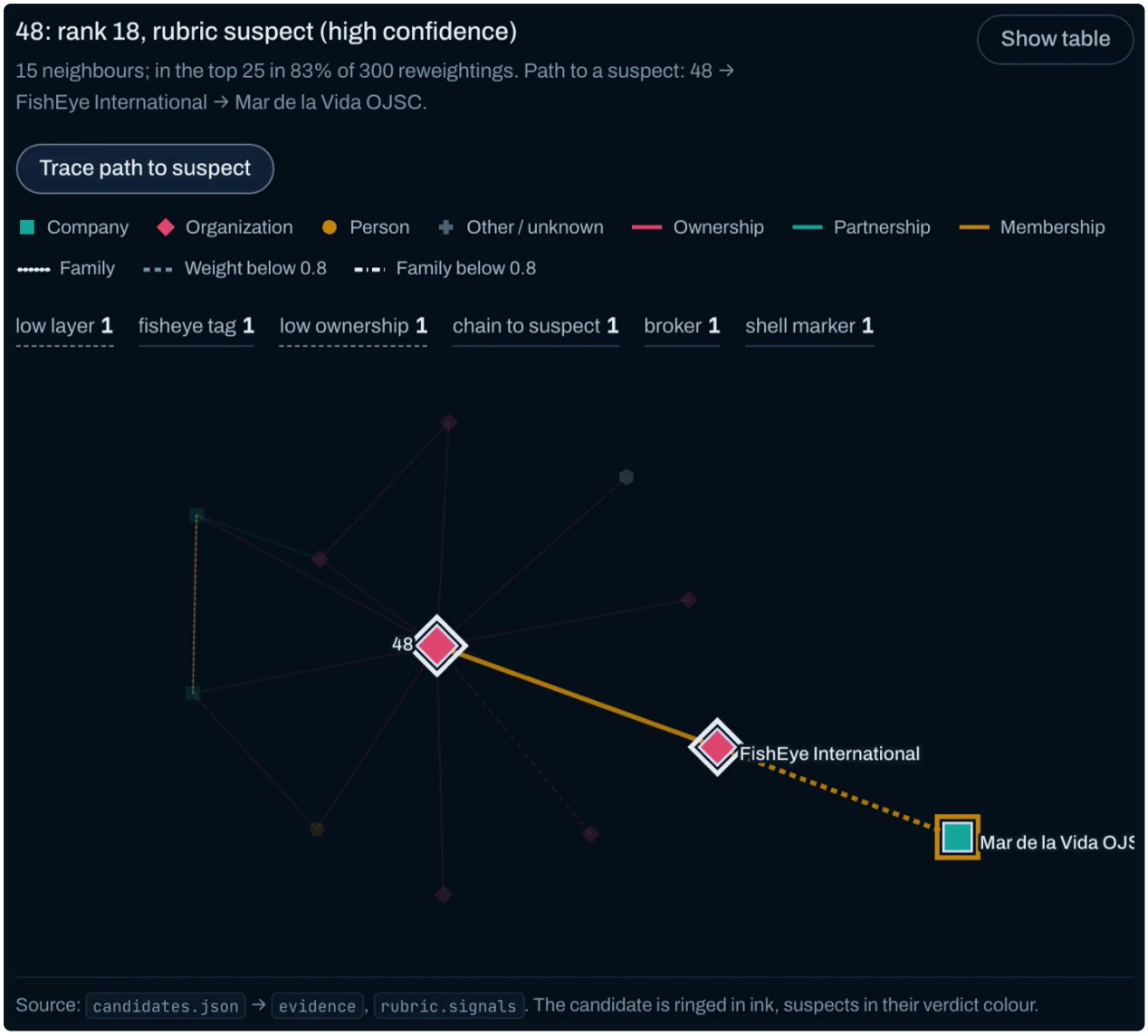


Figure 3.5. #48, the only high-confidence candidate, path traced through FishEye to Mar de la Vida.

Next tier: Suspect with Low confidence, or Possible. Seabreeze Foods Ltd. Corporation United (8327's FishEye-tagged owner); Yu gan BV Investment; #1560, #8814 and #923825740; Ocean Fisheries Llc, Sea Harvest, Sea Products and #55674.

Structures, not just entities.

- Two circular-ownership loops (Figure 3.6): #21538 → #52193 → #958085 → #55674 → #21538, and American → #5421 → #81 → "l y S.A. de C.V. International" → #59 → American.
- A five-hop chain: Darin Torres → Federated → #1354 → Ht Shipping → Yu gan BV Investment → #4942.
- Nine people in the layer, including Daniel Henderson, Traci Shields and Joseph Smith. Weak family edges tie them together and several own layer companies, so they may be the beneficial owners.

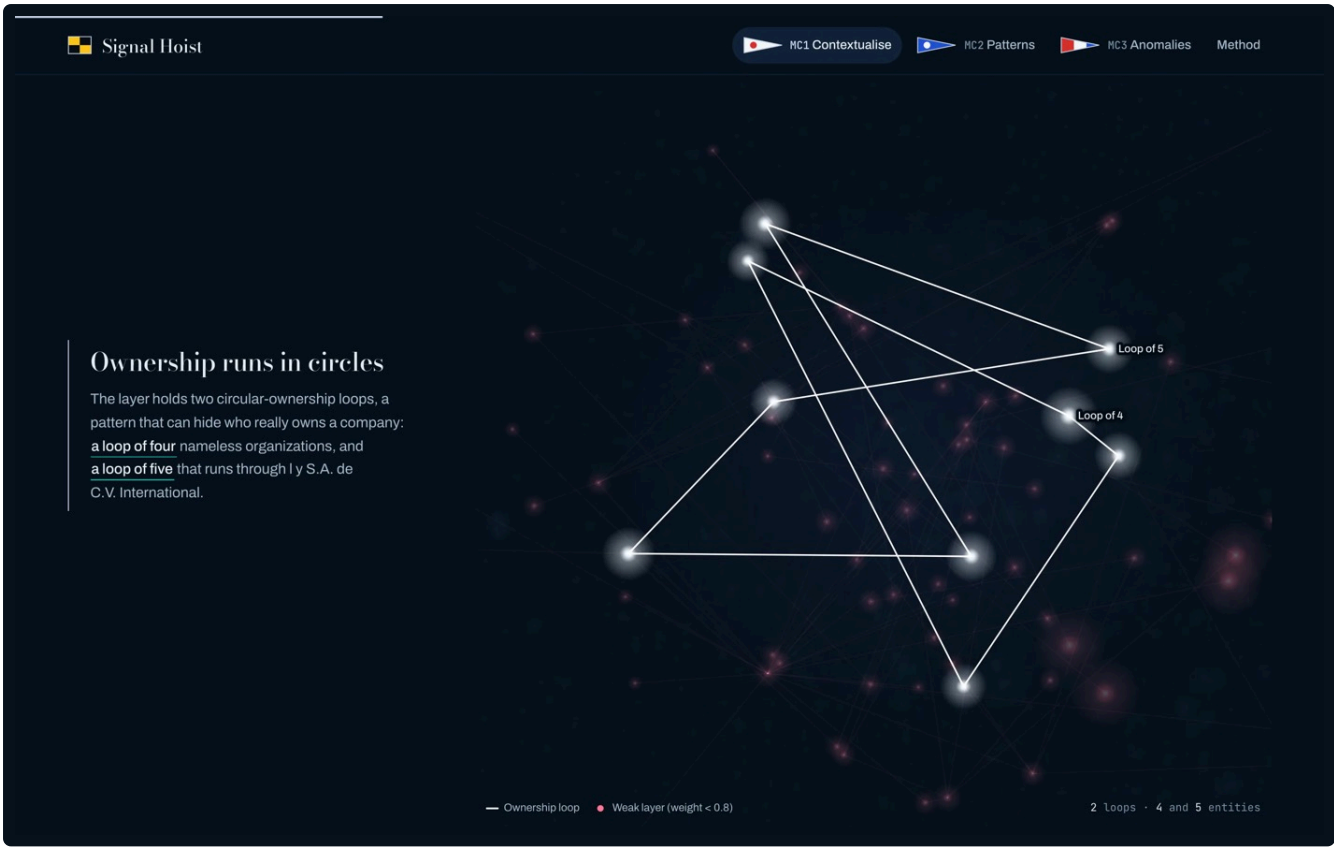


Figure 3.6. Story step: the layer's two circular-ownership loops.

Confidence. Medium for the layer as a whole, which is statistically clear. **Low to Medium** for each individual company, because the strongest edges in that evidence are also the least reliable ones.

Question 4

> *Reflection: What was the most difficult aspect of working with this knowledge graph? Did you have the tools and resources you needed to complete the challenge? What additional resources would have helped you? Limit your response to 300 words*

The hardest part was deciding which parts of the graph to trust. The extraction errors are not random; they form structure that can look like evidence:

- About 10% of nodes are NLP debris (garbled text, words like "weeks" labelled as vessels), and 605 nodes have no type.
- Family edges join two non-people 46% of the time.
- 322 edge rows repeat the same source, target and type; 124 pairs carry several relation types at once.
- A 10-node hub core holds 3.4% of all edge rows; Ocean Fisheries Llc and SeaSpray Wave SRL Solutions share 40 edges of all four types.
- Integer and string ids collide (77 vs "77", 11 cases), so every id needs a type-aware key.
- The data notes declare 3,721 nodes and 7,422 edges; the file has 3,428 nodes and 11,069 rows, and one row misspells its weight field.

The most important signal, edge weight, is undocumented, and so is ownership direction. So the weak layer was treated as a hypothesis: tested against a random-edge baseline, checked for predicting held-out suspects better than degree, and every verdict records the weight of the edges it rests on. Hubs also hide structure, because many

shortest paths run through them and community detection is unstable; the tool compares paths with and without hubs and runs Louvain with several seeds.

Tools and resources. Open-source Python and D3 were sufficient. What would have helped most is a data dictionary defining `weight` and the direction of ownership edges, and ground truth for even a few entities to calibrate the rubric.

VAST Challenge 2023: Mini-Challenge 2

Entry Name: Signal Hoist

Team Member(s): [Name] — [Affiliation], GitHub @amukherjee1991

Tools Used: Python (pandas, NetworkX, scikit-learn, SciPy, PyArrow), React, D3, Vite, Playwright

Approximately how many hours were spent working on this submission in total? [AUTHOR TO FILL: hours]

May we post your submission in the Visual Analytics Benchmark Repository after VAST Challenge 2023 is complete? [AUTHOR TO FILL: Yes / No]

Video: Not provided

Interactive version: https://amukherjee1991.github.io/Vast23/#/mc2 (every figure below is a screenshot of this page in its night-sea design: either a step of the scroll story that opens the chapter, or an evidence view from the question sections; all are interactive there)

Questions

Figures are numbered *question.figure*: Figure 2.3 is the third figure in the answer to Question 2. Confidence is stated as **High**, **Medium** or **Low**.

Data preparation. The graph has 5,464,378 shipment edges; 155,291 (2.8%) exactly duplicate another and are dropped, leaving 5,309,087 across 34,576 companies, 245,807 shipper→receiver pairs and 84 months (2028-01 to 2034-12). HS codes for chapters 01–09 lost their leading zero (306170 is 0306.17, frozen shrimp), and the bundles store them with five digits (30729); appending a "0" matches the graph's codes 100% of the time. "Fish" means HS headings 0301–0309 plus 1604/1605: 13.9% of edges.

Question 1

> *Use visual analytics to identify temporal patterns for individual entities and between entities in the knowledge graph FishEye created from trade records. Categorize the types of business relationship patterns you find. Limit your response to 600 words and 6 images.*

The market as a whole (Figures 1.1, 1.2). Yearly volume peaks in 2030 at 831k shipments and falls to 688k in 2034. Fish holds steady at about 96k–116k a year, so its share rises from 12.7% (2028) to 15.6% (2034). Fish is mildly seasonal: a calendar index of 0.84–0.89 in February–April and 1.05–1.08 in July–December, against 0.91–1.09 for all goods.



Figure 1.1. Story step: each light is ~200 shipments on a current between countries; most flows end in Oceanus.



Figure 1.2. Story step: fish (amber) holds steady while total trade ebbs.

Individual companies (Figure 1.3). The two roles behave very differently:

- **Shippers are durable.** Median span 59 months; 4,875 of 12,217 are active for 73–84 months.

- **Receivers are fleeting.** Median span 3 months; 44% appear in a single month.
- **Fish shippers churn more.** 2,139 of 4,299 fish-majority shippers are *transient* (they appear after the data starts and disappear before it ends), against 4,740 of 12,217 shippers overall.

The 150 per-company series online show flat hubs, step-ups, step-downs and bursts.

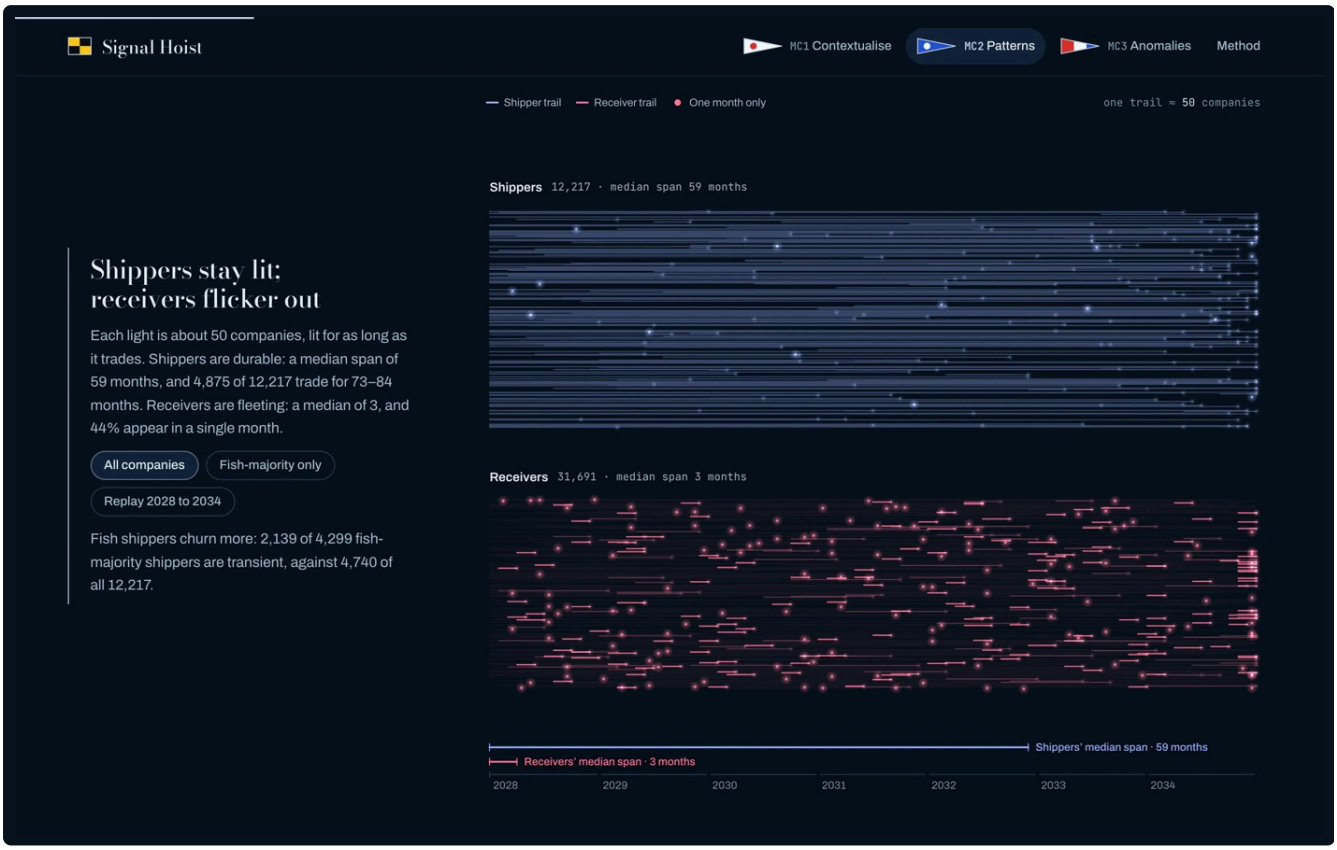


Figure 1.3. Story step: shipper trails (above) last years; receivers (below) flicker out.

Relationships between companies (Figure 1.4). Each pair's monthly series is classified by span, coverage, trend, seasonality and how strongly it ends; rules apply in order:

Pattern	Pairs	Share of shipments	Signature
One-off	108,067 (44%)	2.6%	a single month
Intermittent	60,968	14.5%	recurring but irregular
Short burst	31,604	3.0%	≤ 6 months, then nothing
Fading	16,729	21.9%	second half ≤ ½ of the first
Ramping	13,385	16.0%	second half ≥ 2× the first
Abrupt stop	5,994	8.3%	regular trade that stops at full strength ≥ 12 months before the data ends
Steady	4,634 (1.9%)	32.6%	≥ 24 months, ≥ 60% of months active
Seasonal	4,426	1.2%	≥ 75% of trade in the same 3 calendar months

The network is a small core of steady pairs carrying a third of all shipments, ringed by a large fringe of one-off links. Only 737 pairs trade in both directions, so relationships run one way. Fish pairs follow almost the same mix; abrupt stops are slightly *rarer* among them (1.7% vs 2.4%), so an abrupt stop alone is not a sign of illegal fishing. As a cross-check, k-means (k = 8) on 101,115 recurring pairs agrees with the rules at a weighted purity of 0.57: steady, fading and ramping separate cleanly; the catch-all "intermittent" does not.

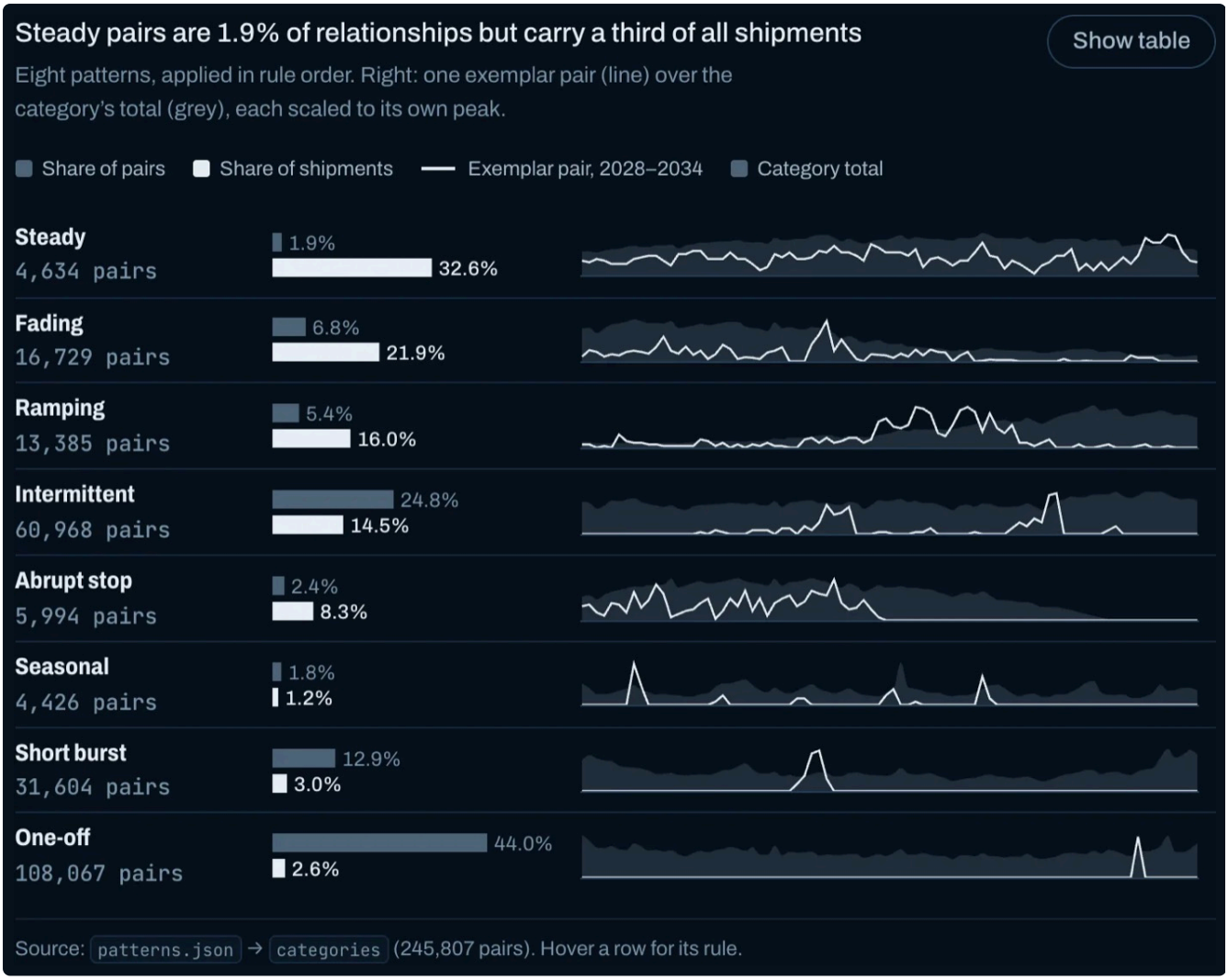


Figure 1.4. Steady pairs are 1.9% of relationships but carry 32.6% of shipments.

Between entities: successors (Figure 1.5). We took company pairs with very similar partner and product profiles (partner and HS cosine ≥ 0.7, ≥ 3 shared partners) and counted them by the gap between the earlier company's last month and the later one's first month:

- **Twins.** Most such pairs *coexisted* for 12–36 months (31 shipper pairs fall in −36..−6). These are sister firms, not replacements.
- **Restarts.** A second, narrower peak sits at gaps of 0–6 months: 1.7 pairs per gap-month, against 0.3 in the −6..0 window just before.

That excess is the signature of a company stopping and a look-alike starting straight afterwards: real but small, so successor pairs are rare and specific (see Question 4).

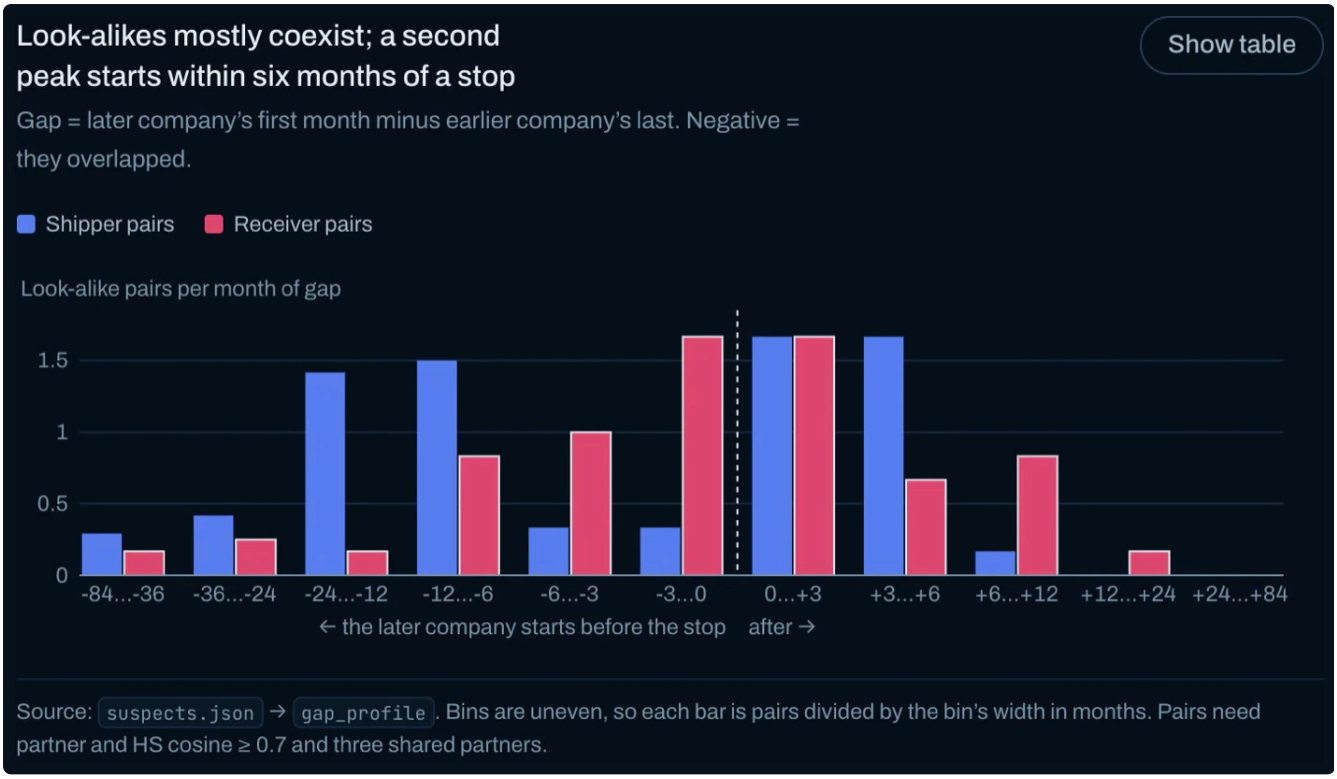


Figure 1.5. Look-alikes mostly coexist; a second peak starts 0–6 months after a stop.

Confidence: High for the aggregate patterns, which are counts over the full graph. The category boundaries are analyst choices, documented with each category in the tool.

Question 2

> *Evaluate the sets of predicted knowledge graph links FishEye has provided using visual analytics. Which sets are most reliable for completing the graph? Limit your response to 600 words and 6 images.*

Each bundle is scored against the real graph on 11 components, each 0–1: repetition, novelty, schema, dates, month mix, pair and product history, HS mix, concentration, weight, value per kg, and node countries (Figures 2.1, 2.2). Distribution tests compare each bundle with **200 same-size random samples of real 2034 edges**, so small bundles are not punished for noise. Gates override the score:

- **Critical gates force "Unreliable":** under 25% of links distinct, fewer than 20 distinct links, or over 95% exact copies of graph edges.
- **Major gates cap the verdict at "Questionable":** under 50% of links carry a weight, or over 5% fall outside the graph's date window.

Novelty is two-sided. An exact match with a real edge (same shipper, receiver, HS code and day) is almost impossible by chance: some matches anchor a bundle to real records; all matches mean it adds nothing. The five diverse bundles independently converge on 42–59% matches.

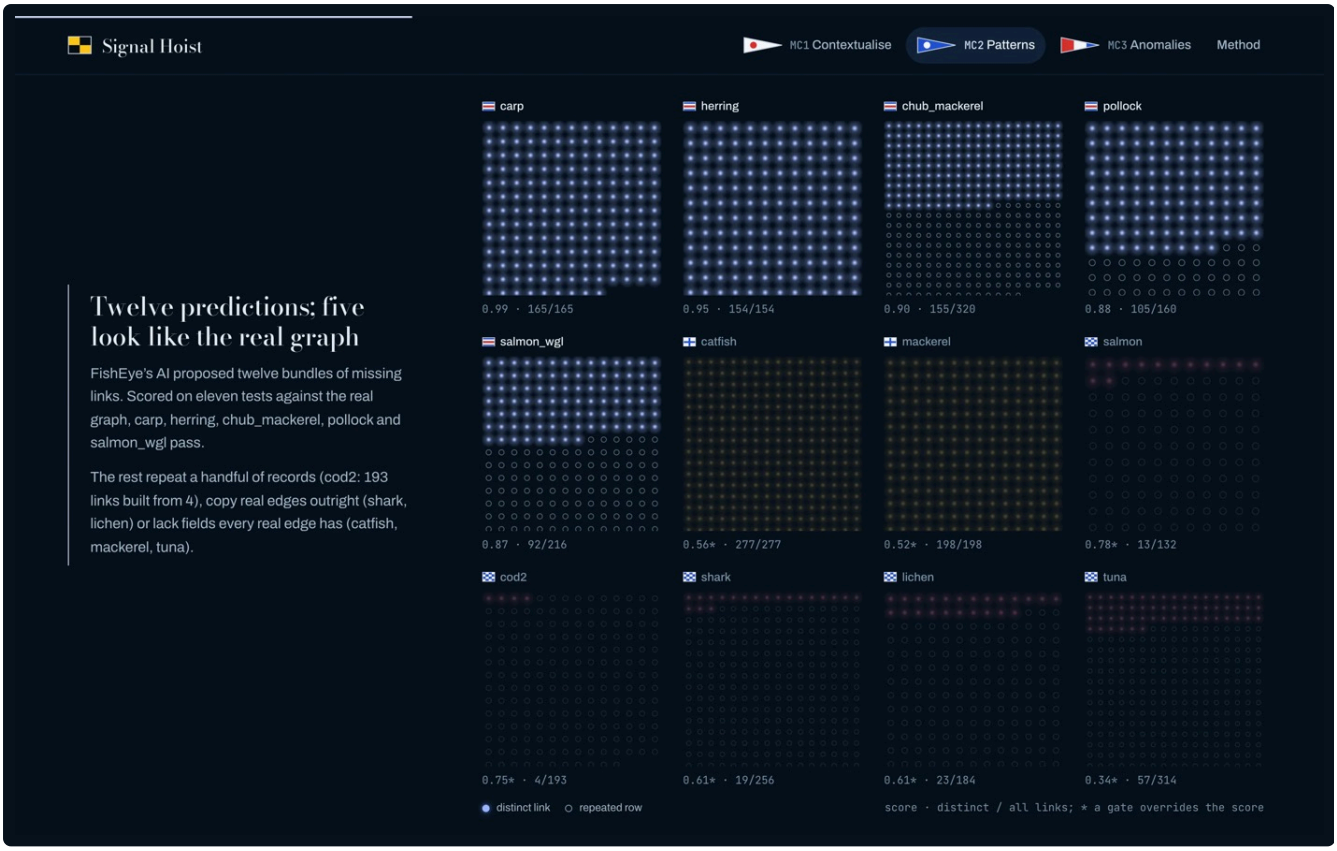


Figure 2.1. Story step: bundles as link grids; repeats dim, five stay bright.



*Figure 2.2. Scorecard: 11 tests per bundle; flag = verdict, * = gate override.*

Bundle	Score	Verdict	Decisive evidence
carp	0.99	**Reliable**	165 distinct links, 44% anchored, value bias +0.09 dex, 95% country match (Figure 2.3)
herring	0.95	**Reliable**	clean; month mix slightly off (JSD 0.068 vs null p95 0.026)
chub_mackerel	0.90	**Reliable**	52% of rows repeat, but 155 distinct links pass every realism test
pollock	0.88	**Reliable**	34% repeats, somewhat concentrated (top target 17%)
salmon_wgl	0.87	**Reliable**	57% repeats; 92 distinct links are realistic
catfish	0.56	**Questionable**	no weight at all, 8.7% of dates in 2035, 49% country match, 8% anchored
mackerel	0.52	**Questionable**	the same problems; 32% of dates in 2035
salmon	0.78*	**Unreliable**	132 links are 13 records (9.8% distinct)
cod2	0.75*	**Unreliable**	193 links are **4** records; top target "-53" never receives anything in the graph (Figure 2.5)
shark	0.62	**Unreliable**	19 distinct records, all exact copies of graph edges
lichen	0.61	**Unreliable**	23 distinct records, all copies; value bias +0.26 dex

Bundle	Score	Verdict	Decisive evidence
tuna	0.34	**Unreliable**	no weights, 0% anchored, 18% distinct, 22% country match, 33% of links to one receiver (Figure 2.4)

*A gate overrides the score; on 4–13 distinct records the distribution tests have almost no power.



Figure 2.3. carp (Reliable, 0.99) against real 2034 edges (white ticks): 165 distinct links, months, weights, prices and HS headings all in line.

Three findings back the verdicts:

1. **Repetition separates the bundles most sharply.** Real data repeats 2.8% of records; the Unreliable bundles repeat 82–98%.
2. **Missing schema identifies the fabricated bundles.** catfish, mackerel and tuna have no weightkg (present on 100% of graph edges). They also disagree on node countries (22–49% vs 93–97%) and rarely match a real record.
3. **Prices.** Reliable bundles' OMU/kg sits within 0.12 dex of the graph's typical price per HS code; lichen and cod2 run high.

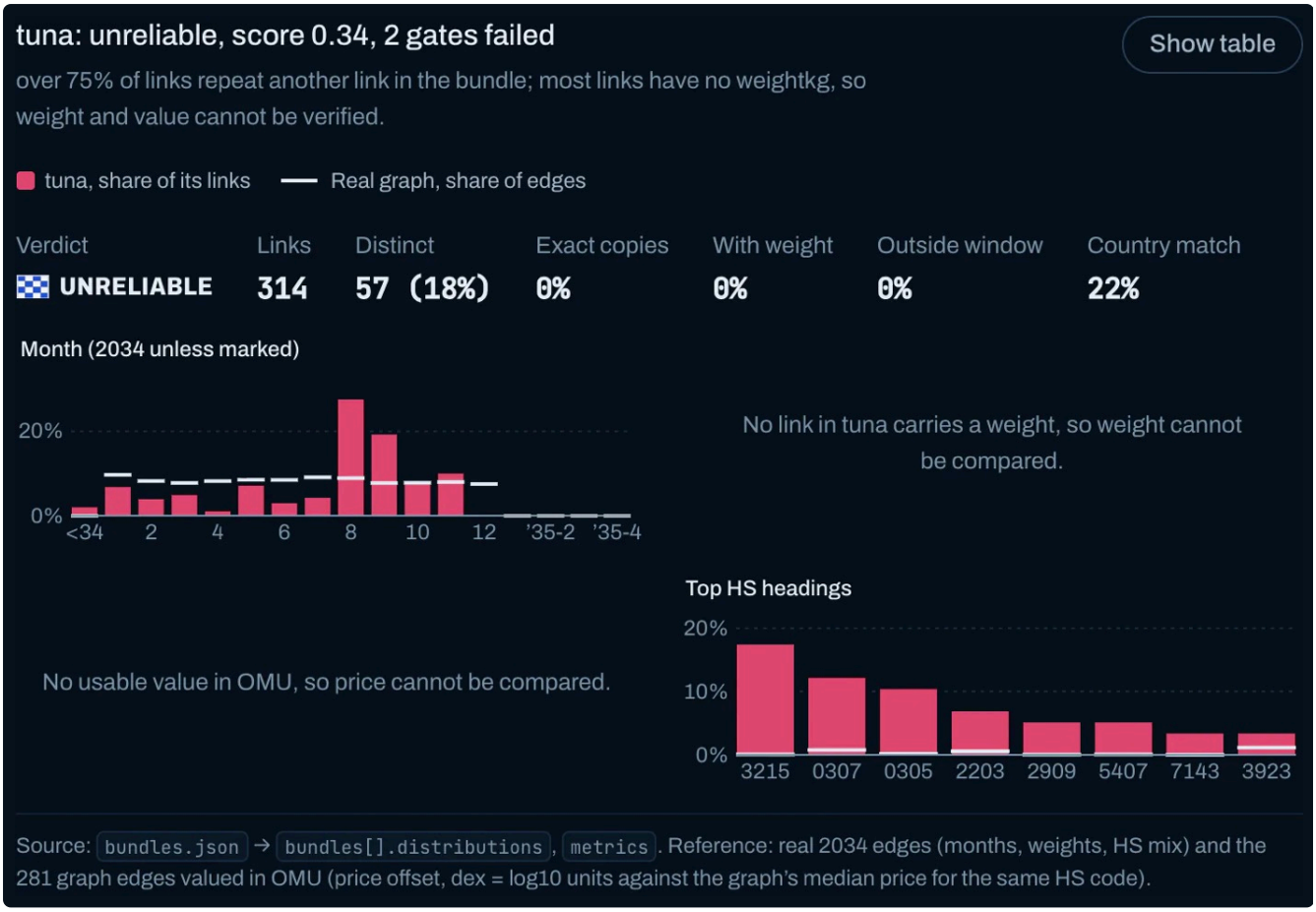


Figure 2.4. tuna (Unreliable, 0.34): 314 links but 57 distinct, no weights or OMU values to compare, 22% country match.

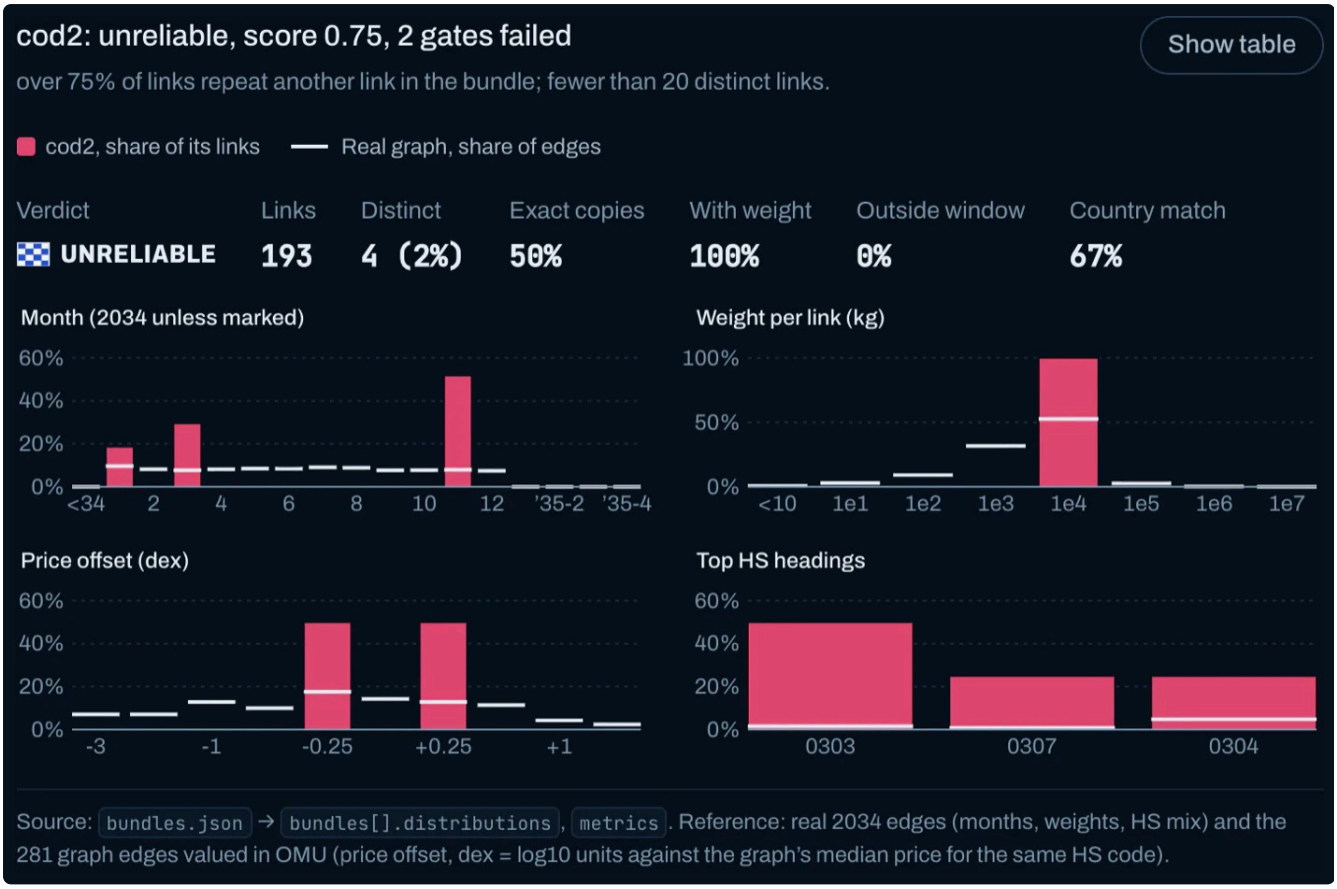


Figure 2.5. cod2 (Unreliable, 0.75

*): 193 links collapse to 4 distinct records, so every distribution is a few spikes.**

The bundles' pair sets barely overlap (Jaccard ≤ 0.05), so corroboration across bundles is impossible.

Most reliable for completing the graph: carp and herring, then chub_mackerel, pollock and salmon_wgl.

Confidence: High for the five Unreliable bundles and for carp and herring. **Medium** for chub_mackerel, pollock and salmon_wgl (realistic distinct links, unexplained repetition) and for catfish and mackerel (weight and value unverifiable).

Question 3

> *Illustrate how your visual analytics approach can be used to identify new patterns and/or anomalies that are present in the knowledge graph after you have added the links you deemed reliable in question 2. Limit your response to 300 words and 4 images.*

- **The links add little (Figure 3.1).** The five Reliable bundles hold 1,015 links; only **347** are usable (distinct, new, inside the window): under 0.1% of 2034 volume. The 2034 fish share stays at 15.6%.
- **Pair patterns barely move (Figure 3.3).** 261 existing pairs gain links and 58 new pairs appear. Only 15 pairs change pattern, mostly one-off or short burst → intermittent, and fading or ramping → steady.
- **Three apparent stops are data gaps (Figure 3.2).** Predicted links keep three "stopped" companies trading: *Chhattisgarh Salt spray OAO United* (receiver, 174 shipments; last real month 2034-03, predicted to 2034-11), *Bahía Azul Ltd Holdings* (shipper, 2034-05 → 2034-12) and *Okavango River S.A. de C.V. Solutions*. Late-2034 stops are partly missing data, so the successor detector requires 6 silent months and marks a pair Unlikely when predicted links revive the predecessor.

- **The flags are stable (Figure 3.4).** All 68 flagged successor pairs (8 Suspect, 60 Possible) keep their verdicts on the augmented graph, and also after adding the two Questionable bundles (701 links). That robustness is the most useful result.
- **Hubs get richer.** Added links go mostly to existing hubs: *hăi dăn Corporation Wharf* (33), *Sailors and Surfers Incorporated Enterprises* and *Mar del Este CJSC* (11 each). The predictions densify the core rather than reveal hidden companies.

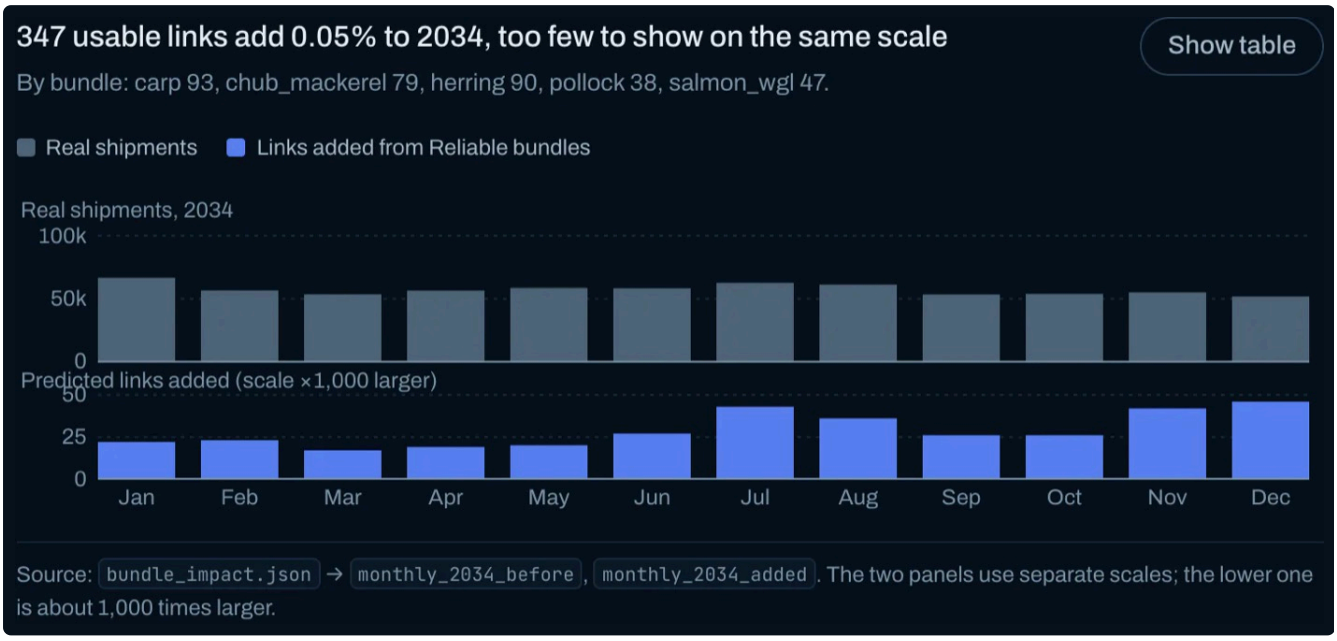


Figure 3.1. 347 usable links (bottom, ×1,000 scale) add 0.05% to 2034.

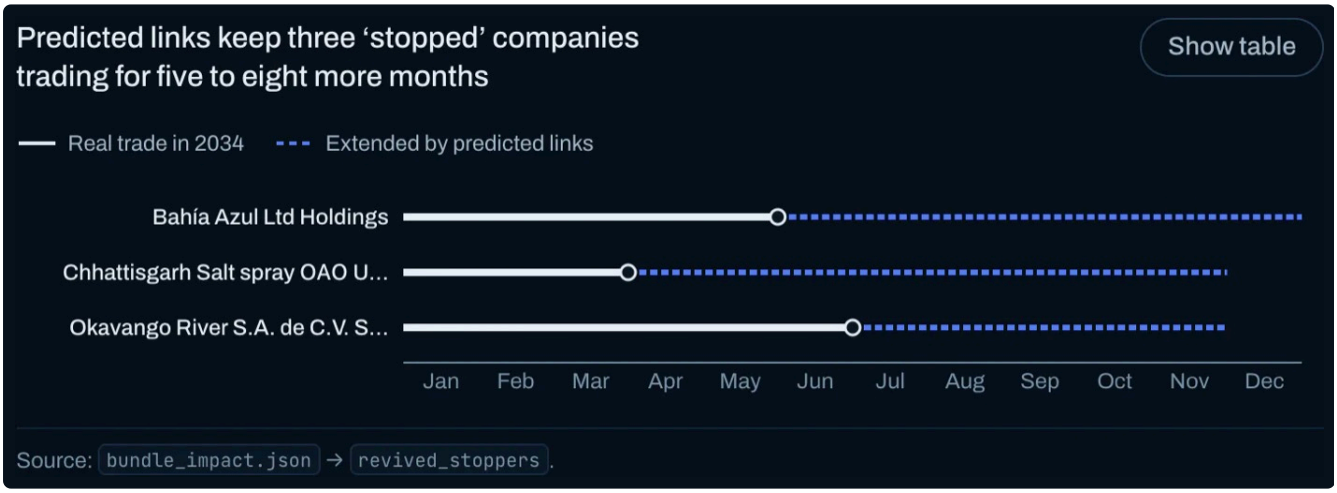


Figure 3.2. Predicted links (dashed) keep three "stopped" companies trading.

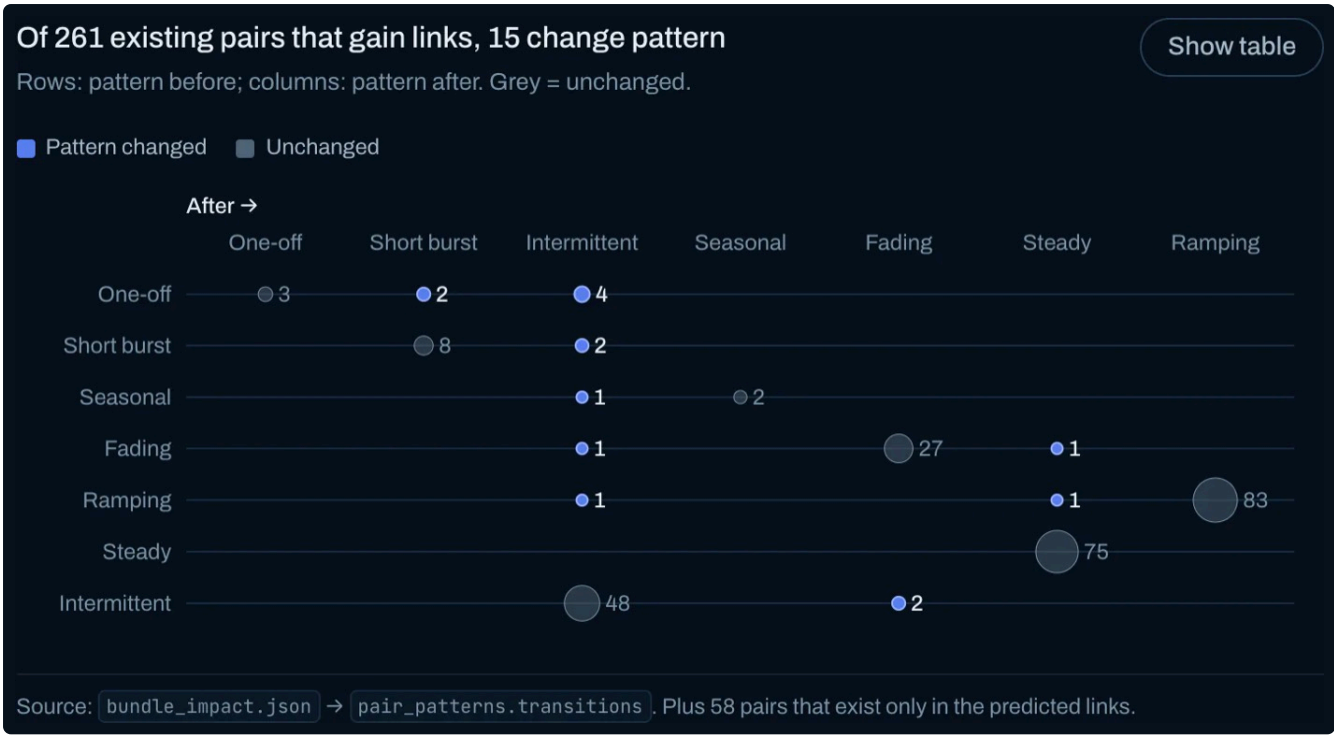


Figure 3.3. Of 261 existing pairs gaining links, 15 change pattern.

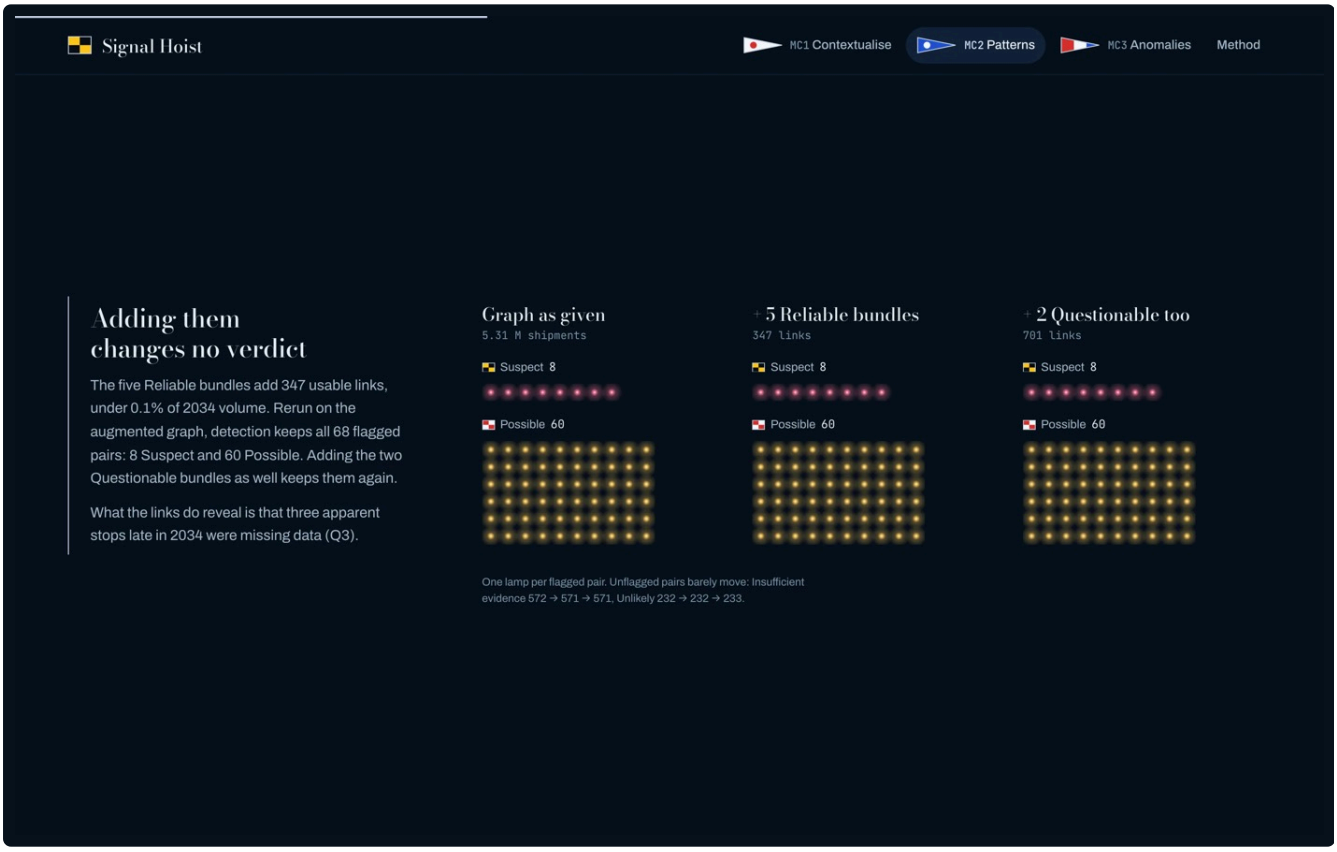


Figure 3.4. Story step: the same 8 Suspect and 60 Possible lamps with and without added links.

Confidence: High that the impact is small. **Medium** on the revivals: three cases, each from one bundle.

Question 4

> *Identify companies that fit a pattern of illegal fishing. Use visualizations to support your conclusions and your confidence in them. Limit your response to 300 words and 4 images.*

We matched 2,780 shippers and 3,888 receivers that stop ≥ 6 months before the data ends with companies starting –6..+12 months from the stop. Similarity is TF-IDF cosine plus Jaccard on partners and HS codes, so shared hubs count little; shared partners must switch over. Of 872 candidate pairs: 8 Suspect, 60 Possible, 572 Insufficient evidence, 232 Unlikely (Figure 4.3).

Suspect pairs (all trade fish on both sides):

Predecessor → successor	Confidence	Evidence
receiver **-811 → -206**	High	fillets (0304); 2032-12 → 2033-01; 5 shared shippers supply 78% of -206's trade (Figures 4.1, 4.2)
Beachcomber's Bounty S.A. de C.V. Express → Karelians Limited Liability Company International	Medium	Karelians also succeeds five more fish shippers that stopped in 2030 (Figure 4.4)
Saltwater Surf Club Sp → Chhattisgarh GmbH & Co. KG Sea spray	Medium	dried fish (0305); 2028-12 → 2029-02; all 3 shared partners switch
Asveja Limited Liability Company → Marine Mavericks Marine biology	Medium	molluscs (0307); 95% of successor's trade inherited
receiver **-812 → -556**	Medium	prepared crustaceans (1605); 5 shared shippers
receiver **-513 → -690**	Medium	identical partner set
The Salty Dog Corporation → Gulf of Guinea Sp	Low	13-shipment predecessor
Estrella del Sol CJSC → Mandalore S.A. de C.V. Logistics	Low	12-shipment predecessor

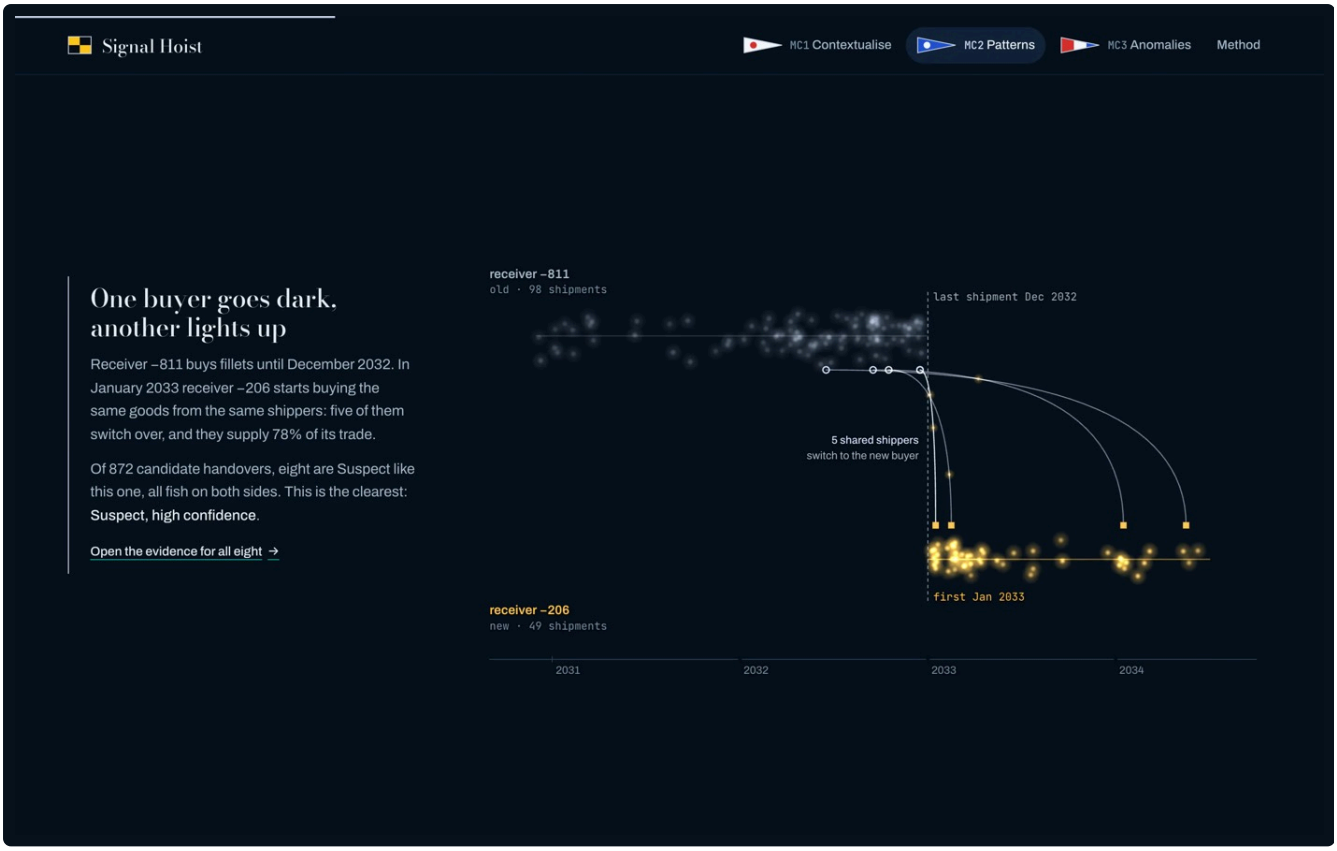


Figure 4.1. Story step: –811 goes dark; five shippers arc to –206.

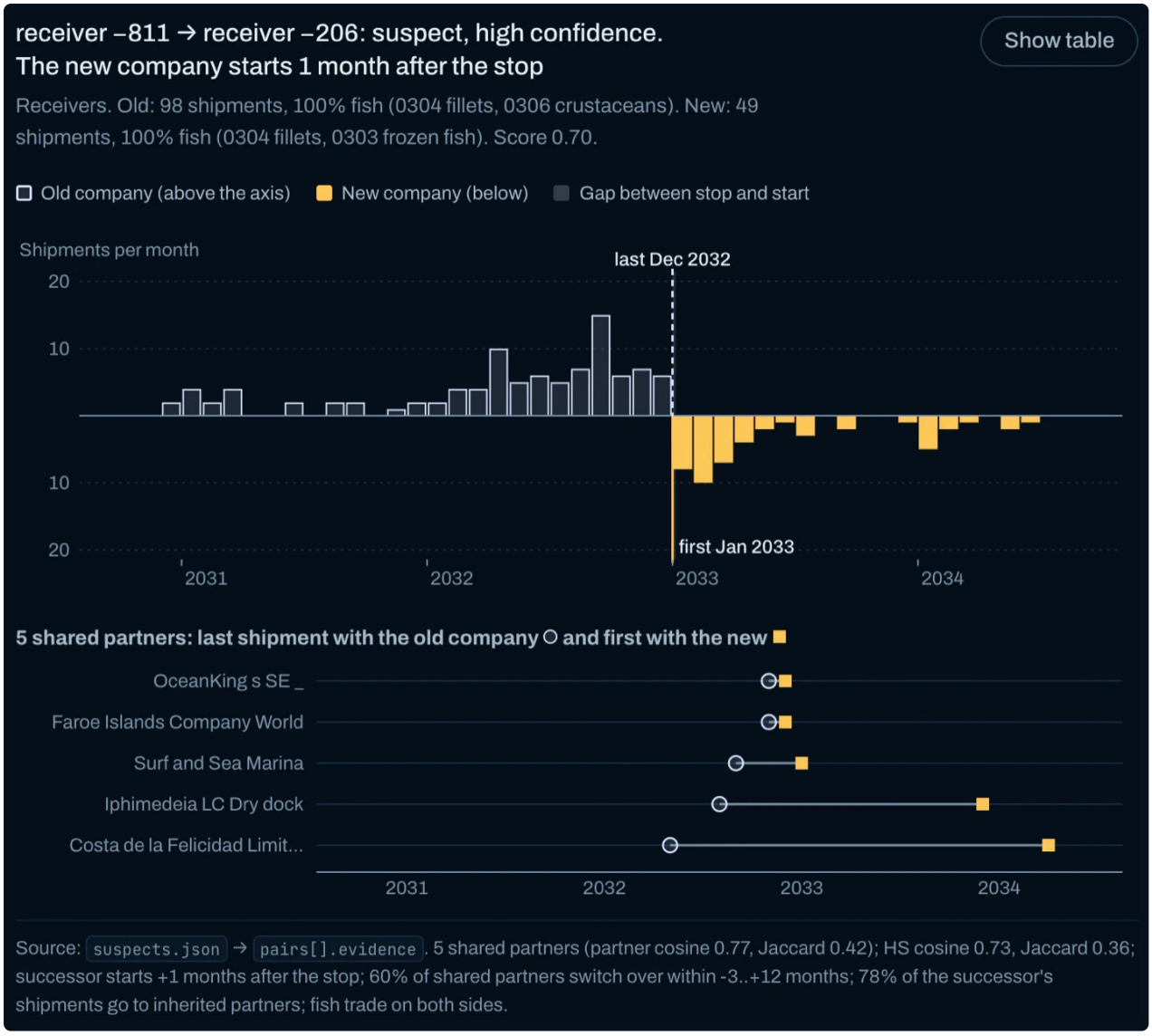


Figure 4.2. –811 → –206: mirrored monthly trade and each shared shipper's switch.



Figure 4.3. 8 Suspect pairs of 872 candidates, with confidence pennants.

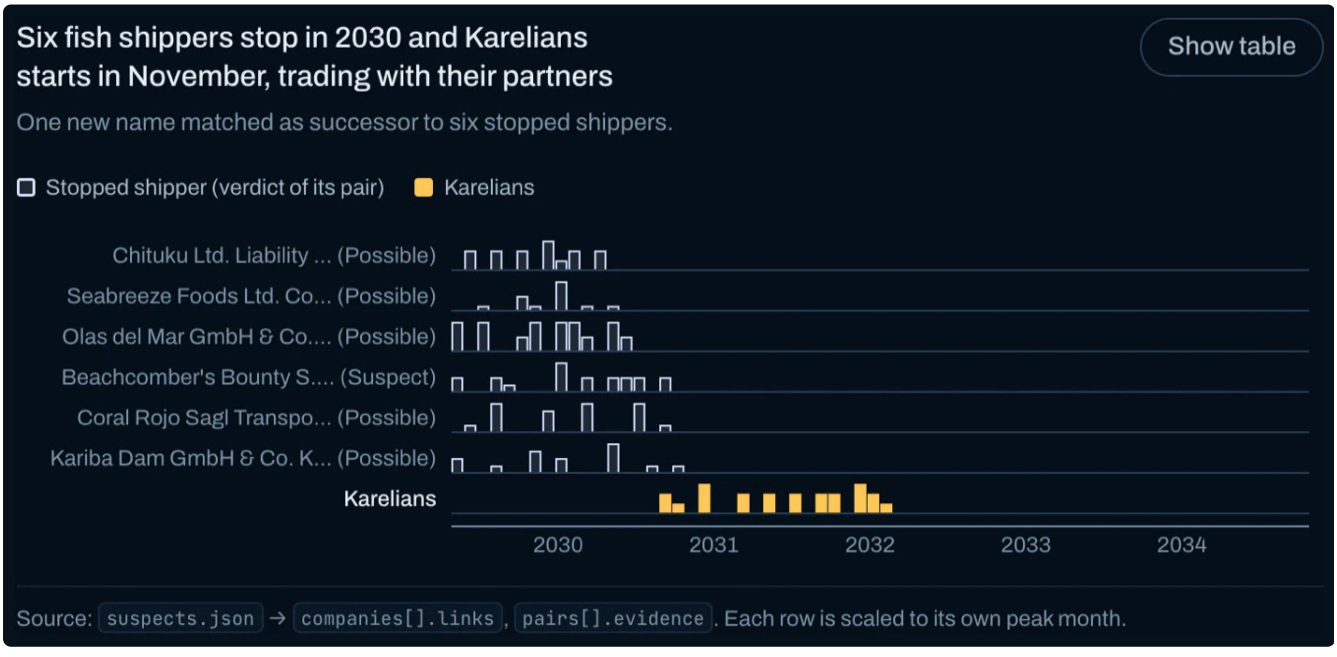


Figure 4.4. Karelians starts after six fish shippers stop in 2030.

Confidence: Medium overall: only 3–5 shared partners per pair, no ground truth, and 24 high-similarity pairs proved Unlikely because partners did not switch.

Question 5

> *Reflection: What was the most difficult aspect of working with this knowledge graph? Did you have the tools and resources you needed to complete the challenge? What additional resources would have helped you? Limit your response to 300 words*

The data needed the most care.

- **HS codes:** the two code formats would have silently zeroed out chapter-03 fish links in every bundle.
- **Duplicates:** 2.8% of graph edges are exact duplicates.
- **Values:** the graph is in USD and the bundles in OMU, and only 281 graph edges carry OMU, so prices are compared as per-HS-code offsets against those 281 edges.
- **Names:** 22,004 company ids are bare negative numbers, so name similarity is useless as evidence.

Null models changed the conclusions. A first pass scored successors by raw cosine and rated catfish and mackerel "reliable". Weighting partners by rarity and comparing with same-size random samples of real 2034 edges reversed both. The gap profile showed most look-alikes coexist, so similarity alone would flood the list with false positives.

Some evidence is ambiguous. An exact copy of a real edge looked like a defect until five independent bundles each showed 42–59% exact matches; it is better read as evidence that a tool is anchored to real data.

Limitations. Scoring weights and thresholds are judgement calls, published so they can be challenged. Detection needs ≥ 2 shared partners, so very small companies cannot be judged. Stops near the end of 2034 cannot be told from missing data.

Tools and resources. Python with PyArrow (Parquet) made the 5.4M-edge graph tractable, and D3 covered the views. Documentation of the HS-code and currency conventions, and ground truth for a few known restarts, would have helped most. Next steps: compare against the MC1 and MC3 graphs, and let an analyst confirm or reject each pair with scores updating live.

VAST Challenge 2023: Mini-Challenge 3

Entry Name: Signal Hoist

Team Member(s): [Name] — [Affiliation], GitHub @amukherjee1991

Tools Used: Python (pandas, NetworkX, scikit-learn, SciPy, PyArrow), React, D3, Vite, Playwright

Approximately how many hours were spent working on this submission in total? [AUTHOR TO FILL: hours]

May we post your submission in the Visual Analytics Benchmark Repository after VAST Challenge 2023 is complete? [AUTHOR TO FILL: Yes / No]

Video: Not provided

Interactive version: https://amukherjee1991.github.io/Vast23/#/mc3 (every figure below is a screenshot of this page in its night-sea design: either a step of the scroll story that opens the chapter, or an evidence view from the question sections; all are interactive there)

Questions

Figures are numbered *question.figure*: Figure 1.3 is the third figure in the answer to Question 1. Verdicts use **Suspect**, **Possible**, **Insufficient evidence** and **Unlikely**; confidence is **High**, **Medium** or **Low**.

How the data is modelled. Every link joins a company (source) to a person (target). We resolve the raw records into 16,697 companies, 21,265 persons and 24,931 distinct company–person edges, after splitting 807 links that list several source companies and merging 374 company names with duplicate records (145 of them in more than one country). 8,568 linked company names have no registry record. A *business group* is a connected component: 13,587 of them, with the one giant component split by seeded Louvain into 86 slices. The pipeline is deterministic (byte-identical output across runs).

Question 1

> *Use visual analytics to identify anomalies in the business groups present in the knowledge graph. Limit your response to 400 words and 5 images.*

1. Name collisions create the giant component (Figures 1.1, 1.2). Persons are identified only by name. Of persons linked to one company, 21.9% have a top-50 surname; of those linked to 5+ companies, 92.6% do (Figure 1.3). The giant holds 7,093 nodes (1,506 companies). Removing the 651 multi-company persons with common surnames shrinks it to 306 nodes; removing as many random multi-company persons leaves 495–1,918 (10 trials). "John Smith", with 12 companies, is likely several people. Confidence: **High**.

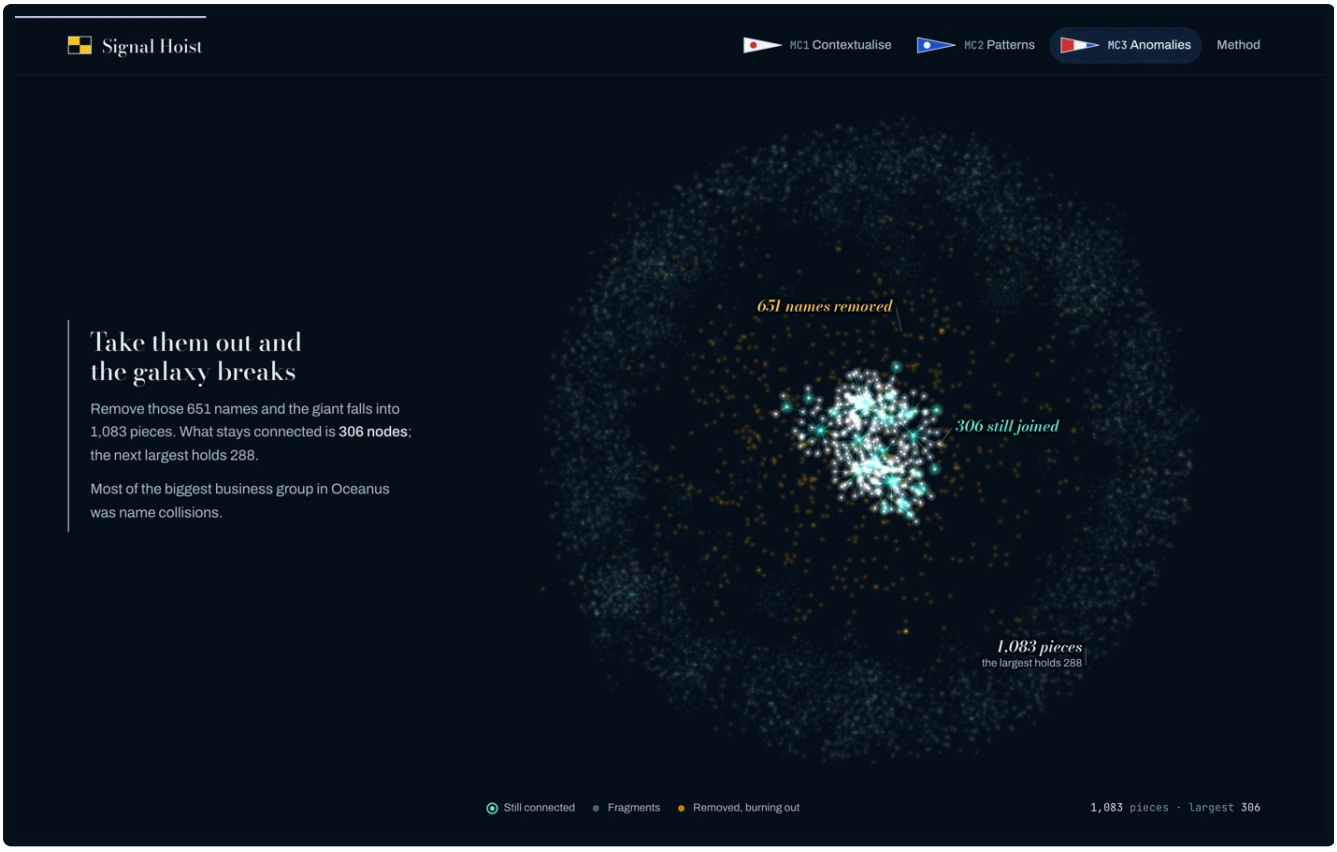


Figure 1.1. Story step: without the 651 common-surname people, the giant breaks into 1,083 pieces; 306 nodes stay joined.

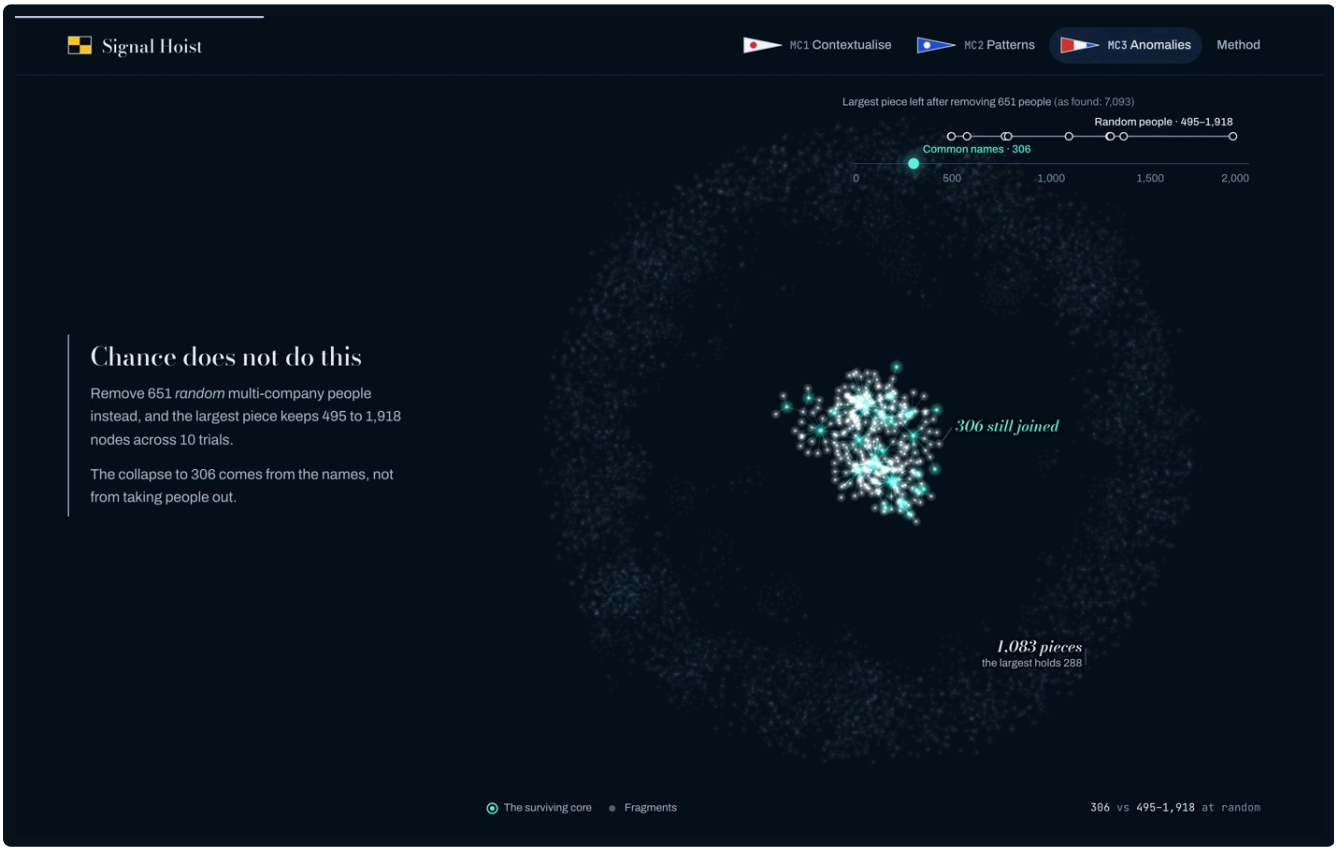


Figure 1.2. Story step: random removals leave 495–1,918 nodes joined (strip, top right), never 306.



Figure 1.3. 13 of the 14 people with the most companies have a top-50 surname.

2. Ownership fragmentation. Most companies have 0–2 beneficial owners, but 107 have 20+ and 38 have 30+. The largest are *Vespuci Sandbar Sp Brothers* (119), *Dutch Oyster Sagl Cruise ship* (90) and *Niger Bend AS Express* (71). 82 groups carry this flag. Confidence: **High**.

3. Serial owners with uncommon names (least likely to be collisions; Figures 1.4, 1.5). *Anna Wheeler* owns 7 companies and *Jason Carrillo* is the contact for 6 of them (BG00148). All 7 are unregistered or declare no product. BG00148 has the highest anomaly score of 2,347 scored groups. Confidence: **Medium**: two people sharing an uncommon name remain possible.

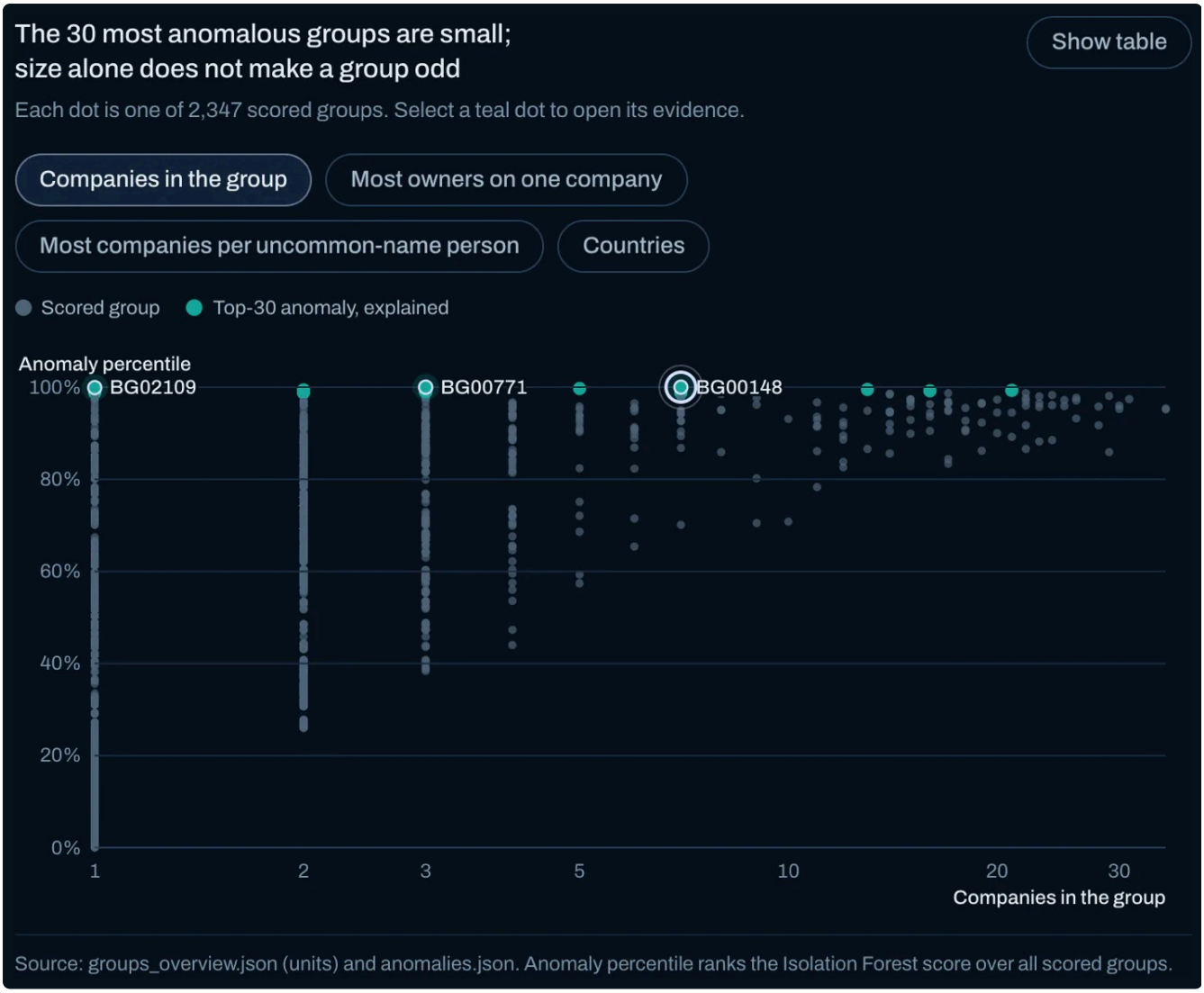


Figure 1.4. The 30 most anomalous groups (teal) are small.

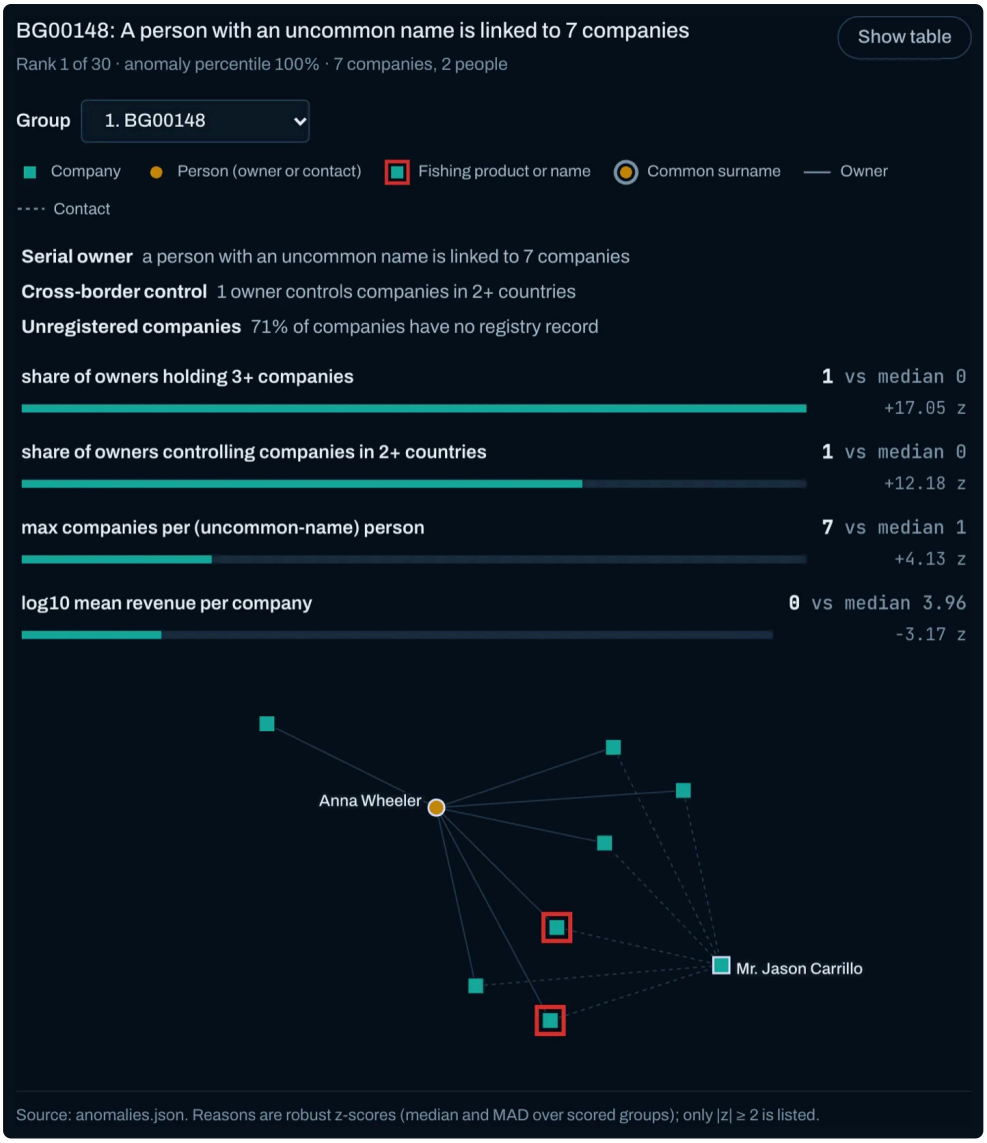


Figure 1.5. BG00148: Anna Wheeler owns 7 companies; Jason Carrillo is the contact.

4. Registry inconsistencies. 112 groups contain a distinctive company name registered in several countries. 166 groups (274 companies) hold a fishing-named firm declaring an unrelated product, such as *Dutch Oyster … Cruise ship*, which sells textiles. *Adriatic Squid* has both "Footwear" and "Seafood" records. Confidence: **Medium**.

5. Data-generation artefacts, not IUU signals: 1,391 names contain a double space where a word was stripped ("Greek ermen"); 113 product texts end in "source: freelance researcher"; 11,420 Faker-style names ("Smith LLC") all sit in country "ZH".

Method. Each group with ≥ 3 nodes is scored by an Isolation Forest over 16 size-normalised features, explained by robust z-scores (median/MAD), plus 8 rule flags.

Question 2

> *Develop a visual analytics process to find similar businesses and group them. This analysis should focus on a business's most important features and present those features clearly to the user. Limit your response to 400 words and 5 images.*

Process.

1. **Clean and resolve** records by name, as above.
2. **Classify product text** with keyword rules: *fishing* (1,043), *adjacent* (food, cold-chain or shipping; 623), *other* (2,209), *unknown* (12,822). Spot checks look right: "Exotic Coldwater Prawns" is fishing, "Freight forwarding" adjacent, "Gasket" other. Fishing words in company names are a second, independent signal on 1,201 names, including pinyin such as *yú* (fish) and *diao yu* (angling).
3. **Use two similarity spaces (Figure 2.1).** 77% of companies have no product text; in one shared space the binary "no text" columns dominate the clustering and inflate the silhouette. So:
    - **Text space** (3,875 companies): LSA of TF-IDF on 1–2 word terms (60 dimensions), weight 0.8, plus structural features, weight 0.2 (Figure 2.2).
    - **Structure space** (12,822 companies): structural features only (Figure 2.3).
4. **Choose k** by silhouette × bootstrap stability, because silhouette alone keeps rising with k on sparse text: k = 20 for text, k = 13 for structure (Figure 3.3).

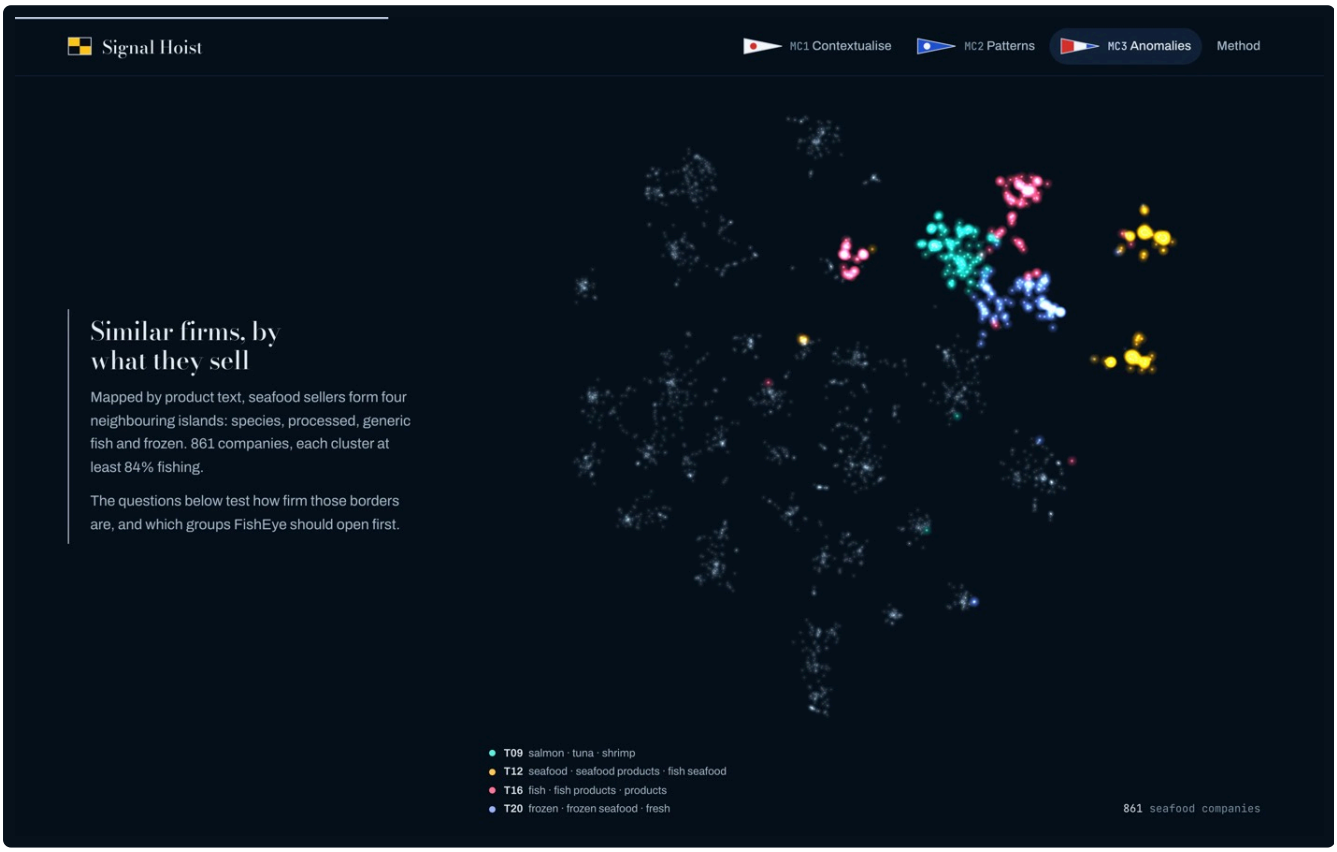


Figure 2.1. Story step: four seafood clusters lit on the text map, 861 companies.

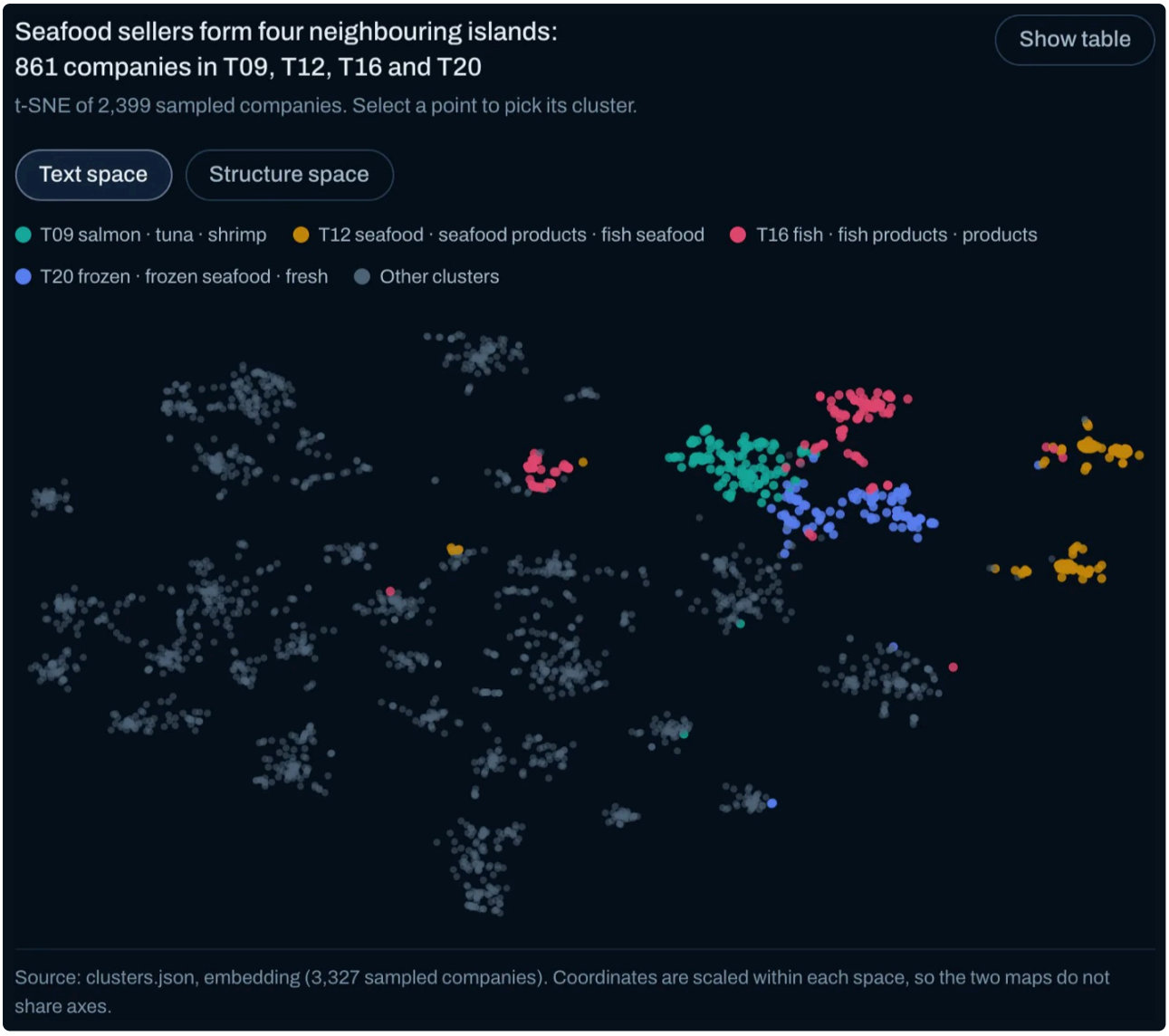


Figure 2.2. Text space: seafood sellers form four neighbouring islands.



Figure 2.3. Structure space: firms without product text group by registry, ownership and name words.

Most important features, presented with each cluster's top terms and key features (Figure 2.4):

- **Product terms** drive the text clusters. Four are seafood segments: species (T09: salmon/tuna/shrimp), processed seafood (T12), generic fish products (T16) and frozen seafood (T20). Together they hold 861 companies, each at least 84% fishing-labelled.
- **Ownership structure**: log owners, log contacts and *owner_reuse* (the most other companies held by any of a company's persons). It separates ownership-hub clusters such as S10 from single-contact shells such as S01.
- **Registry status and countries**: unregistered, number of countries and duplicate records give S07 (96 companies with duplicate records in several countries).
- **Fishing or maritime words in names** give S09 and S12, fishing-named firms with no product text; for unregistered companies this is the only fishing signal.
- **Revenue** presence and level separate S03 and S13, high-revenue firms with no product.

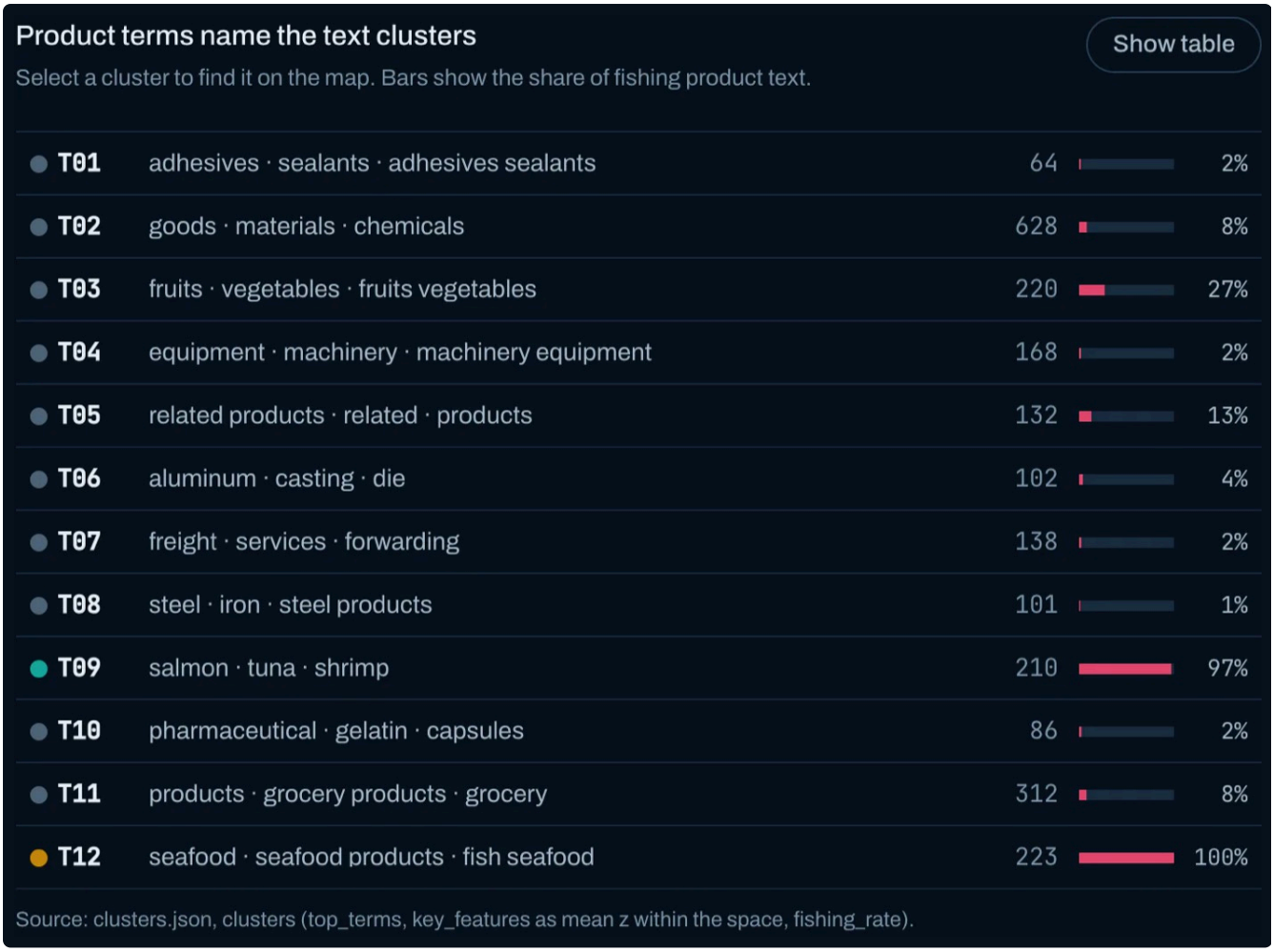


Figure 2.4. Top terms name each text cluster; bars give its fishing share.

Business groups are compared the same way: each group's 16 robust z-scores form a vector, and cosine similarity finds its nearest groups (listed beside each group in Figure 4.1).

Question 3

> *Measure similarity of businesses that you group in the previous question. Express confidence in your groupings visually. Limit your response to 400 words and 4 images.*

Similarity inside a grouping is measured three independent ways:

1. **Cohesion:** mean pairwise cosine between members, and mean silhouette per cluster.
2. **Stability:** clusterwise Jaccard (Hennig). We refit k-means on ten 80% subsamples and record how well each cluster is recovered.
3. **Membership certainty per company:** margin = $(d_2 - d_1) / d_2$, where d_1 and d_2 are the distances to the nearest and second-nearest centroid; 0 means the company sits on a boundary.

Confidence rule. *High*: stability ≥ 0.85 and silhouette ≥ 0.25. *Medium*: stability ≥ 0.6 and silhouette ≥ 0.05. Otherwise *Low*.

- **Text space:** 2 High, 10 Medium, 8 Low (mean stability 0.60), so the text segmentation is only moderately reliable (Figure 3.1). Processed seafood (T12; silhouette 0.49, stability 0.83) is solid. Generic "fish products" (T16; stability 0.47) blurs into the other seafood clusters.

- **Structure space:** 8 High, 2 Medium, 3 Low (silhouette 0.63, mean stability 0.82). Tighter, but much of that comes from binary features: easy to cluster, not rich.



Figure 3.1. Clusters sorted by stability, with bullet bars for stability and silhouette against the Medium and High ticks, and confidence pennants: only 2 of 20 text clusters reach High.

How confidence is shown visually.

- A t-SNE map per space (3,327 sampled companies), coloured by cluster, with opacity encoding margin so boundary companies fade (Figure 3.2).
- The cluster table with bullet bars for stability and silhouette against the Medium and High thresholds, plus confidence pennants (Figure 3.1).
- A k-selection chart of silhouette, stability and their product, showing why k was chosen (Figure 3.3).
- For business groups, the investigation view shows a confidence badge with the reasons behind it (Figure 4.1).

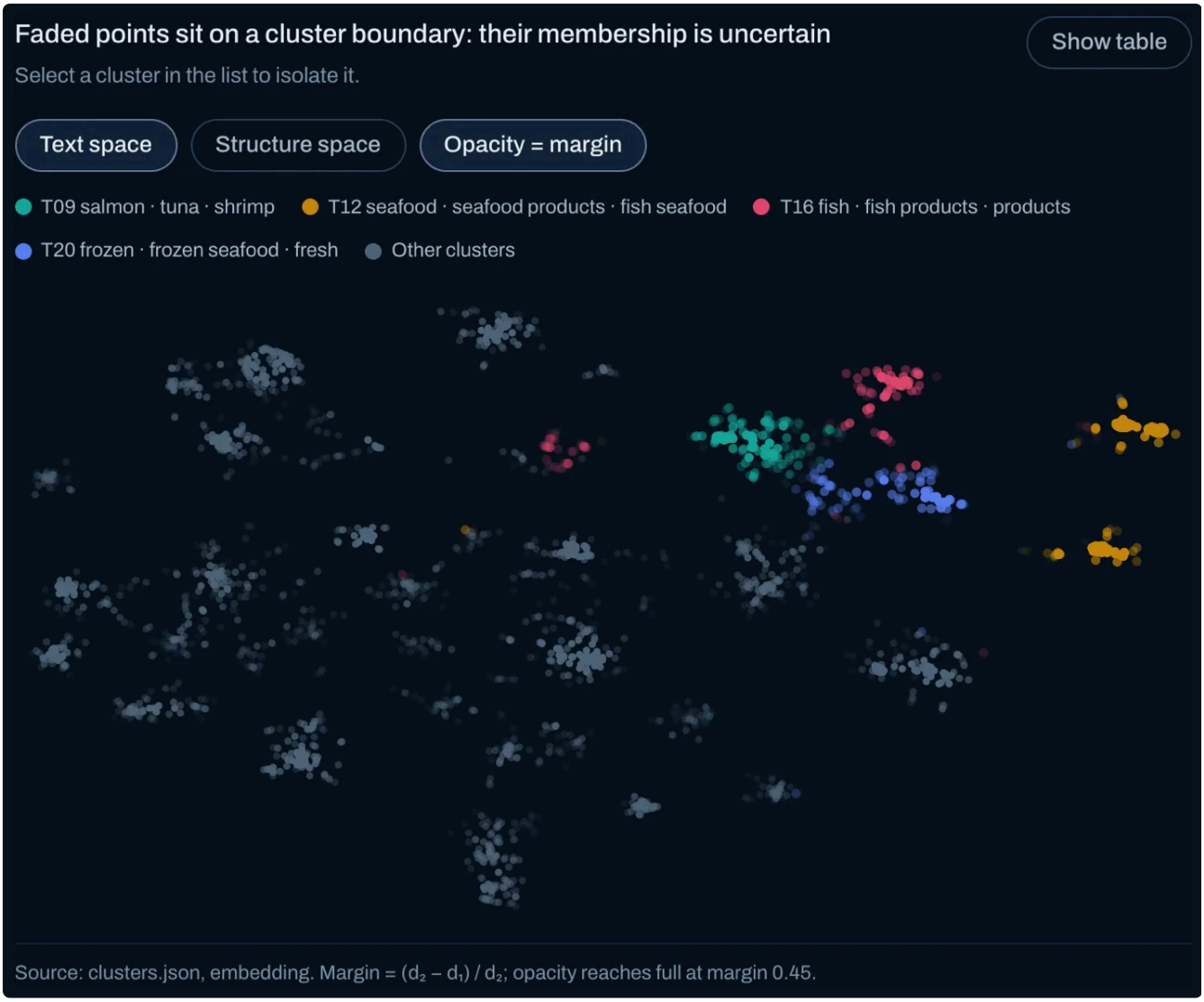


Figure 3.2. Opacity = margin: faded points sit near a cluster boundary, so their membership is uncertain.

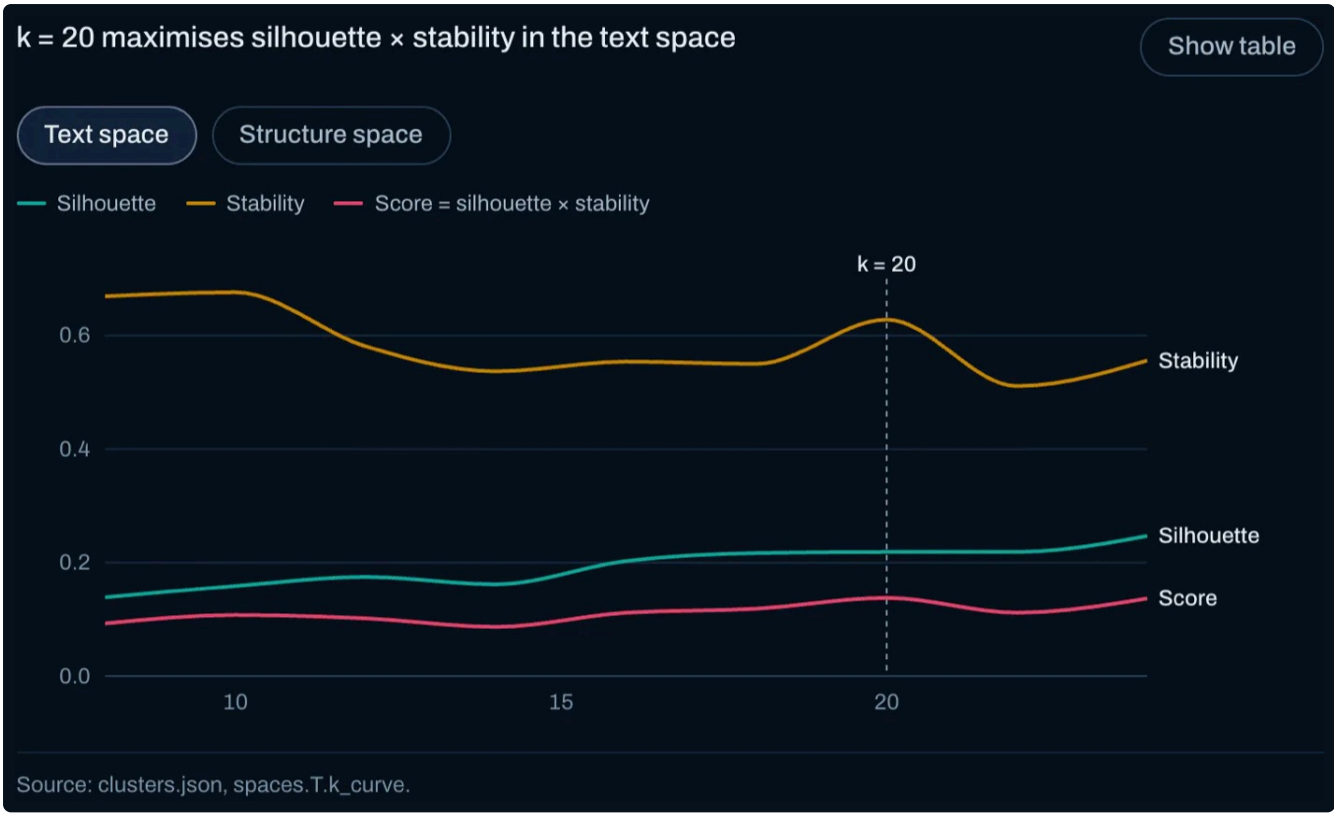


Figure 3.3. Silhouette, stability and their product against k in the text space; k = 20 maximises the product.

Confidence that the clusters are useful for segmentation: **Medium**. Confidence in fine-grained text clusters: **Low**.

Question 4

> *Based on your visualizations, provide evidence for or against the case that anomalous companies are involved in illegal fishing. Which business groups should FishEye investigate further? Limit your response to 600 words and 6 images.*

Short answer. Structural anomaly and fishing activity overlap only moderately. Of the 119 most anomalous groups (top 5%), 13% have fishing relevance of 0.5 or more, against 17% of the rest (Figure 4.2): anomaly alone does not point to fishing. The data cannot prove IUU fishing; it can only rank where to look.

The ranking score is 0.45 × anomaly percentile + 0.30 × flag score + 0.25 × fishing relevance (Figure 4.1). Fishing relevance combines the fishing share of product text with fishing words in names; the product share is shrunk towards the corpus rate of 26.9% (a prior worth 2 companies), so "1 of 1 informative companies is fishing" does not count as 100%. We list 30 groups, with at most 15 Louvain slices of the giant component, each with a rule-based verdict: **2 Suspect, 20 Possible, 7 Insufficient evidence, 1 Unlikely**.

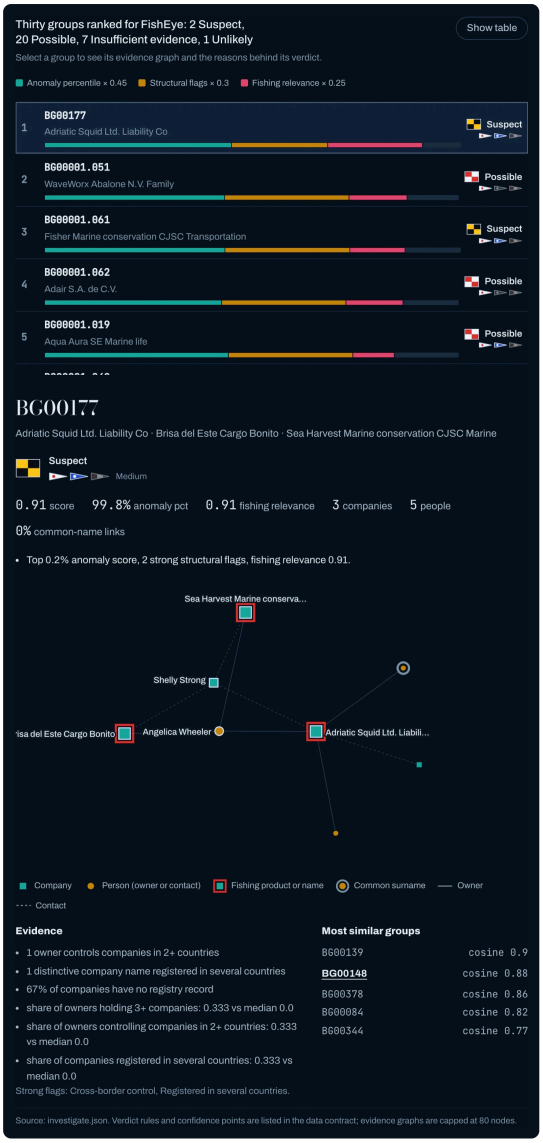


Figure 4.1. Thirty groups ranked for FishEye (bars split by the three score terms), with BG00177 open: verdict, evidence graph, reasons and most similar groups.

Investigate first:

1. **BG00177: Suspect, Medium confidence (Figures 4.1, 4.3).** A whole component of 3 fishing-named companies (*Adriatic Squid*, *Brisa del Este Cargo Bonito*, *Sea Harvest Marine*). One owner (*Angelica Wheeler*) and one contact (*Shelly Strong*) are shared across all three. *Adriatic Squid* is registered in both Marebak and Oceanus, with conflicting "Footwear" and "Seafood" records. Two of the three are unregistered. Anomaly in the top 0.2% of groups, fishing relevance 0.91, no name-collision exposure.
2. **BG00001.061: Suspect, Medium confidence.** A 16-company slice around *Fisher Marine conservation CJSC* and *Adriatic Tuna Seabass BV*. One company has 26 beneficial owners, a person with an uncommon name links 4 companies, and 62% of the companies are unregistered. As a Louvain slice, its boundary depends on community detection.
3. **BG00148: Possible, Medium confidence (Figures 1.5, 4.3).** The highest-anomaly group in the data. The *Anna Wheeler* / *Jason Carrillo* pair controls 7 companies across Oceanus and Coralmarica (*Baltic Sprat*, *jīn qiāng yú AG*, *Náutica del Mar*...). None declares a product, so the fishing link comes from names only.
4. **BG00069 and BG00071: Possible, Low confidence.** Small multi-jurisdiction clusters around *Yenisei Eel GmbH* (3 countries) and *Diao yu BV Logistics*. BG00069 would be Suspect, but half of the persons tying each group together have very common surnames.

5. **Single firms registered in several countries:** BG00861 *Diao yu bi sai SRL* and BG00386 *Delhi ers AS Holdings*, both Possible, Medium confidence.

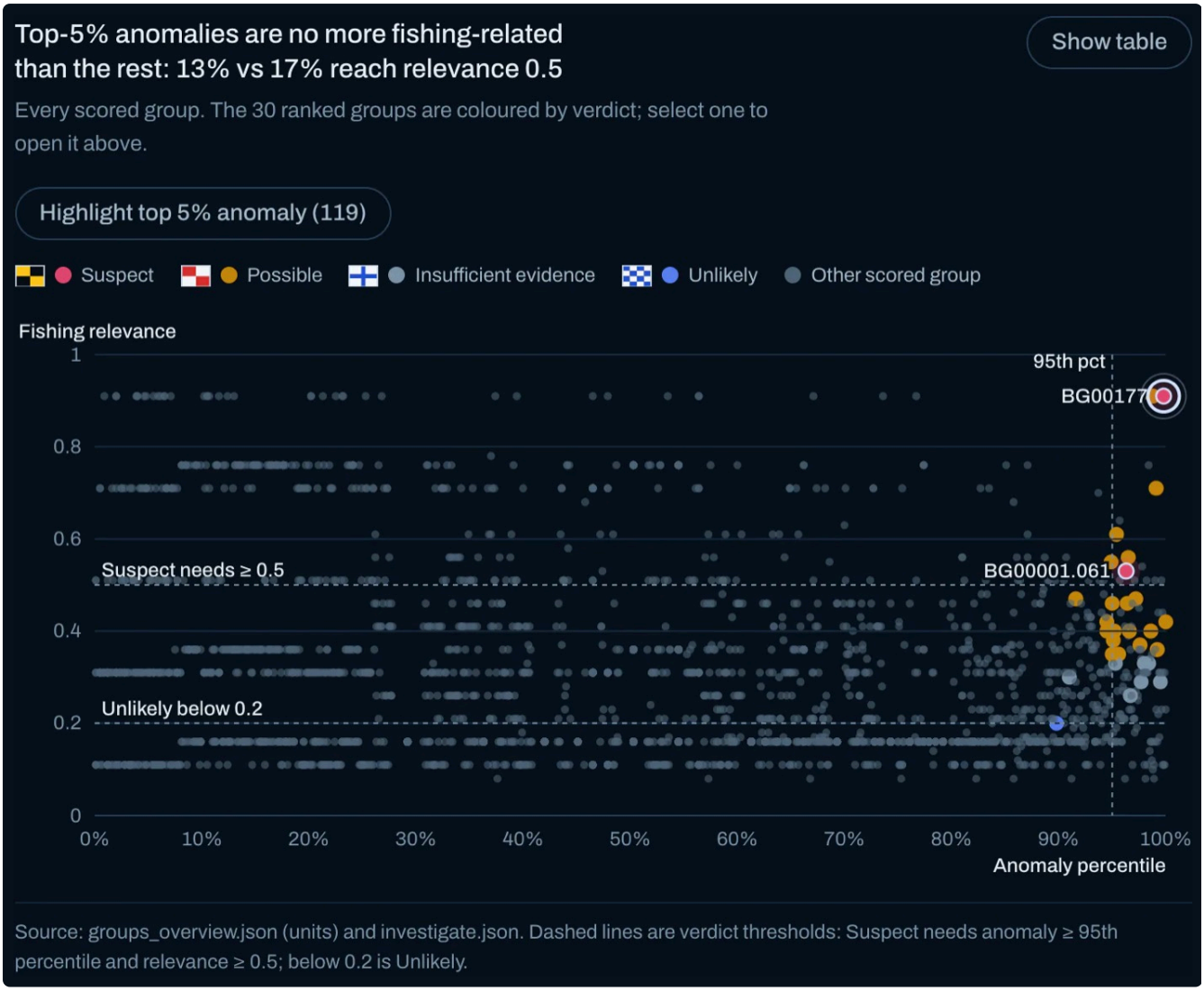


Figure 4.2. Every scored group by anomaly percentile and fishing relevance; the 30 ranked groups take their verdict colour. Top-5% anomalies are no more fishing-related than the rest.

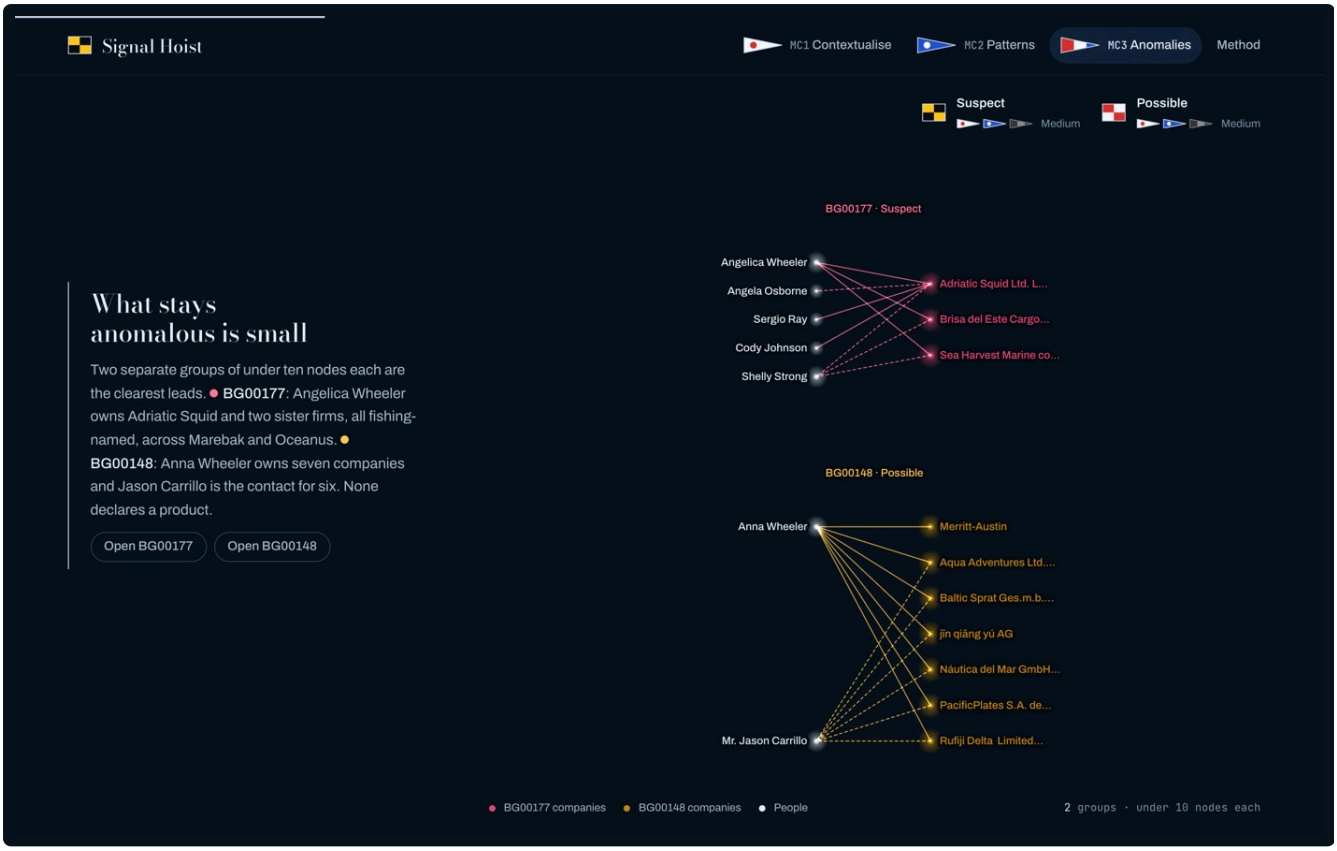


Figure 4.3. Story step "What stays anomalous is small": BG00177 (Suspect; five people around three fishing-named firms) and BG00148 (Possible; Anna Wheeler and Jason Carrillo around seven companies).

Evidence against, or where we cannot conclude. Several large groups have strong structure but no verifiable fishing link. BG00003 (23 companies, 113 persons, top company with 27 owners) is *Insufficient evidence* with **High** confidence. The ownership-hub slices (BG00001.001 around *Vespuci Sandbar*, 119 owners) sell wind-power equipment, textiles or beer: they look like shell or nominee structures, but not fishing ones. BG00016 is *Unlikely* (fishing relevance 0.20).

Caveats. 77% of companies have no product text, so most fishing relevance is inferred from names. Person identity is name-only, so collisions make shared control look larger than it is. Weights and thresholds are judgement calls, published with the tool so a reviewer can re-weight them.

Confidence. That the Suspect and Possible groups are a better-than-random IUU lead list: **Medium**. That any single group is engaged in IUU fishing: **Low**.

Question 5

> *Reflection: What was the most difficult aspect of working with this knowledge graph? Did you have the tools and resources you needed to complete the challenge? What additional resources would have helped you? Limit your response to 300 words*

The hardest problem was identity, not anomaly detection. Persons are bare names, so a naïve connected-component analysis produces one 7,093-node "business group", mostly a product of common names such as "Michael Johnson". Testing surname removal against a random control turned that suspicion into a measurement. Every later signal (serial owners, cross-border control) then had to be restricted to uncommon names, and confidence drops wherever a group depends on collision-prone links.

The second lesson was about clustering mixed data. A single feature space reported a reassuring silhouette of 0.49, only because 77% of rows had an all-zero text vector. Splitting text from structure, and choosing k by silhouette × bootstrap stability, gave lower but more honest numbers.

Reproducibility needed explicit work. `Graph.subgraph(set)` iterates in hash order, so Louvain returned 85 or 87 communities depending on the run. Rebuilding every graph with sorted insertion and breaking ties by id made the outputs byte-identical.

Limits. The fishing classifier uses keywords, so it misses euphemisms. Names with stripped words ("Greek ermen") show the generator removed tokens; fragment rules only partly recover them. The verdict rules are transparent but hand-set.

Tools and resources. The open-source Python stack (NetworkX, scikit-learn) and D3 were sufficient. Unique person identifiers, or any attribute beyond a name, would have helped most, along with a hand-labelled sample of fishing companies. With more time we would resolve persons by co-occurrence (two "John Smith"s who share no company are probably different people), validate the fishing labels against that sample, and test how sensitive the investigation ranking is to its weights.